55

Hematologic Aspects of Kidney Disease

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CHAPTER OUTLINE

KEY POINTS

- Anemia is relatively uncommon in earlier stages (stages G1–3) of CKD.
- The prevalence begins to increase significantly with an eGFR below 60 mL/min/1.73 m^2, but anemia is generally not a frequent or severe complication of CKD until the GFR is below 30 mL/min/1.73 m^2.
- Anemia is a more significant problem for younger women, older men, and blacks.
- Anemia occurs earlier in the course of disease and is often more severe among patients with CKD and diabetes mellitus.
- Screening for anemia (measurement of Hgb) should generally begin at CKD stage G3.
- Serum EPO concentrations are generally equal to or higher than those in patients without CKD.
- Mean serum EPO concentrations increase with worsening anemia in mild to moderate CKD, although to an insufficient degree.
- Mean serum EPO concentrations become more a function of GFR than of Hgb concentration when the GFR drops below around 40 mL/min/1.73 m^2.
- Even with advanced CKD, the ability to produce EPO is preserved. and some degree of responsiveness to lower Hgb is retained.

ANEMIA OF KIDNEY DISEASE

Reduced erythrocyte mass, or anemia, is one of the regular consequences of chronic kidney disease (CKD), because of the central role played by erythropoietin (EPO) in the regulation of erythropoiesis. Anemia can manifest itself early in the course of CKD, and its severity and prevalence go together with the progression of kidney disease. Given the significant effect of severe anemia on quality of life among patients with kidney failure, anemia is considered as one of the most clinically significant complications of this disease. Nevertheless, the direct consequences of CKD-related anemia and the degree to which anemia should be corrected in patients with CKD remain controversial.[1]

DEFINITION AND PREVALENCE OF ANEMIA IN CHRONIC KIDNEY DISEASE

Anemia is a state characterized by a reduced mass of red blood cells (RBCs) and hemoglobin (Hgb) concentration in blood, resulting in reduced oxygen-carrying capacity and delivery to the body's tissues and organs.[2–4] Because direct measurements of red cell mass are cumbersome and not readily available, anemia is defined as a reduction below the normal range for Hgb concentration and hematocrit (Hct); these values depend on gender, race, and age, with an increased prevalence of anemia in older adults.[5–8] The definition of anemia is somewhat arbitrary. The World Health Organization (WHO) defines anemia as an Hgb concentration below 13.0 g/dL for adult men and below 12.0 g/dL for adult women.[9] This definition has been adopted in the clinical practice guideline for anemia in CKD developed by Kidney Disease: Improving Global Outcomes (KDIGO),[10] whereas previous guidelines proposed a slightly higher threshold in men (13.5 g/dL).[11,12] Persons living at higher altitudes are characterized by a larger red cell mass and reduced Hgb oxygen affinity, compensatory changes required to maintain tissue oxygen delivery, given the reduced ambient oxygen tension at such altitudes.[13–19] Normal ranges for subjects living at high altitude are not well developed, resulting in potential misclassification of anemia.[20]

The prevalence of anemia in patients with CKD has been widely studied. In general, anemia is more frequent at lower levels of kidney function, becoming almost universal in those with end-stage kidney disease (ESKD; Figs. 55.1 and 55.2).[21,22] The prevalence reported in different studies depends on the definition of anemia and the target population. The most useful analyses are those that were community-based, avoiding biases inherent in studies of clinic-based populations. Hsu and coworkers have studied 12,055 adult ambulatory subjects from health clinics in Boston using the Cockcroft-Gault equation to estimate creatinine clearance and the Modification of Diet in Renal Disease (MDRD) formula to estimate the glomerular filtration rate (GFR) indexed to body surface area.[23] They found that mean Hct values were progressively lower with creatinine clearance below 60 mL/min/1.73 m^2 in men and below 40 mL/min/1.73 m^2 in women. Moderately severe anemia (Hct <33%) was present in more than 20% of patients when the GFR was below 30 mL/min/1.73 m^2 in women and below 20 mL/min/1.73 m^2 in men.[23] Similar results have been obtained in different populations. In Japan, a study of 54,848 subjects identified an estimated GFR (eGFR) threshold of 60 mL/min/1.73 m^2 for both genders, below which anemia prevalence increased significantly and a threshold of 45 mL/min/1.73 m^2 for the association of anemia with complications.[24] Hsu and coworkers conducted a second study, using the third National Health and Nutrition Examination Survey (NHANES III; 1988–1994) of 15,971 adults older than 18 years, with measurements of serum creatinine and Hgb values and iron indices. Creatinine clearance was estimated using the Cockcroft-Gault formula.[25] A statistically significant lower mean Hgb was found in men and women with creatinine clearances below 70 mL/min and 50 mL/min, respectively, than in those with creatinine clearances more than 80 mL/min. However, a mean decrease of 1.0 g/dL was found only for those with a creatinine clearance less than 30 mL/min. Astor and colleagues[22] studied the same NHANES III data as Hsu and coworkers but restricted analysis to a different age range, selecting 15,419 participants 20 years of age and older. Anemia according to the WHO definition was present in 7.3% of all participants (see Fig. 55.1). Functional iron deficiency and absolute iron deficiency were found to be important predictors of anemia.[22] Another study on the same dataset has shown that Hgb values below 11 g/dL were present in 42.2% of subjects with an eGFR below 30 mL/min/1.73 m^2 and in 3.5% of subjects with an eGFR between 30 and 60 mL/min/1.73 m^2 (MDRD formula).[26] Stauffer and Fan, using the 2007–2008 and 2009–2010 NHANES dataset, have estimated that 14.0% of the US adult population have CKD, with anemic individuals having a twofold greater prevalence of CKD than the general population (15.4% vs. 7.65%).[27]

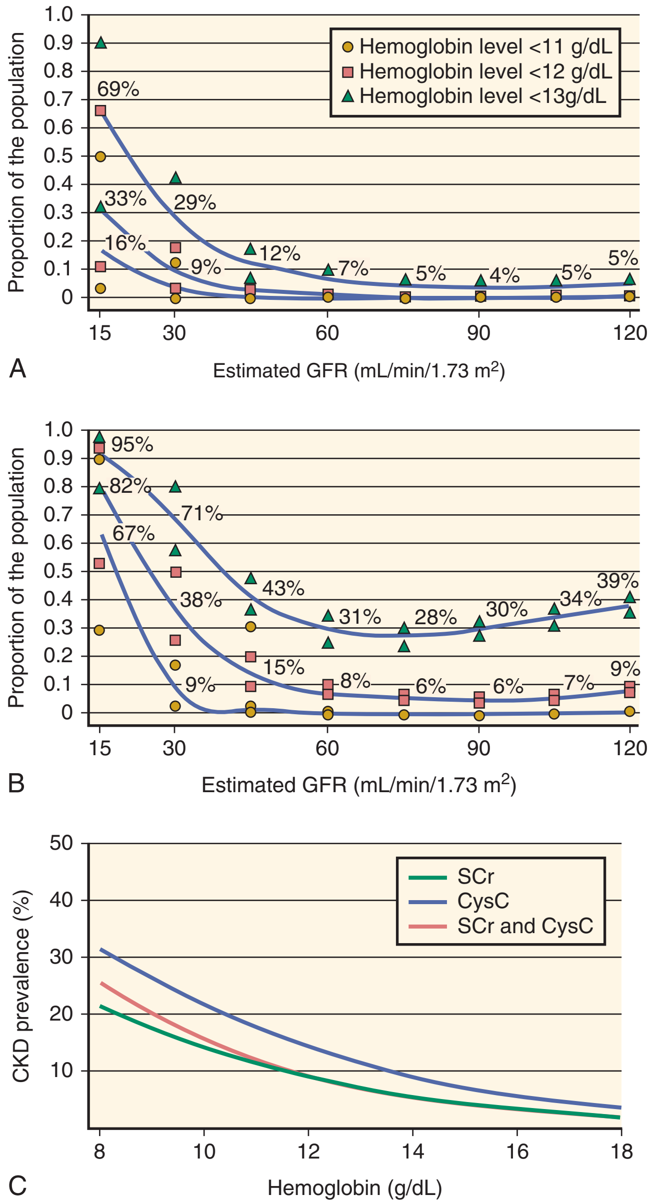

Fig. 55.1 Prevalence of hemoglobin (Hgb) value less than 11 g/dL, 12 g/dL, and 13 g/dL among men (A) and women (B) 20 years of age and older from the Third National Health and Nutrition Examination Survey (NHANES III; 1988–1994). All values are adjusted to the age of 60 years. (C) Predicted prevalence of chronic kidney disease *(CKD)*. This is defined as an estimated glomerular filtration rate *(eGFR)* of 1 to 59 mL/min/1.73 m^2 using different GFR-estimating methods in US adults age 20 years or older by hemoglobin. The eGFR values are based separately on serum creatinine *(SCr)*, serum cystatin C *(CysC)*, and combined serum creatinine and cystatin C (SCr and CysC). Prevalence curves are truncated when the number of relevant participants is less than 30. (A and B modified from Astor B, Muntner P, Levin A, et al. Association of kidney function with anemia. Arch Intern Med. 2002;162:1401–1408; C from Estrella MM, Astor BC, Köttgen A, et al. Prevalence of kidney disease in anaemia differs by GFR-estimating method: the Third National Health and Nutrition Examination Survey [1988-94]. Nephrol Dial Transplant. 2010;25:2542–2548.)

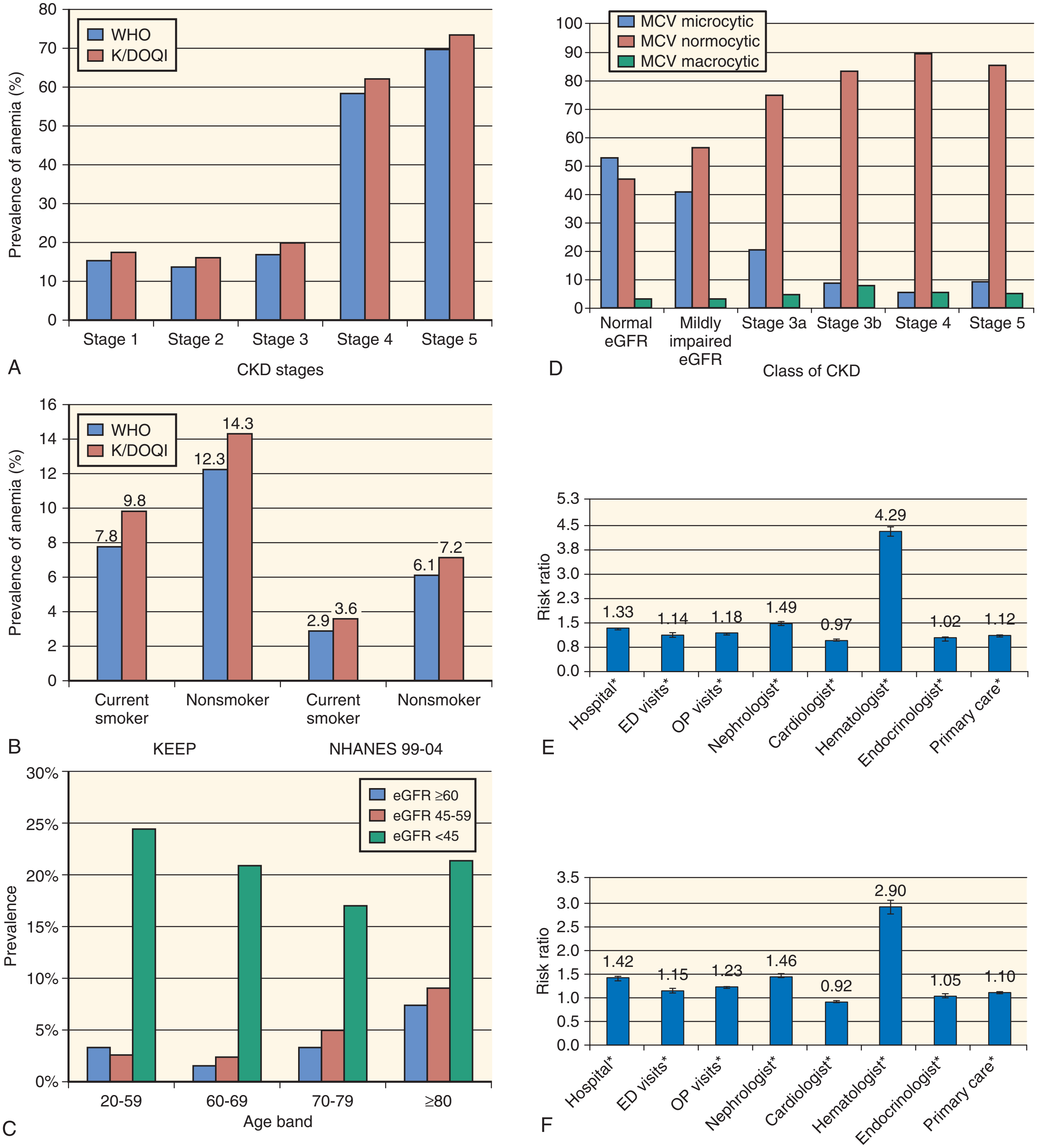

Fig. 55.2 (A) Prevalence of anemia by stage of chronic kidney disease *(CKD)* in the Kidney Early Evaluation Program *(KEEP)*. (B) Prevalence of anemia by smoking status. (C) Age-specific prevalence of anemia in CKD according to estimated glomerular filtration rate *(eGFR)*. (D) Rates of microcytic, normocytic, and macrocytic anemia for each stage of CKD. Shown are adjusted hazard ratios of health care utilization in stage 3–5, non–dialysis-dependent CKD patients with and without anemia. (E) Medicare-covered patients aged 66–85 years. (F) Commercially insured patients aged 18 to 63 years. Results were from Poisson regression models, with the number of each type of health service as the dependent variable adjusting for patient demographics, baseline comorbidities, inflammatory conditions, and CKD stage. *$P < .05$. *ED,* Emergency department; *K/DOQI,* Kidney Disease Outcomes Quality Initiative; *MCV,* mean corpuscular volume; *NHANES,* National Health and Nutrition Examination Survey; *OP,* outpatient; *WHO,* World Health Organization. (A and B from McFarlane SI, Chen SC, Whaley-Connell AT, et al. Prevalence and associations of anemia of CKD: Kidney Early Evaluation Program [KEEP] and National Health and Nutrition Examination Survey [NHANES] 1999–2004. Am J Kidney Dis. 2008;51[Suppl]:S46–S55, 2008; C and D from Dmitrieva O, de Lusignan S, Macdougall IC, et al. Association of anaemia in primary care patients with chronic kidney disease: cross-sectional study of quality improvement in chronic kidney disease [QICKD] trial data. BMC Nephrol. 2013;14:24; E and F from St Peter WL, Guo H, Kabadi S, et al. Prevalence, treatment patterns, and healthcare resource utilization in Medicare and commercially insured non-dialysis-dependent chronic kidney disease patients with and without anemia in the United States. BMC Nephrol. 2018;19:67)

The worldwide prevalence of anemia associated with CKD rose from 1990 to 2010. CKD is currently the sixth leading cause of anemia in women and the ninth in men worldwide.[28]

Anemia is more common among women and non-Hispanic blacks. In the latter population, the risk for anemia was generally more than twice that in non-Hispanic whites in one study.[25] Another study reported a 3.3-fold higher prevalence of anemia in blacks than in whites, with CKD being less common in anemic blacks than in anemic whites (22% vs. 34%), suggesting a higher prevalence of non–CKD-related anemia as well.[29] The investigators extrapolated the data to the general US population and estimated that approximately 1,590,000 Americans with a creatinine clearance less than 50 mL/min are anemic, with an Hgb concentration lower than 12 g/dL.[25] A targeted community-based screening program for CKD has confirmed a threefold higher likelihood of anemia in blacks than in whites, as well as a twofold higher prevalence of anemia in this higher risk population compared with NHANES population survey[30]; the same study reported a lower prevalence of anemia in smokers than in nonsmokers, which has been attributed to enhanced stimulation of erythropoiesis due to relative hypoxia (see Fig. 55.2).

It has been advocated that the traditionally accepted eGFR threshold of 60 mL/min/1.73 m^2 to define CKD should be modified to higher values in blacks, in whom metabolic abnormalities and anemia are more common and are present at higher eGFR values than in whites.[31] In a large retrospective analysis, anemia was associated equally among blacks and whites with ESKD.[32] However, in patients on dialysis, the Hgb threshold below which higher mortality rates are observed is higher in blacks than in whites (11 g/dL vs. 10 g/dL).[33] In any case, observations on race and ethnicity cannot be interpreted in isolation, but must take into consideration socioeconomic status and cultural and behavioral differences.[34]

For people with an eGFR of 30 to 59 mL/min/1.73 m^2, low concentrations of 25-hydroxyvitamin D—25(OH)D—and elevations in C-reactive protein (CRP) were independently associated with Hgb concentrations below 12 g/dL.[35] A similar association has been described in ESKD patients who had undergone kidney transplantation.[36] Other studies have also reported an independent association of high-sensitivity CRP results with anemia in patients with an eGFR less than 60 mL/min/m^2.[37]

Anemia develops earlier in the course of CKD in patients with diabetes mellitus, and its magnitude tends to be more severe in patients with diabetes mellitus than in patients without diabetes.[8,38–46] El-Achkar and colleagues have studied 5380 community-dwelling patients surveyed as part of the Kidney Early Evaluation Program (KEEP), a community-based screening initiative for patients at high risk for kidney disease.[44] Anemia was more prevalent among patients with diabetes and developed earlier than in patients without diabetes mellitus. In patients with CKD stage G3 (GFR, 30–59 mL/min/1.73 m^2), 22.2% of those with diabetes were anemic; in patients with CKD stage G4 (GFR, 15–29 mL/min/1.73 m^2), the prevalence was 52.4%. The difference among patients with and without diabetes was most prominent in patients with CKD stage G3, in whom the prevalence of anemia was nearly threefold greater in those with diabetes. Men with diabetes were particularly prone to anemia, more so than women.

The prevalence of anemia in persons with diabetes and normal kidney function can be as high as 32%, with aggravating factors being advanced age and thiazolidinedione (glitazone) therapy.[47] Symeonidis and associates have explored the mechanism for anemia in diabetes by studying 694 anemic individuals, of whom 237 had diabetes.[45] Serum EPO concentrations were found to be lower in subjects with diabetes, particularly in relation to the degree of anemia present, with a significant inverse correlation between serum EPO value and the fraction of glycosylated Hgb. Thomas and associates have studied the contribution of proteinuria to anemia in 315 Australian patients with type 1 diabetes.[43] The prevalence of anemia was found to be higher in patients with macroalbuminuria than in those with microalbuminuria or with no albuminuria (52% vs. 24% vs. 8%, respectively).

A large study of 79,985 adults with diabetes mellitus has shown a higher risk of anemia in black subjects and a lower one in Asian subjects in comparison with white subjects.[48] A recent study in a large number of Medicare and commercially insured US patients has confirmed the greater prevalence of anemia in older patients, women, blacks (Medicare only), in the presence of comorbidities, and with increased CKD severity.[49] In anemic patients, RBC transfusion, erythropoiesis-stimulating agents (ESAs), and IV iron were used in 11.7%, 10.8%, and 9.4%, respectively, in younger patients (ages 18–63 years; $n = 15{,}716$); corresponding figures were 22.2%, 12.7%, and 6.7% in older patients (ages 66–85 years; $n = 109{,}251$), highlighting the fact that in this 2012 dataset, RBC transfusion was the most common therapy for anemia.[49] Anemic patients were also more likely to use health resources (see Fig. 55.2).

With aging, kidney function tends to decline progressively. The interaction of aging and loss of kidney function might be expected to raise the prevalence of anemia. In actuality, the relation is more complex.[50] Men with CKD tend to have a higher prevalence of anemia with older age, but among women with CKD, anemia is more frequent at a younger age.[25] It is likely that the high prevalence of iron deficiency in menstruating women accounts for this difference. However, if the analysis is limited to older men and women, the association between older age and anemia is clearer. Ble and colleagues have studied 1005 community-living older adults in Italy (InCHIANTI study).[51] The prevalence of anemia was found to increase with age in both genders. By multivariable analyses, much of the risk for anemia segregated to individuals with a creatinine clearance less than 30 mL/min, who also had lower mean serum EPO concentrations. Another InCHIANTI analysis has shown that lower than normal total and bioavailable testosterone concentrations resulted in a significantly higher risk for the development of anemia at 3-year follow-up for men and women.[52] In a study of 6200 nursing home residents (mostly white women), the prevalence of anemia and CKD were 60% and 43%, respectively.[53] Age was an important determinant of anemia in the absence of CKD, whereas this effect was lost in the presence of CKD, which became the strongest determinant of anemia.[53,54] One-third of the anemias found in older adults (>65 years) may be unexplained, but significant associations are present between anemia and low EPO concentrations and low lymphocyte counts.[55]

Hemoglobin values in patients with advanced CKD are frequently confounded by the use of ESAs (see later) and iron. Although there had been an increase in ESA use before

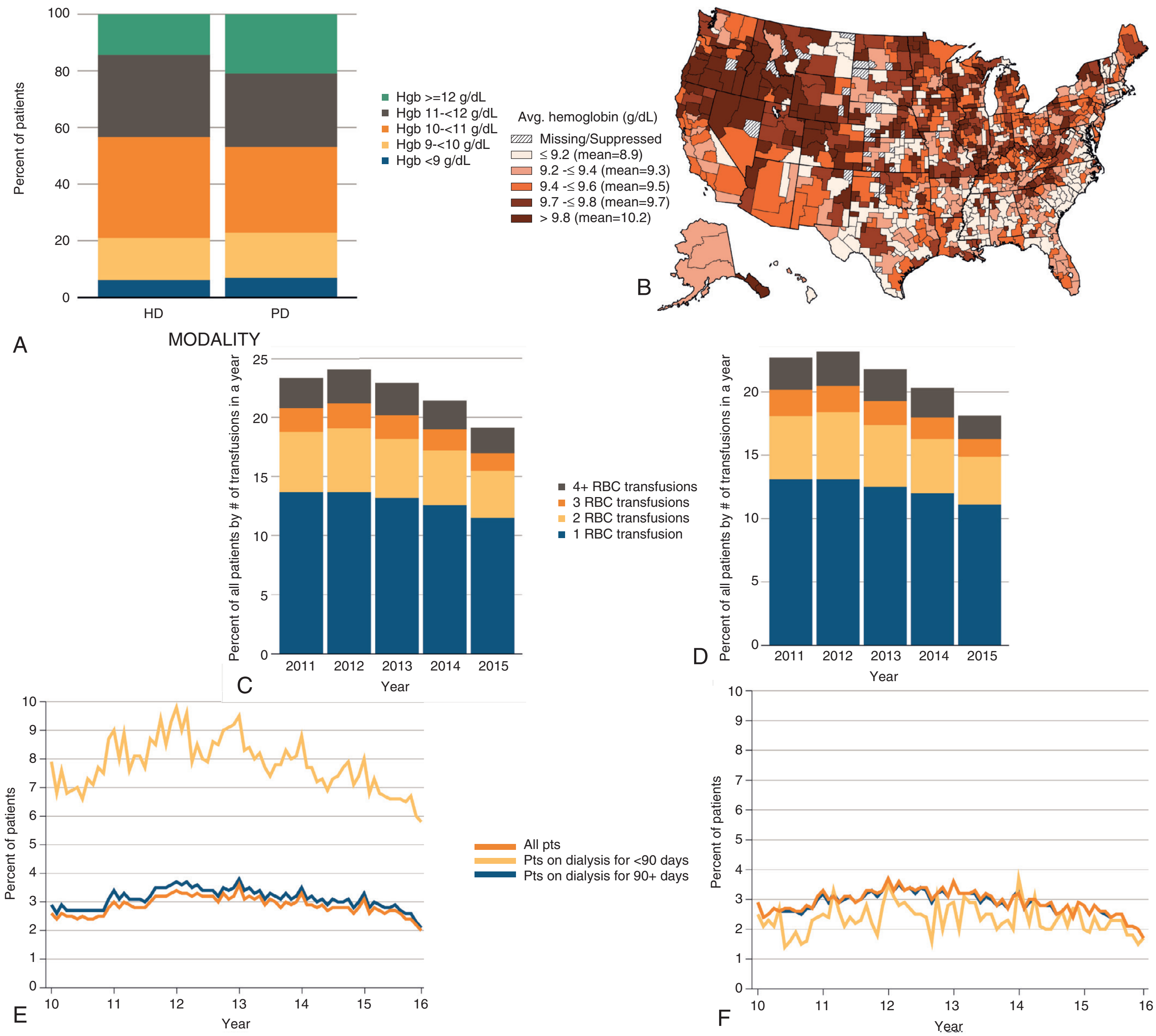

Fig. 55.3 Anemia and treatment in end-stage kidney disease. (A) Percentage distribution of Hgb levels among prevalent hemodialysis and peritoneal dialysis patients. (B) Map of average hemoglobin level at initiation of renal replacement therapy, by health service area, 2011–2015. (C) Percentage of all adult hemodialysis patients by number of red blood cell transfusions received in a year. (D) Percentage of all adult peritoneal dialysis patients by number of red blood cell transfusions received in a year. (E) Percentage of all patients, patients on dialysis less than 90 days, or patients on dialysis 90 days or more who had one or more claims for a red blood cell transfusion in a month. (F) Percentage of all adult peritoneal dialysis patients with one or more claims for a red blood cell transfusion in a month. (From United States Renal Data System, 2017 annual data report. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2017. A from CROWNWeb data, May 2016 vol 2 Figure 2.1.b; B from vol 2 Figure 1.20; C from Medicare claims data 2011-2015, vol 2 Figure 2.7.a; D from Medicare claims data 2010-2015, vol 2 Figure 2.13.a; E from Medicare claims data 2011-2015, vol 2 Figure 2.7.b; F from Medicare claims data 2010-2015, vol 2 Figure 2.13.b.)

the initiation of dialysis and in patients on dialysis in the past, this trend has reversed after publication of the results of randomized controlled trials (RCTs)—changes in prescribing instructions and new guidelines (see later). In fact, there has now been a steady decline in the use of ESAs before the initiation of dialysis (Fig. 55.3). The Hgb concentration at the initiation of dialysis has been declining since 2007 and, in most patients beginning hemodialysis in the United States, the Hgb value is now below 10 g/dL.[56,57]

PATHOBIOLOGY OF ANEMIA IN CHRONIC KIDNEY DISEASE

NORMAL ERYTHROPOIESIS

The delivery of oxygen to peripheral tissues is a highly regulated process; a crucially important determinant is the red blood cell mass, which is determined by the dynamic balance between the removal of older cells from the circulation and the production of newer cells by the bone marrow.

Under normal conditions, approximately 1% of circulating erythrocytes is replaced daily, corresponding to about 250 billion erythrocytes, with 2.5 to 3.0 million erythrocytes being produced each second.[58] The control of red blood cell mass is based on a classic negative feedback loop, mediated by changes in the production of the hormone EPO. EPO is mainly produced in the kidney and regulates the production of erythrocytes by interaction with specific EPO receptors (EPO-Rs) on bone marrow erythroid progenitors. For this mechanism to function properly, several other cofactors such as iron, vitamin B_{12}, and folic acid are also required.

Erythropoietin

EPO, the major regulatory hormone of erythrocyte production, is a 30.4-kDa glycoprotein. Its production in the kidney is modulated by the delivery of oxygen from the circulating erythrocytes. When the mass of the circulating erythrocytes decreases because of decreased production, enhanced destruction, or loss of erythrocytes, the reduction in oxygen delivery results in increased production of this hormone. The first recognition of the linkage between hypoxia and erythrocyte quantity arose from astute 19th century observations on the effects of living at a higher altitude.[59,60] Carnot and Deflandre first postulated that a humoral factor (a so-called hemopoietin) might regulate erythropoiesis.[61] They injected serum from anemic rabbits into normal animals, resulting in increased reticulocyte counts; they termed the circulating factor "hematopoietin." However, in retrospect, this observation was probably an artifact because the amount of serum transferred was too low, and attempts to confirm their results were unsuccessful.[62]

44 years later, Reissmann rekindled interest in the field with ingenious experiments in parabiotic rats (artificial conjoined animals).[63] In this model, rats were joined by skin and muscle, ear to tail, living for 3 months in parabiosis. When one animal breathed air with low oxygen tension and the other breathed normal air, both animals demonstrated increased bone marrow erythropoiesis. This finding provided strong evidence that a humoral factor was the stimulus for erythropoiesis. In 1953, Erslev definitively demonstrated the erythropoietic role of the serum factor, now termed "erythropoietin."[64] He infused 100 to 200 mL of plasma from bled anemic rabbits into normal rabbit recipients. The reticulocyte count increased rapidly, with a fourfold increase in cell count within 4 days of infusion.[64,65] In 1957, Jacobson and coworkers provided indirect evidence to suggest that the kidneys were the primary source of EPO.[66] After demonstrating that removal of a variety of different organs did not affect EPO production after phlebotomy, they showed that nephrectomized rats and rabbits failed to increase EPO production (incorporation of iron-59 into erythrocytes) after blood loss.[66] Further studies by Koury and Lacombe and associates have demonstrated that the cells responsible for EPO production are peritubular interstitial cells,[67,68] which were subsequently identified as peritubular fibroblasts, located in the renal cortex.[69–71] Although the nature of these cells has still not been fully clarified, they have been suggested to be derived from the neural crest[72] and share characteristics of pericytes.[73,74]

It has also been shown that with the growing severity of anemia, the number of EPO-producing peritubular cells increases. This recruitment was found mostly in the inner cortex, but appeared also throughout the renal cortex when the anemia was particularly severe.[75,76] EPO production is regulated by a specific hypoxia-sensing mechanism based on transcription factors stabilized by hypoxia, called "hypoxia-inducible factors" (HIFs; Fig. 55.4).[74] This regulatory mechanism is not unique to EPO and is based on the capability of two separate helix loop–helix components, HIF-α and HIF-β, to bind as a complex to specific hypoxia-responsive DNA elements, which regulate the transcription of hypoxia-inducible genes.[77] The concentrations of the beta subunit do not respond to hypoxia.[78,79] The alpha subunits (1α, 2α, and 3α) are produced constitutively but are rapidly degraded in the presence of oxygen by the ubiquitin-proteasome system.[80–83] In hypoxic conditions, degradation of the alpha subunits is inhibited, leading to rapid increases in HIF-α concentrations and to the formation of the HIF transcription complex. For EPO regulation HIF-2 appears to be the important HIF isoform.[84–89] Renal HIF-2 is required for hypoxia-driven EPO production; in the absence of renal HIF-2, hepatic HIF-2 becomes the main regulator of EPO production.[74,84] HIF-2α, together with HIF-ß, hepatocyte nuclear factor-4 (HNF-4), and p300 bind to a 120-bp enhancer, which is located at the 3' end of the human EPO polyadenylation signal.[58,79,90–92] This interaction results in rapid EPO transcription, followed by translation and secretion of the EPO glycoprotein.[93–96]

The rapid degradation of HIF-α in the presence of oxygen depends on binding of the tumor suppressor protein von Hippel-Lindau (VHL), a process that results in tagging of the molecule for proteasomal degradation via polyubiquitination by ubiquitin ligase.[97,98] This regulatory mechanism is based on the hydroxylation of two proline residues, which are critical for the recognition of HIF-α by VHL. An additional hydroxylation of one asparagine residue is required for HIF binding with p300.[99–103] Hydroxylation at these three sites depends on the presence of oxygen as a molecular substrate for specific hydroxylase enzymes, placing these enzymes in a central role for sensing oxygen and detecting hypoxia. A mutation in the VHL protein that impairs the degradation of HIF-α, and increases EPO production causes the so-called Chuvash congenital polycythemia, an autosomal recessive disorder that is endemic in the mid–Volga River region.[104] HIF-2α but not HIF-1α displays a typical iron response element (IRE) in its 5′ untranslated region (UTR), which has been shown in mice to constitute an important regulatory loop for HIF-2α messenger RNA (mRNA) translation in the presence of hypoxia or iron loading.[105] This regulatory system in conditions of reduced iron availability would allow iron response protein 1 (IRP1) to bind with high affinity to IRE, inhibit mRNA translation, and decrease HIF-2α synthesis and EPO production. When cellular iron is abundant, IRP1 loses its RNA binding activity and becomes a cytosolic aconitase, resulting in derepression of mRNA translation, which increases HIF-2α synthesis and EPO production.

Purification and identification of EPO is credited to Miyake and colleagues.[106] From 2550 L of urine from patients with aplastic anemia, and using multiple isolation steps, they obtained a small quantity of pure glycoprotein.[106] Purification of human EPO led to successful cloning of the gene, reported in 1985 by Lin and associates.[107] They found that the gene encodes a protein of 193 amino acids, including a 27–amino acid leader sequence and a terminal single amino acid that are cleaved during processing, resulting in a 165–amino acid mature EPO molecule. When these investigators introduced

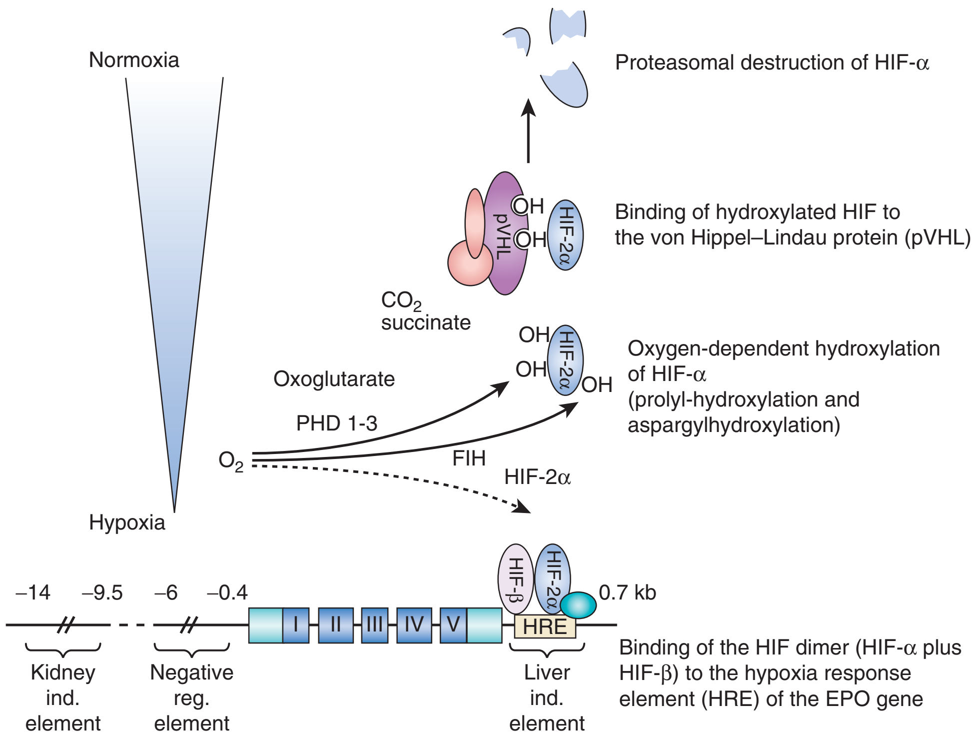

Fig. 55.4 Schematic presentation of hypoxia-inducible factor *(HIF)* signaling and the oxygen-dependent control of erythropoietin *(EPO)* gene expression. HIF consists of one of two oxygen-dependent alpha subunits, HIF-1α and HIF-2α, and a constitutive beta subunit. For EPO regulation, HIF-2α is the relevant isoform. In the presence of oxygen (normoxia), HIF-α is hydroxylated at two prolyl and one asparagyl residues through prolyl hydroxylases (PHDs 1–3) and an asparagyl hydroxylase (factor-inhibiting HIF *[FIH]*), enzymes that require oxoglutarate as a cosubstrate. Hydroxylation of the asparagyl residue inhibits binding of the transcriptional coactivator p300, and hydroxylation of the prolyl residues enables binding to the von Hippel-Lindau protein, which represents the recognition component of an E3 ubiquitin ligase. Thus, hydroxylated HIF-α is targeted for proteasomal destruction, and hydroxylated HIF that escapes destruction is not transcriptionally active. Under hypoxia, there is no substrate (oxygen) for the hydroxylation reactions, and thus HIF-α is stabilized, can bind to hypoxia-responsive elements of its target genes, and can induce or enhance their transcription. The hypoxia response element *(HRE)* of the EPO gene is located at 5′ of the gene; other regulatory elements determine the tissue specificity of its expression, limiting EPO expression mainly to liver and kidneys. *ind.,* Inducible; *reg.,* regulatory.

the gene into Chinese hamster ovary cells, EPO with full biologic activity was produced.[107] These findings were confirmed by an almost simultaneous report by Jacobs and colleagues.[108] The results led in short order to the development of techniques to produce recombinant human EPO (rhEPO). By 1989, clinical trials of rhEPO had demonstrated its remarkable efficacy,[109–112] leading to regulatory approval and routine clinical use of EPO as replacement treatment.

EPO itself is a member of the family of class 1 cytokines.[113] The carbohydrate moiety is important for molecular stability, whereas the 165–amino acid protein component is critical for receptor binding.[114–116] There are four discrete carbohydrate chains, three *N*-linked and one *O*-linked, each of them having two to four branches, most of which end with a negatively charged sialic acid.[114,117–121] The physiologic role of the carbohydrate chains is complex. They seem to be required for the in vivo biologic activity of EPO, but are not essential for in vitro receptor binding or growth stimulation of cells in culture.[122] There is considerable heterogeneity in the glycosylation of EPO, resulting in multiple isoforms with different numbers of sialic acid residues. Isoforms with higher sialic content have a prolonged half-life in the circulation and induce greater stimulation of erythropoiesis, despite having a lower affinity for the EPO receptor.[123] A hyperglycosylated recombinant EPO, called "novel erythropoiesis-stimulating protein" (NESP) or darbepoetin, carries two additional *N*-linked carbohydrate chains with up to 22 sialic acid residues; the endogenous EPO has a maximum of 14.[124] Despite a lower (approximately fivefold) affinity for the EPO-R, darbepoetin exhibits a half-life in the circulation approximately three times longer than that of EPO.[125,126] Gross and Lodish have developed an in vitro model accounting for the prolonged bioactivity of darbepoetin.[123] They found that darbepoetin and EPO have similar rates of internalization when bound to the EPO-R, with similar degradation and resecretion, but that EPO dissociates at a much slower rate from the EPO-R than darbepoetin, so more EPO is internalized and degraded.

EPO is produced primarily by the liver in the fetal period; after birth, the kidneys become the major source of production.[69,70,113,127–130] Clearance of circulating EPO occurs by mechanisms that have not yet been fully elucidated. The liver, kidneys, and bone marrow have all been studied as possible sources of EPO elimination. A small fraction of endogenous or exogenous EPO appears to be cleared by filtration into the urine.[131] EPO degradation products can be found in urine, but the location and mechanisms responsible for this degradation are not known.[118]

An important determinant of the fate of the circulating EPO is its binding to the EPO-R on erythroid cells[132]; the relative abundance of erythroid precursors (i.e., the size of the pool of erythroid progenitors) is known to modulate serum EPO concentrations.[133] The EPO-R is a 55-kDa transmembrane protein that belongs to the cytokine receptor superfamily.[134–139] It is present on erythroid progenitors from the colony-forming unit erythroid (CFU-E) stage to late basophilic erythroblasts.[134] The number of receptors has been estimated to be around 1000/cell. The molecular signaling cascade activated on binding of EPO to EPO-R has been studied in great detail. The first event seems to be the homodimerization of the receptor, which also undergoes a conformational change. This is followed by the generation of the intracellular signal by clathrin-mediated endocytosis and proteolysis of the whole ligand receptor complex, which ultimately determines the clearance of EPO from the circulation.[123,140–145]

Rather than intrinsic enzymatic activity, the EPO-R signal transduction pathway depends on the activation of Janus tyrosine kinase 2 (JAK2), which is physically associated with the receptor, and become phosphorylated when the conformation of the receptor is changed by the binding of EPO.[146–153] Activated JAK2 phosphorylates several of the eight tyrosine molecules of the cytoplasmic side of the EPO-R, exposing SH2 (src homology 2) binding sites for key signaling proteins.[154,155] The result is a cascade of signal transduction, with the activation of multiple pathways, including Ras/MAP kinase, JNK/p38 MAP kinase, JAK/STAT, the p85 regulatory subunit of phosphoinositide 3-kinase (PI3K), and protein kinase B (AKT).[156–161] Both JAK2 and the tyrosines on the cytoplasmic portion of the EPO-R seem to play a role in the internalization process.[162] Familial and genetic forms of polycythemia due to truncations in the EPO-R with absence of key tyrosines such as Y429 and Y431 have been shown to result in defective internalization of the EPO-R complex, prolonged signal transduction, and increased EPO sensitivity.[162] The EPO-R endocytic machinery is critically dependent on a Cb1/p85/epsin-1 pathway, which ultimately leads to receptor downregulation.[163]

The interaction of JAK2 with important intermediaries in signal transduction, STATs (signal transducers and activators of transcription), has been extensively studied. After phosphorylation, STAT5 becomes activated and undergoes homodimerization; it may translocate to the nucleus, where it activates EPO-inducible genes.[164] In transgenic mice lacking both STAT5a and STAT5b, fetal anemia develops, with increased apoptosis of erythroid progenitors due to decreased survival of early erythroblasts.[165,166]

The signaling induced by the binding of EPO to the EPO-R eventually results in an increased number of erythroid progenitors and precursors, in particular CFU-E[167] and, at the same time, in the activation of parallel molecular pathways that eventually suppress the signaling of the receptor via tyrosine phosphatases, which dephosphorylate and inactivate JAK2, downregulate the EPO-R on the cell surface, and induce negative regulators such as CIS/SOCS (cytokine-inducible, SH2-containing protein-suppressor of cytokine signaling).[168–176]

An important aspect of the overall mechanism of action of EPO has been elucidated by Koury and Bondurant,[167] who demonstrated that EPO does not directly stimulate erythroid proliferation but rather prevents the programmed death (apoptosis) of the erythroid progenitors (Fig. 55.5).[177–179] Burst-forming units-erythroid (BFU-Es), named for their capacity to generate multiclustered colonies of cells, are the earliest cell type exclusively committed to the erythrocyte line.[180] It is believed that these cells are produced stochastically from pluripotent stem cells. Only a minority of BFU-Es, 10% to 20%, are in a cell cycle at any given time; the rest remain an inert reserve of progenitor cells. Cells then begin to take on the characteristics of CFU-Es.[181] BFU-Es contain only small quantities of GATA-1, a key transcription factor for erythroid development, whereas CFU-Es have much higher concentrations.[182,183] CFU-Es begin to express some attributes of mature erythrocytes, including blood group and Rh antigens.[184,185] It is at the CFU-E stage that EPO exerts its greatest influence; CFU-E cells express the highest surface concentration of EPO-Rs of any erythrocyte precursor.[180,186,187] Without EPO present, these cells are rapidly lost to programmed cell death.[188–190] EPO is an essential survival factor for erythroid progenitors, beginning from the CFU-E stage all the way to basophilic erythroblasts. There is substantial heterogeneity in the responsiveness of erythroid progenitors to EPO in a certain tissue and differentiation stage, possibly related to the number of EPO-Rs, their functional status, or both.[191] This diversity in EPO-responsiveness corresponds to the 3-log units' range of serum EPO concentrations that can be measured in human patients.

The quantity of EPO is traditionally expressed in units, with 1 unit (U) representing the same erythropoietic effect in animals as occurs after stimulation with 5-mmol cobalt chloride.[113] Steady-state production of small amounts of EPO maintains the serum EPO concentration at approximately 10 to 30 U/L, enough to stimulate sufficient production of erythrocytes to replace those lost to senescence.[90,113] When anemia or hypoxia is present, serum EPO concentrations increase rapidly, to as much as 10,000 U/L.[93–95] Human studies have indicated a sustained increase in serum EPO concentrations after phlebotomy, with values remaining elevated for several weeks.[192,193] With chronic anemia, as occurs with pure red cell aplasia (PRCA) and aplastic anemia, serum EPO values remains chronically elevated, with values as much as 1000-fold higher than normal in those with very severe aplastic anemias.[194–198]

Koury and Bondurant and other associates have incorporated these basic physiologic concepts into a model, which explains how EPO regulates erythropoiesis in a variety of pathologic conditions (see Fig. 55.5).[58,167,199] In this model, the EPO-dependent phase of erythropoiesis encompasses three generations (from CFU-E through early erythroblasts), with each generation having a certain proportion of surviving cells and the remaining cells being lost by apoptosis. Owing to their reduced EPO responsiveness (or greater EPO dependence) most of the cells at the CFU-E stage become victims of apoptosis, so the erythropoietic production flow in a normal subject is produced by a relatively small fraction of progenitors that have escaped apoptosis. When EPO concentrations increase in response to hypoxia, blood loss, or hemolysis, additional progenitors are allowed to escape apoptosis and, a few days later, result in the generation of an increased absolute number of reticulocytes and ultimately of RBCs. If this response is sufficient to compensate for the decreased oxygen-carrying capacity of blood, EPO concentrations decline and so does erythropoiesis. When EPO production is impaired, such as in CKD, a much greater number of cells become apoptotic, and EPO concentrations are

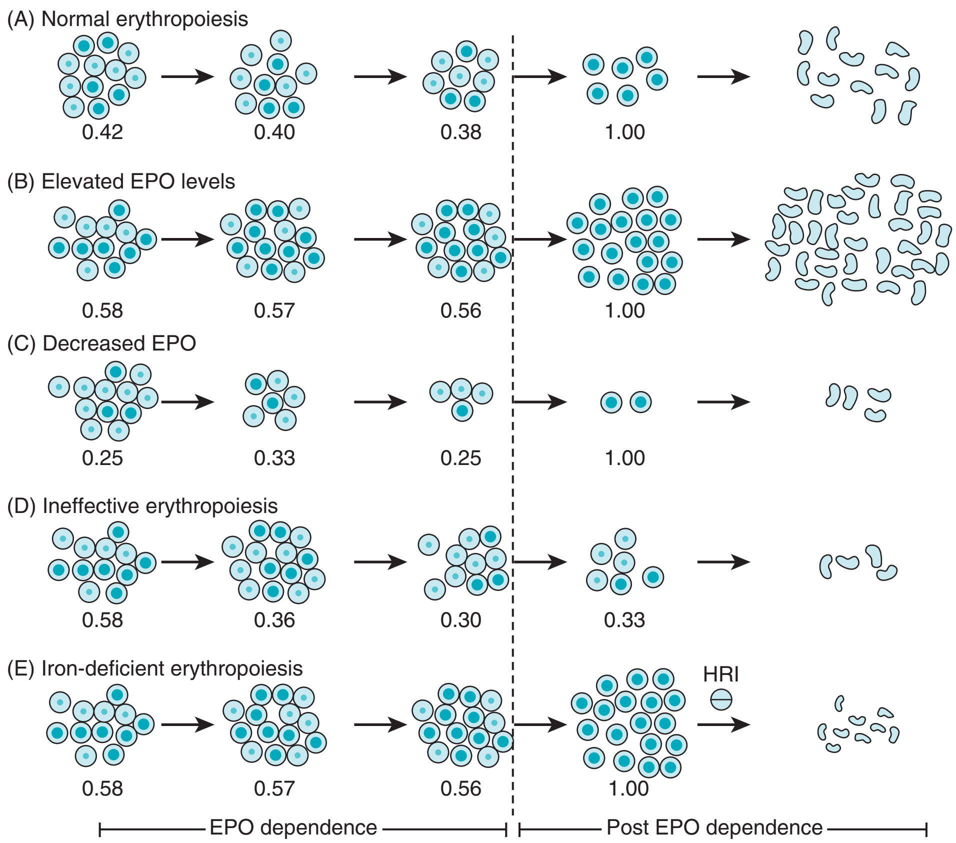

Fig. 55.5 Model of erythropoiesis based on suppression of programmed cell death (apoptosis) by erythropoietin *(EPO)* and heterogeneity in EPO dependence among erythroid cells. (A) Normal erythropoiesis with an average survival rate of 40% in each of the EPO-dependent generations. Normal erythropoiesis produces about 250 billion new erythrocytes daily, even though a minority of all potential erythroid cells survive the EPO-dependent period. (B) Elevated EPO values as found after acute blood loss or hemolysis, increase average survival rates to 55% in each EPO-dependent generation. Daily erythrocyte production increases to three times the normal amount. (C) Decreased EPO values as found in renal failure, decrease the average survival rate to 28% in each EPO-dependent generation. Daily erythrocyte production is one-third of normal. (D) Ineffective erythropoiesis with high EPO values increases rates of apoptosis caused by a pathologic process such as folate or vitamin B_{12} deficiency. High EPO values are the response to decreased erythrocyte production and expand surviving cells in the early EPO-dependent generation, but the increased rates of apoptosis in the late EPO-dependent and post–EPO-dependent stages decrease daily erythrocyte production to one-third normal. (E) Iron-deficient erythropoiesis with elevated EPO Values resulting in a similar increase to an average of 55% survival, as seen in B, but in the post–EPO-dependent period, when hemoglobin is synthesized, heme-regulated inhibitor *(HRI)* prevents apoptosis by inhibiting protein synthesis. The inhibited protein synthesis decreases the size of the erythrocytes produced and reduces daily erythrocyte production to 75% of the normal numbers. (From Koury M: Red cell production and kinetics. In Simon T, Snyder EL, Solheim BG, et al, eds. Rossi's Principles of Transfusion Medicine. Hoboken, NJ: Blackwell Publishing; 2009.)

insufficient to maintain an adequate pool of differentiating progenitors, with resulting impaired reticulocyte production and ultimately anemia.

It has become apparent that erythropoiesis does not happen in a vacuum and thus critically depends on the interaction of erythroblasts with the macrophages at the center of the erythroblastic island in the marrow.[200] A still undetermined fraction of basal and EPO-stimulated erythropoiesis requires contact between erythroblasts and central macrophages.[201,202] Macrophages play a role not only in the proliferation and final enucleation of erythroblasts, but also possibly in supplying ferritin and iron to the erythroblast.[203] In chronic inflammatory conditions, negative regulation of erythropoiesis may be mediated locally by macrophage-produced cytokines such as tumor necrosis factor-α (TNF-α), transforming growth factor-β (TGF-β), interferon-γ (INF-γ), and interleukin-6 (IL-6). Osteoblasts are another important component of the hematopoietic microenvironment in bone. They are able to produce EPO, and this local production can modulate the response to systemic anemia.[204]

The Roles of Iron, Folate, and Vitamin B_{12} in Erythropoiesis

Because of the continuous proliferative activity of the erythroid tissue and the associated production of large amounts of hemoglobin, adequate nutritional supplies of folate, vitamin B_{12}, and iron are essential for proper erythropoietic function. If any of these three components is inadequate, erythropoiesis becomes unable to meet both baseline and stimulated demands.

Inefficient erythropoiesis is a distinguishing feature of megaloblastic anemias, with the inability of erythroid progenitors to progress through the cell cycle and escape apoptosis owing to impaired DNA synthesis and repair. Studies by Koury and associates have shown that the erythroid differentiation stage that is most affected by folate or B_{12} deficiency is the one coinciding with the end of the EPO-dependent effects and initiation of Hgb synthesis.[205,206] The expansion of erythroid progenitors induced by EPO creates a large pool of progenitors (CFU-Es and pro-erythroblasts), which are extremely susceptible to apoptosis. The inefficient erythropoiesis of megaloblastic anemias is characterized by a reduced number of reticulocytes, increased serum bilirubin and lactic dehydrogenase (LDH), and accelerated iron turnover. In the case of vitamin B_{12} deficiency, thymidine and purine synthesis are impaired because of the unavailability of methylenetetrahydrofolate and formyltetrahydrofolate, respectively, and the trapping of folate as methyltetrahydrofolate.[207] Folate deficiency affects several key coenzymes that are involved in the transfer of single carbon units for the synthesis of pyrimidines and, for purines, and amino acid metabolism.

A regulated iron supply capable of matching the iron needs of the erythroid marrow is key for proper erythropoiesis. Intracellular availability of iron, heme, and globin chains have to be perfectly matched because excess of any of these constituents is toxic for the cell. Iron, 1 mg, can be adsorbed daily from the intestine, approximately 5% to 10% of the 14 mg of iron found in the average daily Western diet. The large majority of iron used for erythropoiesis comes from recycling of iron contained in aged RBCs via macrophages. Each milliliter of blood contains, on average, 0.5 mg of iron. Small, long-term blood losses result eventually in the depletion of the body iron stores and development of iron-deficient erythropoiesis and anemia. Iron deficiency partially suppresses the anemia-induced HIF-2α synthesis, which in turn reduces EPO production, resulting in decreased erythropoiesis and inappropriately low reticulocyte counts for the degree of anemia.[105,208,209] Heme-regulated eukaryotic initiation factor 2 (eIF-2α) kinase (HRI) is a key master controller of globin synthesis based on iron/heme availability.[210] In iron-sufficient states, free heme binds to HRI and inhibits the phosphorylation of eIF-2α, allowing globin synthesis to proceed. In iron-deficient states, HRI phosphorylates eIF-2α and suppresses mTORC1, leading to a numerically increased production of microcytic erythrocytes, without ineffective erythropoiesis.[211] In macrophages, HRI acts as a positive modulator of cytokine and hepcidin production, thus affecting both inflammation and iron metabolism.[212]

ANEMIA OF CHRONIC KIDNEY DISEASE

Anemia in CKD can develop because of any of the diseases or deficiencies that may affect individuals without kidney disease, such as iron deficiency, vitamin B_{12}[213,214] or folic acid deficiency,[213] and chronic blood loss.[215] However, the form of anemia most common in CKD is a normocytic, normochromic, or slightly hypochromic[216] anemia, with insufficient production of erythrocytes (see Fig. 55.2).[217–220] The cause is multifactorial, with contributors such as relative EPO deficiency, iron deficiency, blood loss, hemolysis, chronic inflammation,[221] drugs such as nonsteroidal antiinflammatory drugs (NSAIDs), and other factors, which may include circulating inhibitors of erythropoiesis.[220,222–224] The preponderance of evidence has demonstrated that EPO deficiency is the major cause of anemia in those with CKD.[225–228] Ultimately, the greatest proof of the primacy of EPO deficiency in the pathogenesis of renal anemia has been the consistent success of treatment with rhEPO or its derivatives. Other contributing causes to anemia should be considered if the severity of anemia is much greater than expected, if higher than usual doses of rhEPO are needed, and in the presence of leukopenia or thrombocytopenia.

ERYTHROPOIETIN PRODUCTION AND KIDNEY DISEASE

In normal persons, serum EPO concentrations rise in response to a reduction in red cell mass—in other words, anemia. In patients with CKD, EPO concentrations are inappropriately low for the degree of anemia, but may still be similar to or even higher than those in normal nonanemic subjects.[229–233] The adequacy of EPO production in response to anemia appears to decline in rough proportion to the degree of reduction in nephron mass.[234–236] Radtke and coworkers measured serum EPO in 135 patients with CKD and 59 normal subjects.[237] At all stages of CKD, serum EPO values were found to be higher than in nonanemic normal subjects, but the relationship between Hgb and serum EPO depended on the severity of the CKD. Among patients with mild to moderate CKD, the correlation was inverse, with lower Hgb concentrations being associated with higher serum EPO concentrations. However, among patients with creatinine clearances below 40 mL/min, mean serum EPO concentrations were severely depressed and uncorrelated with the degree of anemia, but directly correlated with creatinine clearance, indicating a parallel loss of renal excretory and endocrine function.[237] Fehr and associates have studied 395 patients undergoing coronary angiography, 84% of whom had reduced creatinine clearance values.[238] Like Radtke and colleagues, they found that serum EPO concentrations were higher in patients with lower Hgb, except when creatinine clearance was below 40 mL/min.[237,238]

Why EPO production by diseased kidneys is inadequately low remains incompletely understood. Some evidence has suggested that EPO production is reduced because of the transformation of peritubular fibroblasts into myofibroblasts.[239,240] On the other hand, it has been demonstrated that pharmacologic stabilization of HIF with inhibitors of the prolyl hydroxylases results in significant EPO secretion from renal and extrarenal tissues, even in patients undergoing dialysis.[241] On the basis of these findings and other circumstantial evidence, it appears that a disturbed oxygen-sensing mechanism rather than destroyed production capacity for EPO is the primary cause of renal anemia. In fact, despite the severely diminished EPO response with advanced CKD, some degree of sustained feedback remains. In the 6 months before starting dialysis, Radtke and colleagues found that as anemia worsened, serum EPO concentrations increased, and in the 6 months after the start of dialysis, the opposite occurred.[237] This continued response to anemia in patients with advanced CKD was also demonstrated by Walle and coworkers, who found that serum EPO values increased after hemorrhage and declined after blood transfusion in patients on dialysis.[242] Others have also reported that hypoxia can increase EPO production significantly in anemic patients with CKD.[243,244] Consistent with such observations and a relevant capacity for endogenous EPO secretion in patients with CKD, a large analysis in the United States has revealed that with increasing altitude—and thus lower blood oxygen content for any given Hgb concentration—higher achieved Hgb values were observed, even though lower doses of ESAs were used.[245]

The pronounced breakdown of EPO production in response to anemia when creatinine clearance is below 40 mL/min fits well with the observation that clinically relevant anemia becomes common only with moderate to advanced CKD (see earlier discussion and Figs. 55.1 and 55.2).[22,25]

SHORTENED RED CELL SURVIVAL

Although there have been several published reports on reduced red cell survival in CKD,[21,218,246] it is not clear how much it contributes to the anemia of CKD. Several abnormalities have been described in uremic erythrocytes, which may result in their increased premature destruction. An abnormal externalization of phosphatidylserine (PS), a phospholipid normally present only on the inside of the RBC membrane, has been associated with increased erythrophagocytosis and anemia in CKD.[247] Uremic red cells have been reported to become more fragile in response to osmotic stimuli,[248] although this finding was not confirmed in pediatric patients undergoing peritoneal dialysis.[249] The rheologic properties of uremic erythrocytes are altered owing to changes in RBC shape and decreased deformability.[250] Uremic erythrocytes may not be able to mount an effective response to oxidative stress,[251–253] possibly because of glutathione deficiency,[254] and may benefit from the antioxidant effects of vitamin E bound to dialysis membranes.[253,255] Carnitine deficiency may also contribute to the reduced survival of uremic erythrocytes.[256,257] An abnormal deposition of complement onto erythrocytes in CKD could also play a role in their premature removal from the circulation.[258] Because the contribution to shortened RBC life span of each of these factors is variable from patient to patient and is not easily quantifiable, and because there are no simple reliable methods to measure RBC survival, it is extremely difficult to identify anemic patients who are particularly affected, unless they have a preexisting RBC disorder.

BLOOD LOSS

Excessive bleeding has long been recognized as a common and significant complication of CKD.[259] The coagulopathy of CKD is discussed in the final section of this chapter and is thought to play a major role in the occult blood loss via gastrointestinal bleeding of patients with CKD.[260] In addition, blood loss due to the dialysis procedure and associated laboratory studies is also significant. A classic paper published 30 years ago estimated the blood loss due to hemodialysis to be between 1 and 3 L/year.[261] Subsequent improvements in dialysis techniques and clinical laboratory testing methodology have reduced this loss considerably. Later estimates of the blood lost in the whole extracorporeal circuit for each dialysis session vary from a range of 0.5 to 0.6 mL[262] to a median of 0.98 mL (range, 0.01–23.9 mL).[263] Each milliliter of blood contains approximately 0.5 mg of iron, so an important consequence of blood loss is the loss of iron and the development of iron deficiency (see below).

UREMIC "INHIBITORS" OF ERYTHROPOIESIS

Although a variety of uremic toxins have been identified in CKD,[264,265] including some with hematologic effects, such as quinolinic acid[266] and *N*-acetyl-seryl-aspartyl-lysyl-proline (AcSDKP),[267] there is no convincing demonstration that any of them plays a significant role in the anemia of CKD. Nevertheless, the response to ESAs can be improved with dialysis, and the EPO doses used to treat patients with anemia in CKD are much higher than the amounts endogenously produced in normal individuals, indicating reduced responsiveness. Apart from factors associated with anemia, it is likely that inhibition of erythropoiesis in CKD also occurs through the concomitant chronic inflammatory state, which is characteristic for the anemia of chronic disease (ACD). Contributing factors include decreased EPO production or responsiveness plus hepcidin-induced reduction in iron availability or absorption.[268,269] No significant reduction in the dose of ESAs used to manage anemia was observed for patients undergoing frequent hemodialysis (six times/week) in comparison with those on the conventional three times/week schedule.[270]

IRON METABOLISM, HEPCIDIN, AND ANEMIA OF CHRONIC DISEASE

During the last decade, we have witnessed an unprecedented explosion of our knowledge on the regulation of iron metabolism.[271] This has not only profoundly affected our understanding of iron metabolism in CKD but has also opened the door to novel therapeutic opportunities.[272] Although iron deficiency and functional iron deficiency have been a major focus of clinical and basic research in the iron metabolism of CKD, more attention is now being paid to patients in a positive iron balance and possible negative consequences of iron overload.[259]

Patients with CKD are in negative iron balance due to increased blood loss (see earlier), which frequently cannot be adequately compensated. In the absence of chronic inflammation, blood loss leads to a reduction of serum ferritin and serum iron values and a progressive increase in the desaturation of transferrin, below the 16% threshold that guarantees a normal supply of iron to the erythroid marrow.[273] Some studies have reported a reduced intestinal iron absorption in patients on maintenance dialysis,[274,275] mostly due to the concomitant inflammation. However, other studies have shown it to be upregulated by EPO administration and not substantially impaired in comparison with that in normal subjects.[276,277] The chronic inflammatory state frequently accompanying CKD creates additional constraints to the proper absorption and utilization of iron.[268]

The identification of hepcidin as a key regulator of iron homeostasis in normal conditions and in the ACD has redefined our understanding of iron homeostasis in CKD (Fig. 55.6).[278–286] Hepcidin, a 25–amino acid peptide produced and secreted by the liver,[287] modulates iron availability by promoting the internalization and degradation of ferroportin,[288] a key iron transporter (so far the only identified mammalian iron exporter) that is essential for iron absorption in the duodenum and recycling of iron/iron efflux by macrophages.[289,290]

High hepcidin concentrations turn off both duodenal iron absorption and release of iron from macrophages; low hepcidin concentrations promote iron absorption and heme iron recycling and iron mobilization from macrophages. Thus, hepcidin concentrations are expected to be high in iron overload states and decreased in iron deficiency states. In normal subjects, an oral iron load produces a measurable increase in hepcidin concentrations.[291] A hepcidin knockout mouse model has shown increased iron absorption, increased liver iron concentration, and decreased reticuloendothelial

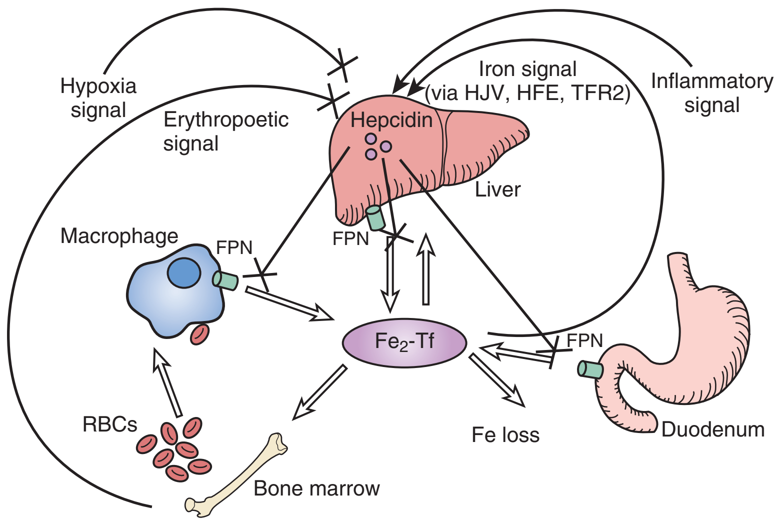

Fig. 55.6 Hepcidin is a central regulator of systemic iron homeostasis. Serum iron concentrations are determined by the balance of iron entry from intestinal absorption, macrophage iron recycling, and mobilization of hepatocyte stores versus iron utilization, primarily by erythroid cells in the bone marrow. A peptide hormone secreted by the liver, hepcidin controls iron release into the plasma by downregulating cell surface expression of the iron export protein ferroportin *(FPN)* on absorptive enterocytes, macrophages, and hepatocytes. Hepcidin production is inhibited by the erythropoietic drive and hypoxia to ensure iron availability for erythropoiesis. Hepcidin production is stimulated by iron (through human hemochromatosis protein *[HFE]*, hemojuvelin *[HJV]*, and transferrin receptor 2 *[TFR2]*) as a negative feedback loop to maintain steady-state iron concentrations. Hepcidin production also is stimulated by inflammation, thereby sequestering iron from invading pathogens in the setting of infection but also causing the hypoferremia of anemia of chronic disease. *FE_2-Tf,* Transferrin-bound iron; *RBC,* red blood cell. (From Babitt JL, Lin HY: Molecular mechanisms of hepcidin regulation: implications for the anemia of CKD. Am J Kidney Dis. 2010;55:726–741.)

iron stores.[279] Hepcidin overexpression in mice leads to severe iron deficiency.[292] Some of the genetic forms of hemochromatosis, like juvenile hemochromatosis, are caused by mutations in the hepcidin gene.[293] Hepcidin production can be induced by type II acute inflammatory reactions, which are mediated by IL-6 but not IL-1 or TNF-α,[294,295] thus providing a mechanism for inflammation to affect iron availability and causing the ACD. Anemia, EPO administration, and hypoxia increase iron absorption and mobilization by decreasing hepcidin production,[296,297] although hepcidin is not a direct target gene of HIF.[298] Erythroferrone (ERFE) has been identified as the key mediator of the erythropoietic regulation of iron metabolism. ERFE production rises with increased erythropoietic activity, leading to hepcidin suppression and mobilization of the iron stores.[299,300]

Urinary and serum concentrations of hepcidin have been measured with mass spectrometry.[301–303] An immunoassay for serum human hepcidin has been developed, with a lower limit of detection of 5 ng/mL, yielding a normal range for serum hepcidin of 29 to 254 ng/mL in men and 16 to 288 ng/mL in women.[291] The assay has enough sensitivity to detect changes in serum hepcidin values due to diurnal variation and in response to oral iron. Measurements of prohepcidin, the precursor of the biologically active 25–amino acid hepcidin, seem to be poorly correlated with those of hepcidin and are unresponsive to known hepcidin regulators.[304] Several studies have measured hepcidin concentrations in CKD, but have at times produced conflicting results.[305] A proper interpretation of these studies' findings must consider the following caveats. First, in the presence of anemia, hepcidin concentrations are reduced in persons with normal kidney function; thus, a normal hepcidin concentration in CKD can be still inappropriately high for the level of anemia. Second, hepcidin concentrations in healthy persons are reduced by 70% to 75% 24 hours after EPO administration.[306]

Residual kidney function, iron stores, erythropoiesis status, and inflammation all seem to be related to the hepcidin concentrations observed in CKD.[307] Hepcidin concentrations were found in one study to be elevated in patients on dialysis, but were not correlated with IL-6 concentrations or responsiveness to treatment, although they decreased after initiation of EPO therapy.[308] Progression or severity of anemia in patients with nondialysis-requiring CKD (ND-CKD) seems to be associated with higher serum hepcidin concentrations.[309,310] Elevated serum hepcidin has also been observed in anemic patients with combined renal (GFR, 20–70 mL/min/1.73 m^2) and cardiac failure[311] and in association with fatal and nonfatal cardiovascular events in patients undergoing maintenance hemodialysis.[312] EPO therapy leads to a reduction in serum hepcidin values that correlates with the bone marrow response.[311] Elevated serum hepcidin concentrations in pediatric and adult patients with CKD have been found to be associated with elevated ferritin and/or CRP concentrations and with stage 5 CKD.[291,313] Serum hepcidin assessment by surface-enhanced laser desorption/ionization–time-of-flight (SELDI-TOF) mass spectrometry did not seem to offer any advantage as a predictor of iron needs over the established traditional markers for adult patients on hemodialysis and maintenance ESA therapy.[314] Serum hepcidin concentrations are reduced following dialysis treatment.[315,316] Hepcidin concentrations also may vary in patients on hemodialysis carrying concomitant *HFE* gene mutations, the presence of which results in reduced hepcidin production.[317] Other studies have reported normal concentrations of hepcidin in CKD, but did not address the caveats mentioned previously.[315,318] Although it has been suggested that hepcidin may improve the identification of iron deficiency in CKD prior to transplantation,[319] more work is needed to define the role of hepcidin in the assessment of iron status in CKD. In iron-deficient persons without CKD, elevated hepcidin values have been shown to predict unresponsiveness to oral iron therapy.[320] These findings have not been confirmed in non–dialysis-dependent CKD.[321] Baseline values or changes over treatment time of serum hepcidin did not predict a response to iron therapy in 61 patients with ND-CKD.[321]

ERFE is an EPO-driven negative regulator of hepcidin production.[299] A study in 148 patients with CKD has shown that endogenous EPO values in serum or weekly rhEPO doses positively correlate with serum ERFE values.[322] However, ERFE values did not seem to influence serum hepcidin values suggesting the influence of other regulatory mechanisms in determining hepcidin values in CKD. In another study, on 59 hemodialyzed patients, treatment with ESAs was associated with increased values of ERFE and reduced values of hepcidin in serum.[323]

Given the central role played by hepcidin in the ACD, pharmacologic modulation of its production or bioavailability has potential as a new therapeutic modality.[324] In particular, the use of antihepcidin compounds may restore iron availability in ACD and improve the effectiveness of ESA therapy.[325] Strategies targeting hepcidin production, neutralizing hepcidin with specific peptides, or interfering with the binding of hepcidin to ferroportin or with the hepcidin-induced endocytosis of ferroportin are under consideration.[324] For conditions of iron overload, hepcidin mimetics (minihepcidin) have shown effectiveness in vivo in animal models,[326] and stimulators or hepcidin production are also under study.[272]

Another regulator of iron homeostasis, growth differentiation factor 15 (GDF15), is induced by hypoxia and iron depletion[327] and downmodulates hepcidin, thus contributing to iron overload in conditions with significant expansions of the bone marrow and inefficient erythropoiesis, such as severe beta thalassemias.[328] However, in patients with ACD, there seems to be no association between GDF15 and hepcidin concentrations, suggesting that this regulatory loop may not be active in the presence of inflammation.[328] Serum values of GDF15 were strongly correlated with the risk of progression to CKD in two separate patient cohorts ($n = 224$ and $n = 297$).[329]

Transmembrane serine protease 6 (TMPRSS6) may act as a cell membrane sensor of iron deficiency, which in turn suppresses hepcidin production and allows increased intestinal absorption of iron.[330] Mutations in TMPRSS6 have been associated with iron-refractory iron deficiency anemia (IRIDA),[331] and population studies have identified a TMPRSS6 allele associated with lower serum iron and Hgb concentrations.[332–337] It is likely that iron metabolism in patients with CKD may be similarly affected by genetic polymorphism of TMPRSS6.

INFLAMMATION AND ANEMIA OF CHRONIC DISEASE

Inflammation may also have an impact on erythropoiesis, independent of its effects on iron metabolism and RBC survival. Responsiveness to EPO declines in patients with CKD in the presence of acute inflammation, bacterial infections, and cancer.[338] A chronic inflammatory state with elevated serum cytokine concentrations and decreased lymphocytes and CD4+ T cell counts has been described in patients on dialysis.[339] Serum CRP has been used to monitor or predict the hematologic consequences of inflammation.[340] Inflammatory cytokines can induce anemia via impaired production of EPO, impaired erythropoietic response to EPO, with suppression of erythroid progenitor differentiation and proliferation, and possibly also reduced RBC survival. TNF-α is known to inhibit erythropoiesis directly[341] and to reduce EPO production.[342] Antibody-mediated blockade of TNF-α produces an improvement of anemia in rheumatoid arthritis[343] and inflammatory bowel disease.[344] Cytokines such as TNF-α, IL-1, IL-8, IL-12, and INF-γ may impair erythroid proliferation via multiple mechanisms, including cytokine-induced apoptosis,[345] downregulation of EPO-Rs, impaired production of other factors, such as stem cell factor, and direct toxic effects on progenitors.[346,347] It has also been proposed that inflammation may promote the release of soluble EPO-Rs, which may inhibit EPO signaling and increase EPO resistance.[348] Sotatercept and luspatercept, two novel therapeutic agents that target activin A and possibly other receptors of the TGF-β superfamily, have been under investigation for prevention of vascular calcification and improvement of anemia in CKD.[349,350]

FOLIC ACID, VITAMIN D, AND ZINC DEFICIENCY

A net loss of folate is associated with dialysis, although the deficit is typically compensated for by a normal diet and/or routine supplementation of water-soluble vitamins. Folate status is best assessed by measuring RBC folate because the plasma assay is affected by recent dietary intake and overestimates the true prevalence of folate deficiency.[351] Changes in RBC parameters (e.g., increases in mean corpuscular cell volume [MCV] and mean corpuscular cell hemoglobin [MCH] from baseline) are helpful in identifying folate deficiency in patients on dialysis.[352,353]

Vitamin D deficiency is an independent predictor of anemia in early CKD,[35,354] as well as in normal subjects,[355] but whether vitamin D directly affects erythroid proliferation[356] or is just a marker remains to be demonstrated.[357] Vitamin D supplementation (50,000 IU/month) in patients on dialysis did not result in changes in Hgb concentrations in one study, although a possible EPO-sparing effect was reported.[358] Similar negative findings were reported for adult patients in stage 3 or 4 CKD,[359] ESKD,[360] and pediatric ND-CKD patients.[361]

Low plasma zinc (Zn) concentration has been reported, with variable incidences in patients undergoing hemodialysis.[362–364] A Zn supplementation trial has shown measurable improvements in Hgb concentrations.[365]

ALUMINUM OVERLOAD

Formerly, aluminum was commonly used in patients on dialysis for its effects as a potent intestinal binder of phosphate. Although calcium-containing and non–calcium-containing phosphate binders have largely supplanted aluminum, the effects of aluminum toxicity on hematopoiesis are of historic interest. Parenteral aluminum exposure, either via dialysate contamination[366] or another route,[367] is still observed. The erythropoietic effects of aluminum toxicity are characterized by altered iron metabolism,[368] direct inhibition of erythropoiesis,[369,370] and disruption of red cell membrane function and rheology.[371–373] In dialyzed patients, the most notable hematologic effect of aluminum overload is microcytic anemia,[374,375] which improves with the use of deionized water to reduce the aluminum content of the dialysate[376] or chelation therapy with desferrioxamine.[377] EPO responsiveness is reduced in patients undergoing dialysis who have higher serum aluminum concentrations at baseline or after a deferoxamine challenge[378] and can be restored with deferoxamine treatment.[379] Improvement of anemia has also been shown in patients with the use of chelation therapy with deferoxamine, even in the absence of overt aluminum toxicity.[380,381] Interestingly, HIF destabilization requires iron as a cofactor, and iron chelation with deferoxamine can induce HIF, providing a possible alternative explanation for an improvement of anemia.

HORMONES, PARATHYROID HORMONE, AND MARROW FIBROSIS

Elevated serum values of adiponectin have been reported in CKD; a large prospective study has shown a negative correlation of adiponectin with Hb values and a significant association with the risk of developing anemia in 1227 nonanemic patients at baseline.[382] Because multiple associations have been reported between adiponectin and a variety of markers of disease severity in CKD, the precise relevance and mechanistic implications of this finding remain unexplained.

Intact fibroblast growth factor has been shown to be an important determinant of anemia and iron deficiency in mouse models of renal failure.[383] Fibroblast growth factor 23 (FGF23) values in serum exhibited a negative correlation with Hb values in 3869 subjects with mild to severe CKD,[384] with high values being associated with a risk of developing anemia in a cohort of 1164 nonanemic patients[385] and a reduction in serum values after IV iron therapy.[386] However, FGF23 values showed no association with anemia in a cohort of 282 patients on hemodialysis.[387]

The inhibitory effects of parathyroid hormone (PTH) on erythropoiesis are primarily indirect and are a consequence of myelofibrosis.[388] Secondary hyperparathyroidism is associated with diminished responsiveness to EPO.[389] Moreover, PTH values were identified as effect modifiers of the erythropoietic response to EPO in adult CKD patients on hemodialysis.[390] In 979 patients with nondialysis CKD, a significant inverse association has been reported between PTH and hemoglobin.[391] However, in pediatric patients, no association was found between serum intact PTH and Hgb concentrations.[392] Racial differences have been reported for serum PTH levels, with higher values in blacks versus whites.[393] An increase of EPO values and improvement of anemia have been reported after parathyroidectomy.[394,395]

DRUGS

Use of renin-angiotensin-aldosterone system (RAAS) inhibitors may induce or worsen anemia[396] for several reasons. Angiotensin II has direct facilitating effects on erythroid progenitor cells, which are inhibited by these compounds.[397] AcSDKP, an endogenous inhibitor of erythropoiesis, accumulates in patients treated with angiotensin-converting enzyme (ACE) inhibitors.[267] Endogenous EPO production may also be reduced through the hemodynamic effects of angiotensin II inhibition. Because angiotensin II leads to preferential constriction of efferent glomerular arterioles, it increases the ratio of filtered sodium—the main determinant of renal oxygen consumption—to peritubular blood flow and thus oxygen supply, thereby presumably lowering peritubular oxygen tension. RAAS inhibitors reverse these effects, and therefore have the potential to mitigate renal hypoxia and the signal for EPO production.[398] It has also been postulated that RAAS inhibitors may promote anemia and EPO resistance via a reduction in testosterone serum concentrations in men younger than 60 years.[399] Myelosuppressive effects of immunosuppression may further contribute to anemia, especially in the posttransplantation setting.[400–402]

ASSOCIATION OF ANEMIA WITH ADVERSE OUTCOMES

The availability of rhEPO greatly increased interest in the role that anemia plays with respect to health-related quality of life (HRQOL) and prognosis of patients with CKD. A large number of observational studies have consistently shown that even modest reductions in Hgb concentrations are associated with adverse outcomes. This statement applies to mortality in patients on dialysis,[11,403] and patients with CKD not on dialysis,[404] as well as individuals in the general population[405] or with another complex chronic disease, such as heart failure.[406] A large study of 159,720 patients undergoing hemodialysis and receiving epoetin therapy has shown that the duration of anemia, rather than the Hgb concentration per se, is the most powerful predictor of short-term mortality, with Hgb concentrations less than 11 g/dL for 3 months or longer being associated with an increased risk of death.[407,408] However, there is no agreement on how best to study Hgb variability effects on mortality, with various methods having been applied to describe Hgb variability[409] and with significant confounding attributable to variations in Hgb concentrations among dialysis centers[410] and the effects of ESA dosing and iron therapy.[411] Hb values less than 10 g/dL and more than 12 g/dL prior to ESKD have been associated with values higher all-cause mortality and cardiovascular mortality, whereas Hb less than 10 g/dL was associated with increased hospitalization rates in a study of more than 31,000 US veterans.[412]

A systematic review has supported the notion that Hgb concentrations below a formerly established reference range (Hgb 9–10 g/dL in some studies and Hgb 11–12 g/dL in others) are generally associated with increased all-cause mortality in dialysis recipients. Similar findings have been reported in pediatric patients on peritoneal dialysis.[413] In ND-CKD patients, the severity of anemia is also associated with the rate of decline in kidney function,[414] consistent with the concept that anemia may aggravate intrarenal hypoxia.[415–417] Moreover, anemia was also found to be a strong risk factor for the development of left ventricular hypertrophy,[418,419] an established surrogate for mortality and cardiovascular events. An increase in cardiac output as part of the compensatory mechanisms that maintain oxygen delivery in anemia has been considered as a possible reason for the link between anemia and cardiac geometry.[418]

Other specific complications, such as proliferative retinopathy in patients with diabetes, were found to be associated with anemia.[420] In a relatively large (21,899) cohort of dialyzed patients in the United States, an Hgb concentration below 8 g/dL were associated with a twofold increase in the odds of death in comparison with those for an Hgb concentration between 10 and 11 g/dL; several laboratory parameters related to iron status and nutrition, as well as dose of dialysis, were also associated with Hgb concentration.[216] There was no association with mortality when Hgb was above 11 g/dL in this cohort study, but other studies have suggested that the relationship among Hgb, comorbidities, and outcomes extends into the normal range of Hgb, leading to the hypothesis that normalization of Hgb might be associated with best outcomes. However, RCTs have failed to confirm this suggestion (see later). Therefore, although it is undisputable that anemia is a sensitive risk marker for adverse outcomes, its role as a causal risk factor has not been established.

ERYTHROCYTOSIS OF PATIENTS WITH KIDNEY DISEASES

Although anemia is a typical complication of advanced CKD, irrespective of its cause, there are few circumstances under which disorders of renal structure and function can also result in abnormally high rates of RBC production—that is, erythrocytosis. The pathogenesis of these disorders remains incompletely understood, but they probably all result from increased production of renal EPO.

(POLY)CYSTIC KIDNEY DISEASE

The degree of anemia in patients with autosomal dominant polycystic kidney disease (ADPKD) is usually somewhat less severe than for other causes of CKD, although patients with ADPKD on dialysis usually require treatment with ESAs. Occasionally patients with ADPKD may become polycythemic.[421] Erythrocytosis may also develop in patients on hemodialysis with acquired renal cysts and single cysts.[422,423] Serum EPO concentrations in patients with ADPKD are, on average, up to twofold higher than in patients with CKD from other causes,[244,424,425] and significant arteriovenous concentration differences for EPO have been found in polycystic kidneys.[426] In the cyst walls of patients with ADPKD, interstitial cells have been shown to express EPO mRNA, and cysts derived from proximal but not distal tubules contain increased concentrations of bioactive erythropoietin.[426] In a later study, continuous activation of HIF was demonstrated in cyst walls of patients with ADPKD and in a rat model of cystic kidney disease.[427] The physiologic distinction between HIF-1α expression in tubular cells and HIF-2α expression in peritubular cells is maintained in the cyst walls. The genetic defects underlying ADPKD do not lead to HIF activation. However, cyst expansion results in pericystic hypoxia, and hypoxic stimulation of pericystic angiogenesis is believed to play an important role in cyst progression.[428,429] Therefore, the enhanced production of EPO in cystic kidneys is probably due to local hypoxia and mediated via HIF activation. It is possible that factors other than EPO, induced through this pathway, contribute to cyst growth. Regional hypoxia also appears to stimulate cyst growth, primarily via increased fluid secretion into the cyst lumen.[430]

POSTTRANSPLANTATION ERYTHROCYTOSIS

Kidney transplantation is usually followed by full correction of renal anemia.[406,431,432] Interestingly, a regular increase of EPO production is not related to the presence of the transplant, but does correlate with the onset of graft function,[433] providing further evidence for the role of excretory kidney function in EPO regulation. Some 10% to 20% of patients manifest overcorrection and demonstrate erythrocytosis, usually within the first 6 months following transplantation.[434–436] Graft failure is associated with anemia, and therefore polycythemia is more likely to occur in patients with normal kidney function.[406,424]

Increased plasma EPO concentrations have been reported in patients with posttransplantation erythrocytosis.[437] Selective venous catheterization studies and the response to removal of the native kidneys have suggested that the native kidneys are the main source of increased EPO production.[437,438] Although this suggestion clearly indicates that a sufficient production capacity for EPO may be preserved in diseased kidneys, it is unclear how the secretion rate is enhanced after transplantation. Improvement of the uremic state has been speculated to play a role. Moreover, given that inflammatory cytokines can inhibit erythropoietin production, the application of immunosuppressive agents could theoretically enhance EPO formation. Interestingly, the prevalence of posttransplantation erythrocytosis seems to be elevated in combined kidney and pancreas transplantation,[439] but whether the erythrocytosis is related to enhanced EPO formation or to insulin-stimulated pathways remains unclear. In some patients with posttransplantation polycythemia, the circulating EPO concentrations are normal or reduced, and it may be that in these cases there is an increased sensitivity of the erythroid progenitor cells to EPO or loss of other feedback control mechanisms.

The most effective therapy of posttransplantation erythrocytosis consists of agents blocking the RAAS.[435,440–442] There is no evidence that angiotensin acts directly on EPO- producing cells, but there are several ways through which RAAS blockade may inhibit erythropoiesis (see earlier). Alternative therapeutic strategies to reduce increased RBC concentrations after transplantation include the discontinuation of diuretics, application of theophylline,[441] and phlebotomy, which, however, can lead to iron deficiency.

RENAL ARTERY STENOSIS

Although renal artery stenosis reduces the oxygen supply to the kidneys, it is only rarely associated with erythrocytosis.[443–446] The data on EPO production in this case are contradictory. In experimental animals, enhancement of EPO production after renal artery stenosis has been demonstrated by some, but not all, investigators.[447] A study performed in rats has shown that graded reduction of renal blood flow to 10% of the control value caused a maximal threefold increase in serum EPO concentrations.[448] Therefore, renal EPO production appears rather insensitive to changes in renal blood flow. Because the ratio of oxygen demand and delivery determines local oxygen tension in the area of EPO-producing cells, it is possible that the two are equally reduced after a reduction in renal blood flow, thus not resulting in sufficient hypoxia to stimulate EPO gene expression. It has been argued that the indirect coupling of oxygen demand to supply makes the kidney an ideal site for the oxygen sensing that controls RBC production.[449]

RENAL TUMORS

Up to 5% of patients with renal carcinomas have erythrocytosis[450] and, conversely, approximately one-third of tumor-associated erythrocytosis is caused by renal cancer.[451] Conflicting data have been reported concerning serum EPO concentrations in patients with renal tumors but, at least in some patients, raised values have been found.[452] Furthermore, overexpression of EPO mRNA has been demonstrated in renal tumors.[453] In situ hybridization has revealed that accumulation of EPO mRNA occurs in epithelial tumor cells, but not in interstitial cells of the tumor stroma.[453] Most clear cell renal carcinomas, the most frequent type of renal cancer, are associated with mutations of the VHL gene that interfere with its ability to target HIF for proteasomal degradation (see earlier).[454] Indeed, clear cell renal carcinomas contain high concentrations of HIF.[455–458]Although stabilized HIF in renal tumors appears to be functionally active in inducing HIF-target genes, it is as yet unclear why overexpression of EPO is confined to about one-third of these tumors.[459] Although activation of HIF appears necessary for EPO gene expression in renal cell carcinoma, it is clearly not the only determinant. The fact that erythrocytosis occurs far less frequently than overexpression of EPO in renal cancer is probably due to a variety of mechanisms causing anemia in patients with cancer, which include inhibition of the effect of EPO and reduced iron availability. There is some albeit controversial evidence suggesting that EPO has autocrine or paracrine tumor growth-promoting effects.[460,461]

Clinical Relevance

When using ESAs, always consider the possibility of functional iron deficiency, especially in women and men with normal or borderline normal iron stores. Serum ferritin values are not indicative of iron stores in the presence of inflammation and in most patients with CKD or ESKD. Functional iron deficiency is less likely to occur when serum ferritin values are >100 for ND-CKD, and >200 for patients on dialysis. Monitor hematologic response and assess iron adequacy based on Hb increase and presence of normal reticulocyte Hb content (CHr or Ret-He).

TREATMENT OF RENAL ANEMIA

ERYTHROPOIESIS-STIMULATING AGENTS

Recombinant human erythropoietin was developed in the 1980s, with support from an orphan drug program. At that time, it was unclear to what extent the anemia of patients with kidney disease could be influenced by application of the hormone, as well as how many patients might benefit from this type of therapy. The initial clinical studies revealed an unexpected efficacy in patients undergoing dialysis, with both high response rates and evidence that hemoglobin concentrations could not just be increased to some extent, but could virtually be normalized.[110,112] In the subsequent years, the use of rhEPO in patients on maintenance dialysis became routine in most parts of the world. The indication was subsequently extended to the much larger group of patients with ND-CKD, as well as to several other patient groups with anemia, including those whose cancer was treated with chemotherapy.

Over the years, the efficacy of the therapy and presumed benefits led to a gradual increase in Hgb concentrations in virtually all treated patient groups (see Fig. 55.3). Data from the US Renal Data System have indicated a substantial increase in the use of these agents (as well as intravenous [IV] iron and blood transfusion) in older (≥67 years) adults with ESKD.[462] Not surprisingly, the expanded clinical use resulted in an extraordinary commercial success. Investigators originally intended to copy the endogenous molecule as closely as possible, but patent and marketing considerations, together with concepts for improving patient management, resulted in the development of a number of derivatives of the EPO molecule with altered pharmacokinetic properties and, later, the development of different molecules that can directly or indirectly stimulate the EPO-R. Discussions about the appropriate terminology for all these compounds have not been settled, but the term "erythropoiesis-stimulating agents" (ESA) is commonly used to describe the heterogeneous class of drugs that stimulate erythropoiesis through stimulation of the erythropoietin receptor.

Epoetin

The term *epoetin* is usually applied to rhEPO preparations, produced by means of overexpression of the human EPO gene in mammalian cell lines. Production in mammalian cells rather than bacteria is required because EPO is a highly glycosylated molecule, and bacteria lack the ability to generate glycoproteins. Epoetin alfa and epoetin beta are the two compounds first developed by two different companies. Both are produced in Chinese hamster ovary (CHO) cells and show a high degree of similarity, with identical protein backbones of 165 amino acids and one *O*-linked and three *N*-linked glycosylation sites each, but with subtle differences in their carbohydrate composition.[463]

Although the amino acid sequence unequivocally determines glycosylation sites, the precise composition of the sugar side chains is also determined by the repertoire and activity of glycating enzymes, which may vary among cell lines and under different tissue culture conditions. Glycosylation of the EPO molecule is not required for binding or activation of its receptor[122]; in fact, the in vitro activity of deglycosylated erythropoietin is enhanced.[464] However, in vivo deglycosylated EPO is inactive because of rapid clearance from the circulation, and the carbohydrate chains are thus responsible for its pharmacokinetic properties.

Early clinical trials in patients on hemodialysis used IV epoetin administered three times weekly; subsequently, the intraperitoneal (IP), subcutaneous (SC), and intradermal routes of administration were also investigated.[465,466] After IV administration, plasma EPO concentrations decay monoexponentially, with an elimination half-life of approximately 4 to 11 hours.[467] The apparent volume of distribution of EPO is about one to two times the plasma volume, and the total body clearance is lower than for other protein hormones, such as insulin, glucagon, and prolactin. The IP route was investigated as a potential means of administering EPO to patients on peritoneal dialysis but the bioavailability of intraperitoneal epoetin is disappointingly low, 3% to 8%. This application has therefore not been pursued.[465,468,469]

With SC administration, peak serum concentrations of about 4% to 10% of an equivalent IV dose are obtained at around 12 hours, and thereafter they decay slowly, so that concentrations greater than baseline are still present at 4 days.[466,467] The bioavailability of SC epoetin is around 20% to 25%. Nevertheless, SC application is even more efficient than

IV application, allowing a dose reduction of approximately 30% to maintain the same hemoglobin concentration.[470,471] Presumably, the early peak concentrations of epoetin after IV injection are inefficient, and the more prolonged elevation of hormone levels following SC application allows a more sustained stimulation of RBC production. Thrice-weekly administration has remained the most popular dosage frequency for both IV and SC administration, although once-weekly,[472] twice-weekly, and seven times weekly (once-daily) dosing have all been used.[473] With IV epoetin, once-weekly administration is associated with much lower efficacy, and twice- or thrice-weekly dosing is required. In 2015, the mean weekly dose of EPO in adult HD and peritoneal dialysis patients in the United States was below 10,000 U/week, a substantial reduction when compared with 2010 values (Fig. 55.7).

A number of additional epoetin preparations have been developed worldwide. Some have distinct differences in the

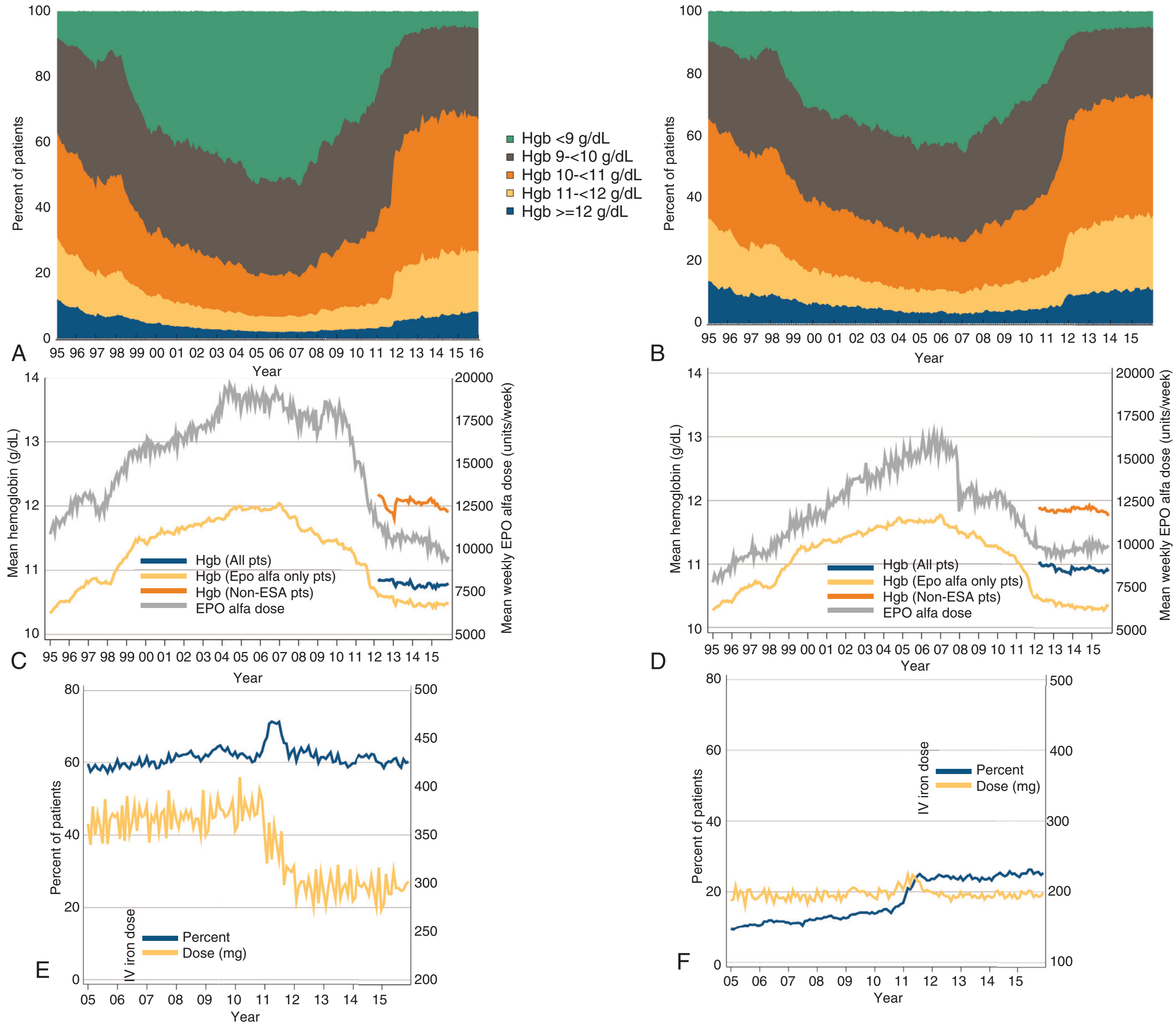

Fig. 55.7 Anemia, erythropoiesis-stimulating agents (ESAs), and IV iron in hemodialysis and peritoneal dialysis patients. (A) Distribution of monthly Hgb values in ESA-treated adult hemodialysis patients on dialysis ≥90 days, from Medicare claims; time trend from 1995–2015. (B) Distribution of monthly Hgb values in ESA-treated adult peritoneal dialysis patients on dialysis ≥90 days, from Medicare claims 1995–2015. (C) Anemia measures among adult hemodialysis patients on dialysis ≥90 days, 1995–2015. (D) Anemia measures among adult peritoneal dialysis patients on dialysis ≥90 days, mean monthly Hgb value and mean weekly EPO alfa dose (averaged over 1 month), from Medicare claims data 1995–2015. (E) Monthly percentages of IV iron use and mean monthly IV iron dose in adult hemodialysis patients on dialysis ≥90 days, from Medicare claims 2005–2015. (F) Monthly IV iron use and mean monthly IV iron dose in adult peritoneal dialysis patients on dialysis ≥90 days, from Medicare claims 2005–2015. *EPO,* erythropoietin. (From US Renal Data System. 2017 Annual Data Report. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

production process—for example, epoetin delta was produced in a human cell line through increased transcription of the endogenous EPO gene,[474] but this product is currently not being distributed. Other epoetins are so-called "biosimilar" generic drugs designed as copies of epoetin alfa or beta; these are being licensed on the basis of a more limited clinical trial program after expiration of the patents for the originator compounds in Europe.[475] Additional epoetins are available in other parts of the world, which are not necessarily produced to the same regulatory standards as the preparations marketed in the United States and Europe and may show variable product characteristics.[476] Various studies in CKD patients have compared biosimilar and originator ESAs, with no significant difference in effectiveness or outcomes.[477–483]

The importance of the formulation of epoetins was highlighted in 2002 with an upsurge in cases of antibody-mediated, pure red cell aplasia (PRCA) in association with the SC use of epoetin alfa marketed outside the United States after a change to an albumin-free formulation. Patients affected by this complication develop neutralizing antibodies against both rhEPO and the endogenous hormone, resulting in severe anemia and transfusion dependence.[484] The cause of this serious complication remains obscure, although circumstantial evidence suggested that rubber stoppers of prefilled syringes used for the albumin-free epoetin alfa formulation may have released organic compounds that act as immunologic adjuvants.[485] Factors such as breach of the cold storage chain may also have played a role. In anti-EPO antibody cases observed so far, the SC application route was usually a prerequisite. Although the unfortunate combination of adverse factors leading to a temporary increase in antibody-induced PRCA was specific for one product, a low baseline rate of PRCA also occurs with use of epoetin beta and darbepoetin alfa (see later).[486]

Darbepoetin Alfa

Darbepoetin alfa is an EPO derivative with a further two *N*-linked glycosylation sites, created by site-directed mutagenesis to prolong its plasma survival time (as discussed earlier).[124] Each of these glycosylation sites can carry an additional four sialic acid residues. Thus, this molecule (called "darbepoetin alfa" or "novel erythropoiesis-stimulating protein" [NESP]) contains five *N*-linked and one *O*-linked glycosylation chains and has the capacity to carry up to 22 sialic acid residues, compared with a maximum of 14 sialic acid residues for original rhEPO. The additional glycosylation on darbepoetin alfa results in a molecule weighing 37.1 kDa compared with 30.4 kDa for epoetin. As intended, darbepoetin alfa has a longer half-life in vivo than rhEPO—25.3 hours versus 8.5 hours after IV administration.[126] The elimination half-life after SC administration is about 48 hours, which is approximately twice that previously reported for epoetin alfa or beta. A number of studies have examined once-weekly and once every other week dosing.[487,488] Darbepoetin alfa can both correct and maintain Hgb at these dosing frequencies, and its side effect profile is very similar to that of epoetin alfa or beta.[465,489] Several "conversion" studies have suggested that an appropriate conversion factor for switching patients on epoetin alfa or beta to darbepoetin alfa is 200 U of epoetin to 1 μg of darbepoetin alfa. In contrast to epoetin alfa or beta, the dose requirements for darbepoetin alfa do not differ significantly between the IV and SC administration routes.

Methoxypolyethylene Glycol Epoetin Beta

Alternative bioengineering techniques to prolong the half-life of EPO further have resulted in the development of methoxypolyethylene glycol epoetin beta (also called "continuous erythropoietin receptor activator" [CERA]), a PEGylated derivative of epoetin beta with an elimination half-life of about 130 hours when administered IV or SC.[490–492] A methoxypolyethylene glycol polymer chain is integrated through amide bonds between the N-terminal amino group or the ε-amino group of lysine (predominantly lysine-52 or lysine-45), with a single succinimidyl butanoic acid linker. The molecular weight of CERA is twice that of epoetin (~60 kDa). Phase III studies have shown that because of the longer half-life time of CERA, less frequent injections are sufficient to maintain stable hemoglobin values CERA given IV once every 2 weeks was found to be as safe and effective as epoetin given thrice weekly for correcting anemia in patients on hemodialysis.[493] A larger study has also shown that CERA given at 4-week dosing intervals was not inferior to epoetin given three times weekly in terms of maintaining Hbg concentrations.[494] So far, no significant differences in outcomes or complications have emerged for CERA compared with other ESAs.[495] Conversion factors have been established and tested for switching from other ESAs to CERA.[496]

Other Erythropoiesis-Stimulating Agents

Several other ESAs have been developed or are currently in clinical development.[497–499] These include EPO polymers, EPO fusion proteins, EPO mimetic molecules and the so-called HIF stabilizers, which induce endogenous EPO formation.[500] The ability of molecules that are structurally unrelated to EPO to dimerize the EPO-R and activate the intracellular signaling cascade was first described 20 years ago.[501] Peginesatide is an EPO mimetic peptide that was subsequently developed for the treatment of anemia. Its amino acid sequence was completely unrelated to that of native or rhuEPO, although it shares the same biologic properties with EPO with respect to EPO-R activation.[502,503] The potential advantages of this compound included greater ex vivo stability, allowing storage at room temperature, prolonged pharmacodynamic action, allowing once-monthly administration, and a simple manufacturing process involving synthetic peptide chemistry. In addition, because peginesatide was structurally unrelated to EPO, it did not cross-react with anti-EPO antibodies, allowing effective treatment of anti-EPO antibody-mediated PRCA.[504] Two phase 3 studies have demonstrated that peginesatide is not inferior to conventional epoetin for correcting anemia in CKD.[505,506] However, for reasons that remain unclear, peginesatide increased the risk of a combined cardiovascular endpoint in patients with ND-CKD.[505] Accordingly, it was only approved in the United States for use in patients on dialysis only. Only slightly more than 6 months after its introduction, the drug was recalled as a result of postmarketing reports of serious hypersensitivity reactions—including fatal reactions in approximately 0.02% of patients—that occurred within 30 minutes after the first IV dose and had not been reported during clinical trials.[507] Currently, no EPO mimetic is available for rescue therapy in patients with anti-EPO, antibody-mediated PRCA.

Hypoxia-Inducible Factor Stabilizers

The HIF stabilizers are competitive inhibitors of HIF prolyl hydroxylases and asparagyl hydroxylase, enzymes involved in the degradation of HIF and suppression of its transcriptional activity, respectively (as discussed earlier).[74,508,509] The HIF stabilizers, therefore, cause an increase in endogenous EPO production.[510] These drugs are orally active. They have been shown to stimulate erythropoiesis effectively in monkeys[511] and are currently being tested in phase 2 and 3 studies in patients with CKD.[512,513] A phase 1/2 single-dose study comparing erythropoietin formation in small groups of patients on dialysis with native kidneys and after bilateral nephrectomy has provided evidence that EPO production can be stimulated in extrarenal sites (presumably the liver) and the diseased, nonfunctioning, fibrotic kidneys.[241] These data provide proof for the concept that a disturbance of the renal oxygen-sensing mechanism, rather than a loss of EPO-producing cells, is the main cause of renal anemia. There has been much discussion about whether HIF stabilizers will upregulate not only EPO gene expression but also other HIF target genes, such as those involved in iron metabolism and neoangiogenesis. Although some of these effects may facilitate an increase in hemoglobin concentration, the long-term consequences, good or bad, of these other effects have not been established.[508] Roxadustat (FG-4592) and vadadustat (AKB-6548) have been tested with positive results in CKD (dialysis and nondialysis), with phase 3 studies ongoing.[514–517] Daprodustat and molidustat are also in advanced clinical development.[517a–517d]

Interestingly, genetic causes of impaired degradation of HIF, potentially comparable to long-term pharmacologic inhibition of HIF degradation, have been identified as causes of rare polycythemias.[104,518,519]

Initiation and Maintenance of Therapy

Following commencement of regular therapy with ESAs, a significant increase in the reticulocyte count to around two to three times baseline is usually evident at 1 week, and an increase in Hgb concentration is seen by 2 to 3 weeks. The increase is dose-dependent, and most physicians aim for an increment of not more than 1 g/dL/month to minimize the risk of adverse effects. In most patients, ESA therapy is initiated at the outset of dialysis therapy and, according to the US Renal Data System report, peak doses of ESA are being administered at month 2 after the initiation of dialysis (see Fig. 55.7). Only 0.3% of US CKD patients in the predialysis stage are treated with ESAs; these patients are usually older, with more advanced disease and significant comorbidities, resulting in high rates of death and cardiovascular complications.[520]

The increase in Hgb concentration following ESA therapy is associated with an increase in RBC count. No significant changes in leukocyte or platelet counts are usually seen, although a moderate increase in the platelet count has been documented in some studies. There is usually a marked decline in the serum ferritin concentration and/or the transferrin saturation value after the initiation of ESA therapy, unless iron stores are being replenished in parallel, because large quantities of iron are used up in the manufacture of new RBCs (see later).

Radioisotope blood volume studies have confirmed that there is an increase in RBC mass after treatment with ESAs, associated with a compensatory reduction in plasma volume so that the whole blood volume remains unchanged. Early ferrokinetic studies have indicated that epoetin therapy induces a twofold increase in marrow erythropoietic activity, as evidenced by a doubling of marrow and RBC iron turnover.[243,521] There is little or no change in mean RBC life span after epoetin therapy; thus, the increased RBC mass is largely accounted for by the production of greater numbers of RBCs, rather than by any significant change in their survival.

IRON MANAGEMENT

Iron is the fourth most common element—after oxygen, silicon, and aluminum—in the earth's crust and the most abundant transitional metal in the human body. Although it plays an essential role in multiple biologic processes, such as transport of oxygen, transfer of electrons, DNA synthesis, and heme-based enzymatic reactions, iron is also highly susceptible to undergoing transition from the ferrous state (Fe^{2+}) to the ferric (Fe^{3+}) state and to generate reactive oxygen species (ROS) via the Haber-Weiss-Fenton reaction, thus requiring the presence of multiple systems to prevent or control this potentially harmful transition.[522]

The metabolism of iron is geared toward conservation and recycling, with the gastrointestinal absorption of iron in adults being tightly regulated to compensate for the daily losses and to keep the total iron pool in the body constant, in the range of 35 to 45 mg per kilogram of body weight. Of the iron pool, approximately two-thirds is contained in the RBC pool as Hgb, with the remaining fraction stored in macrophages and reticuloendothelial system (RES), liver, and muscle (myoglobin).

Multiple factors induce a negative iron balance in patients with CKD, including reduced intake and absorption, chronic losses due to occult and overt blood loss, and reduced bioavailability of iron due to the chronic inflammatory state and increased hepcidin production (see previously).[523] Iron losses in patients with CKD can be up to 5 to 6 mg iron daily (1 mg iron daily in normal subjects) and cannot be adequately compensated with oral iron supplements because gastrointestinal absorption is limited by the chronic inflammatory state and hepcidin. In addition, in patients with CKD treated with ESAs, insufficient amounts of iron are released from the body stores to meet the greater demand of ESA-driven erythropoiesis.[524] Similar evidence has been provided for normal subjects when erythropoiesis is increased by an intensive blood donation schedule and EPO, or EPO alone, despite concomitant oral iron supplementation.[525–527]

Markers of Iron Status

When hematologic signs of iron-deficient erythropoiesis—reduced Hgb value, with abnormally low MCV and MCH values inadequate reticulocyte response, and low reticulocyte Hgb content—are associated with biochemical markers of low iron stores (abnormally low serum ferritin), the diagnosis of absolute iron deficiency is straightforward. However, straightforward determination of iron deficiency is the exception rather than the rule in patients with CKD, in whom iron may be present in storage form but not readily available for erythropoiesis, and serum ferritin concentrations are increased owing to the concomitant inflammatory state. Thus, the diagnosis of iron deficiency in CKD must rely on a variety of markers, biochemical and hematologic and, in the most challenging cases, on the erythroid response to IV iron (see later for the significant limitations of bone marrow biopsy).

These markers are determined in individual patients over time; markers with high biologic and analytic variability, such as transferrin saturation and ferritin, are less suitable to assess iron status than markers with low variability such as Hgb, Hct, and reticulocyte Hgb content.[528,529]

Serum Ferritin. Serum ferritin values higher than 200 μg/L are recommended for patients on dialysis,[11,12] and values of 100 μg/L should be considered the lower limit of normal for ND-CKD patients.[530] The sensitivity for ruling out iron deficiency was reported to be 90% for a ferritin cutoff of 300 μg/L and 100% for 500 μg/L.[531,532] Concerns about iron toxicity and overload have resulted in several guidelines setting an upper limit for ferritin of 500 μg/L,[10–12,530] above which IV iron is not recommended. However, this recommendation is not evidence-based.[533] Additional factors that elevate serum ferritin are hyperthyroidism, liver disease (associated with hepatitis C virus [HCV] or other conditions), alcohol consumption, and oral contraceptives, whereas vitamin C deficiency and hypothyroidism decrease ferritin concentrations.[534] More than 50% of US patients on hemodialysis have serum ferritin values higher than 800 ng/mL, and more than 20% have values greater than 1200 ng/mL (see later; Fig. 55.8E).

Serum Iron, Transferrin, and Transferrin Saturation. The biochemical markers serum iron, transferrin, and transferrin saturation (TSAT) are routinely used in the diagnosis of iron deficiency but have some important limitations. Serum iron concentrations and TSAT values are sensitive to diurnal variations and to dietary intake, with serum iron concentrations being higher early in the day. These are increased by greater iron intake with food or dietary supplements and decreased in the presence of infection and inflammation.[535] Some biochemical methods used to measure iron are sensitive to hemolysis and will produce falsely elevated iron values,[536] whereas other serum iron assays have been shown to perform poorly in patients on dialysis.[537] Serum transferrin can be elevated by the use of oral contraceptives and reduced with inflammation or infection. Several studies have shown that these traditional biochemical iron parameters perform poorly in CKD and are inferior to some of the newer hematologic parameters (described later).[538–541] However, a lower serum iron concentration has been shown to be an independent predictor of mortality and hospitalization in dialysis recipients,[542] and a higher TSAT value has been associated with lower mortality.[543] A TSAT of 20% is generally considered a threshold value below which iron therapy is indicated.[11,12] In one study using data from NHANES, more than 50% of the noninstitutionalized adult US population was found to have values below the CKD threshold for ferritin (100 ng/mL) and TSAT (20%)[544] Overall, women were more likely to have laboratory-based evidence of iron deficiency. Men with CKD had a higher prevalence of iron deficiency than men without CKD, whereas the prevalences in women with and without CKD were similar (Fig. 55.8). Serum transferrin and total iron-binding capacity (TIBC) are dual markers of iron status and of nutritional status and protein balance: A lower baseline TIBC value or its decrease over time in dialyzed patients is associated with higher mortality and with the presence of protein-energy wasting and inflammation.[545] Approximately half of US patients on chronic hemodialysis have TSAT values of 30% or greater (see Fig. 55.8D).

Serum Transferrin Receptor. Serum transferrin receptor (sTfR) concentration is a marker of iron status that has shown promise in the evaluation of iron deficiency in patients with CKD. Transferrin receptors are shed from the membrane of maturing erythroblasts and reticulocytes, either in soluble form or as vesicles.[546–548] The concentration of sTfR is abnormally elevated in iron deficiency and has been shown to be a valuable parameter in several different clinical conditions, including ACD.[549–553] Although the sTfR concentration is not affected by inflammation, it is an expression of the size of the pool of maturing erythroblasts; also, independent of iron status, this parameter increases in hyperproliferative anemias and with the use of ESAs.

The sTfR assay is not yet widely available, and a single reference standard has been established only rather recently,[554] with different methods still reporting different units and normal ranges. A study in patients on dialysis with anemia has shown that an sTfR concentration lower than 6 mg/L (which rules out iron deficiency; normal value, 3.8–8.5) was associated with responsiveness to the initiation of EPO therapy.[555] However, because increased erythropoiesis by itself raises the sTfR concentration, sTfR measurement could not reliably detect functional iron deficiency in patients on maintenance EPO therapy. Other studies have failed to show a predictive value for sTfR in CKD anemia management.[556,557] A decline in sTfR concentration may reflect increases in iron availability when IV ascorbic acid is used to mobilize iron stores.[558] Race and ethnicity, smoking, alcohol consumption, and body mass index have been shown to be associated with sTfR values.[559,560] One can correct the sTfR value for the value of iron stores by also accounting for serum ferritin: The sTfR/ferritin ratio provides an accurate assessment of iron status and the need for iron supplementation.[547,561,562] However, to date, there is limited evidence to support the clinical use of this or similar ratios in patients on dialysis.[563–565]

Clinical Relevance

Although oral iron may be effective in ND-CKD patients, the response is generally suboptimal, and most patients do not display a hematologic response. Adherence is poor due to side effects. Side effects for the different IV iron preparations available in the United States and European markets (e.g., iron sucrose, ferric gluconate, ferric carboxymaltose, ferumoxytol, ferric isomaltoside) are usually rare and limited to mild to moderate hypersensitivity reactions. Many of these reactions are due to premedication with diphenhydramine. In approximately 1:200 administrations of IV iron, arthralgias, myalgias, or flushing appear at the beginning of the infusion in the absence of associated hypotension, tachypnea, tachycardia, wheezing, stridor, or periorbital edema. No intervention is necessary except for a temporary halt in the IV infusion. After symptoms abate, IV infusion can be restarted.

Erythrocyte Ferritin Concentration

Some studies in the 1990s had suggested a potential value for using the erythrocyte ferritin concentration as a marker

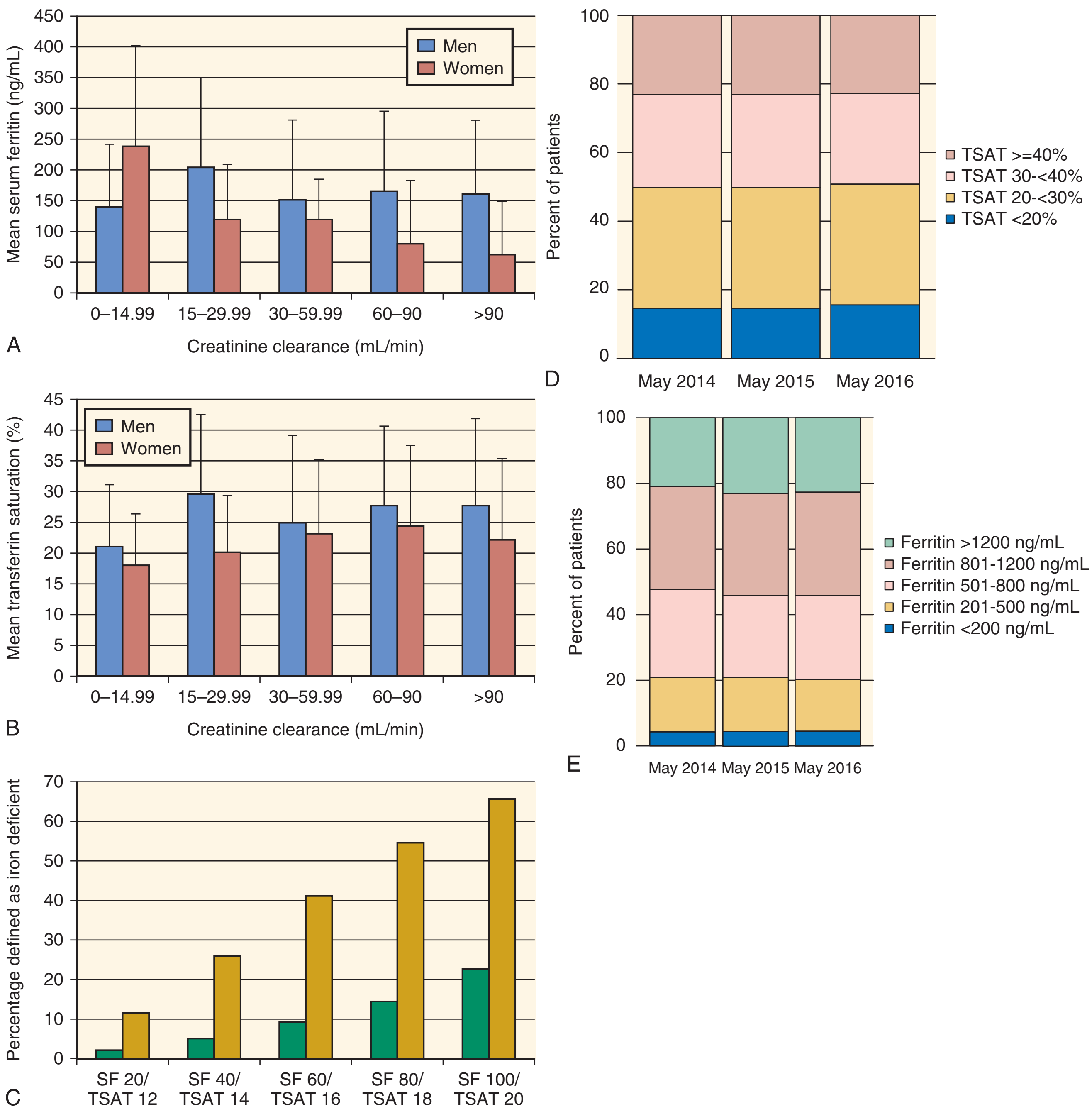

Fig. 55.8 Mean (A) serum ferritin and (B) transferrin saturation *(TSAT)* as a function of creatinine clearance (CrCl) for the combined National Health and Nutrition Examination Survey (NHANES) cohorts (*error bars* are standard deviation [SD]). The trends for both TSAT and ferritin are not significant (NS) for men but, for women, $P < .0001$ for serum ferritin and $P < .02$ for TSAT. National Kidney Foundation (NKF) chronic kidney disease (CKD) stages relative to CrCl (mL/min) are stage 5, 0 to 14.99; stage 4, 15 to 29.99; and stage 3, 30 to 59.99. Patients with CrCl 60 to 90 mL/min and patients with CrCl greater than 90 mL/min may have CKD stage 1 or 2, respectively, if other renal abnormalities are present. (C) Percentage of individuals defined as iron deficient with the use of different threshold combinations of serum ferritin *(SF)* and TSAT. The NKF Kidney Disease Outcomes Quality Initiatives (KDOQI) thresholds of serum ferritin, 100 ng/mL, and TSAT, 20%, are different from indices of iron deficiency in the non-CKD population, in which lower thresholds are generally used. The *green bars* indicate "and logic," with both test results below the specified threshold, and the *yellow bars* indicate "or logic," with either test result being below the threshold. (D) Distribution of TSAT values in adult hemodialysis patients on dialysis for at least 90 days, May 2014, 2015, and 2016. (E) Distribution of the most recent value of serum ferritin taken between March and May in adult hemodialysis patients on dialysis for at least 90 days, 2013–2016. (A–C from Fishbane S, Pollack S, Feldman HI, et al. Iron indices in chronic kidney disease in the National Health and Nutritional Examination Survey 1988–2004. Clin J Am Soc Nephrol. 2009;4:57–61; D and E from US Renal Data System. 2017 Annual Data Report. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.)

of iron status in patients on dialysis.[532,566–568] Although this assay can be run on automated analyzers using the regular serum ferritin methodology,[569] the method is cumbersome, requires complete removal of white blood cells to avoid measuring leukocyte ferritin,[570] is insensitive to dynamic changes in iron status, and is rarely available to clinicians.

Erythrocyte Zinc Protoporphyrin Concentration. The determination of erythrocyte zinc protoporphyrin (ZPP) concentrations has shown some promise to identify patients on maintenance dialysis who require iron replacement therapy.[571–574] This marker is elevated in the presence of iron-deficient erythropoiesis, with Zn replacing iron in the heme precursor protoporphyrin.[575] The erythrocyte ZPP concentration is also elevated in the presence of lead poisoning. However, the diagnostic value of erythrocyte ZPP concentration appears to be inferior to that of RBC or reticulocyte parameters.[539,576] In addition, because the whole-blood ZPP concentration is falsely elevated in patients undergoing dialysis and in the presence of bilirubin and various drugs, careful washing of the RBCs is required to remove these interferences, rendering this assay not suitable for routine clinical care.[577,578]

Percentage of Hypochromic Red Blood Cells. A distinguishing characteristic of iron-deficient erythropoiesis is the production of hypochromic microcytic erythrocytes. Iron-deficient erythropoiesis results in an increase in the percentage of hypochromic erythrocytes (%HYPO), defined as the percentage of erythrocytes with a mean corpuscular hemoglobin concentration (MCHC) lower than 28 g/dL, for hematology analyzers produced by Siemens Medical Solutions.[579,580] Similar parameters (low hemoglobin density, [LHD%], and DF-Hypo XE, respectively) are available when using Beckman-Coulter and Sysmex instruments.[581,582] A classic study by Macdougall and colleagues has shown that functional iron deficiency induced by epoetin treatment and the response to IV iron could be detected by changes in %HYPO.[524]

Several studies have confirmed that an increased %HYPO is a sensitive and early indicator of iron deficiency.[539,583–587] A European study has found %HYPO to be the only independent predictor of mortality among various iron status parameters, with a twofold higher mortality risk for values higher than 10% than for values lower than 5%.[588] According to the European Best Practice Guidelines for the Management of Anaemia in Patients with Chronic Renal Failure, patients with %HYPO values higher than 6% are most likely to show response to IV iron therapy.[530] A clinical study has tested these guidelines for anemia management; they recommended a %HYPO target of less than 10% by prospectively raising the delivered dose of IV iron to 228 patients to achieve a %HYPO lower than 2.5% and a serum ferritin concentration of 200 to 500 ng/mL.[587] In this study, the median %HYPO value decreased from 8% to 4%, the median serum ferritin concentration increased from 188 to 480 ng/mL, and the median rhEPO dose decreased from 136 to 72 IU/kg per week, showing that a strategy aimed at achieving %HYPO values much lower than 10% could be cost-effective. However, it also resulted in serum ferritin values in some patients much higher than those recommended by guidelines.

Contrary to the European studies, North American studies have failed to show value for %HYPO in assessing iron availability in dialysis patients.[589,590] The reasons for this discrepancy are not clear. It is worth noting that %HYPO progressively increases with storage of the blood sample, owing to the concomitant increase in MCV and reduction in MCHC, and is therefore best measured within 4 hours. In addition, %HYPO increases with reticulocytosis because reticulocytes have a lower MCHC than mature RBCs.[591]

Reticulocyte Hemoglobin Content. After being released from the marrow, reticulocytes spend 18 to 36 hours in the circulation before becoming mature erythrocytes. Studies of the cellular characteristics of reticulocytes thus provide a real-time assessment of the functional state of the bone marrow. Automated analyzers can determine not only the absolute number of reticulocytes with great precision, but also their size and Hgb concentration and content.[592,593] The reticulocyte Hgb content[594] (CHr or RetHe; in pg/cell) has been extensively studied, especially in patients treated with rhEPO.[595] A reduction in CHr is the most sensitive indicator of functional iron deficiency. Healthy subjects with normal iron stores who were treated with rhEPO produced a substantial fraction of hypochromic, low Hgb content reticulocytes when their baseline serum ferritin values were below 100 μg/L.[526] When IV iron was used in conjunction with EPO in normal subjects, the production of hypochromic reticulocytes was abolished.[596] Several small studies have described the value of CHr in identifying iron deficiency in dialysis recipients, mostly based on the subsequent response to IV iron.[541,589,590,597] A sensitivity of 100% and specificity of about 70% to 80% were reported in one study,[590] although other studies have reported lower values.[541,597] These initial studies led to additional large clinical trials that tested the values of CHr in managing the dosing of IV iron and rhEPO in dialysis patients. Fishbane and colleagues randomly assigned 157 patients to two different IV iron management strategies, one based on CHr, in which IV iron was started if CHr fell below 29 pg/cell, and one in which IV iron was started if the serum ferritin fell below 100 ng/mL or the TSAT values decreased below 20%.[540] A significant reduction in exposure to IV iron was obtained in the CHr-based management, with no differences in weekly EPO dosing between the two groups.[540] Tessitore and coworkers[539] have compared the diagnostic precision of a variety of hematologic and biochemical markers to identify subjects who exhibit an increase in Hgb values in response to IV iron. A combination of %HYPO higher than 6% and CHr less than 29 pg/cell has shown the best diagnostic efficiency for iron deficiency (80%) based on the Hgb response to IV iron. Other studies have provided additional confirmation of the diagnostic value of CHr,[598,599] although one has questioned its superiority to TSAT,[600] and only one study has shown that the use of IV iron in patients with low CHr resulted in decreased weekly usage of rhEPO.[601] A normal reticulocyte Hgb content may be used to optimize and reduce, if needed, the use of IV iron.[602]

Several studies have also validated reticulocyte Hgb measurements (RET-He and Ret-Hb) generated by analyzers produced by Sysmex.[528,582,603–606] The current availability of the reticulocyte Hgb parameter on several analytic platforms may allow a wider utilization of this parameter. Guidelines on the diagnosis and management of iron deficiency have included reticulocyte Hgb among those recommended.[529,607] A limitation of reticulocyte Hgb is that it cannot be used to assess iron availability in the presence of thalassemia traits (alpha or beta) or megaloblastic erythropoiesis.

Bone Marrow Iron. Although iron staining of a bone marrow biopsy is regarded as the gold standard method of assessing iron stores, widely divergent estimates of the prevalence of iron deficiency have been generated by this invasive, potentially painful procedure.[608–611] A study in 100 patients with ND-CKD has shown that evaluation of iron stores by iron staining of a bone marrow sternal aspirate was no better than TSAT or ferritin in correctly identifying responders to IV iron therapy.[612,613] Some patients with CKD have been found to have no stainable iron evident on sternal bone marrow biopsy, despite the presence of normal to elevated serum ferritin concentrations.[611,614] Bone marrow studies also have no value in identifying patients at risk for the development of functional iron deficiency with EPO therapy.

Liver Magnetic Resonance Imaging. Hepatic magnetic resonance imaging (MRI) provides a noninvasive tool to estimate liver iron deposition and is regarded as the gold standard methodology for monitoring patients with iron overload disorders. However, the number of studies applying this technology in CKD is still very limited.[615–618] In the largest of these studies, conducted in 119 patients undergoing hemodialysis and receiving IV iron in a single center, according to current guidelines, 84% had evidence of hepatic iron deposition, and 30% had hepatic MRI findings consistent with severe iron overload.[617] These data raise concerns that the use of IV iron and current thresholds for laboratory parameters may be too liberal, especially considering the increased use of IV iron in the United States since 2011 (see Fig. 55.7). On the other hand, contrary to cardiac iron MRI assessments,[619] it has not yet been demonstrated that the observed increases in hepatic iron are of any functional significance and/or are associated with clinically relevant adverse outcomes.[259,620]

Iron Balance Considerations. As noted, 1 mL of blood normally contains 0.5 mg of iron and proportionately less when the Hgb concentration is reduced. An estimated annual blood loss of 2 L in a dialysis recipient with moderate anemia (20% reduction in Hgb) therefore corresponds roughly to 0.8 g of iron loss. Irrespective of all parameters of iron metabolism, IV iron supplementation in excess of this amount results in positive iron balance unless blood loss (and thereby iron loss) is more pronounced than anticipated. When patients were categorized according to their level of hepatic iron deposition, the average monthly iron doses were 150 and 283 mg in those with signs of mild and moderate iron overload, respectively, compared with 100 mg in those without.[617] The functional consequences of a positive iron balance, and in particular whether it carries risk, however, are still unclear.[259]

Intravenous Iron Therapy

There is general agreement that oral iron therapy is insufficient to support the functional needs of EPO-stimulated erythropoiesis in patients with ESKD properly. A systematic review and meta-analysis have shown that the Hgb response is much more potent with IV iron than with oral iron, with this effect being more substantial in patients on dialysis and of a lower magnitude in patients with ND-CKD.[621,622] A systematic Cochrane analysis has identified significant associations of IV iron therapy with increased Hgb, ferritin, and transferrin saturation values, as well as reduced ESA requirements, with no differences in mortality.[623]

These findings have been confirmed in a recent meta-analysis, which showed a 23% dose reduction in ESA with the use of IV iron.[624] The cost-effectiveness of IV iron therapy has also been demonstrated under the assumption that a higher mortality risk is associated with Hgb values less than 9.0 g/dL.[625]

Nevertheless, oral iron may be effective with ND-CKD patients.[624] The FIND CKD study (Ferinject assessment in patients with iron deficiency anemia and Non-Dialysis-dependent Chronic Kidney Disease) compared the efficacy and safety of oral iron with the IV administration of ferric carboxymaltose targeting two different serum ferritin ranges, 100 to 200 and 400 to 600 μg/L.[626] Although the IV therapy targeting the higher ferritin range showed greater efficacy, there was no difference in Hgb concentration or the need to switch to another anemia therapy between the IV arm targeting the lower ferritin range and the oral iron therapy arm. However, the response to oral iron was suboptimal in many patients, with only 21.6% showing an Hgb increase of at least 1 g/dL after 4 weeks of oral iron therapy, and more than half of the patients never achieving this level of response throughout the trial.[627] A recent systematic review and meta-analysis of trials comparing oral and IV iron therapy in CKD has shown significant benefit for IV iron use in CKD stages 3 to 5.[628]

Several IV iron preparations are available for clinical use, with most of them containing iron associated with a carbohydrate shell. The strength or lability of this association is crucial for dosing, with the most stable preparations, such as iron dextran, being suitable for large dose replacements, and the more labile preparations, such as iron gluconate, requiring multiple dosing, with a single-dose maximum of approximately 100 mg. Intravenous iron infusion may lead to some immediate binding of the infused iron to transferrin, resulting in its complete saturation and the generation of free iron, which has vasoactive effects and can produce hypotensive and/or anaphylactoid reactions. This risk involves mainly semilabile iron-sugar complexes such as iron sucrose and iron gluconate, and not more stable complexes, such as ferric carboxymaltose, ferumoxytol, and iron dextran.

Several preparations of IV iron are available in the US and European markets (Table 55.1).

Lower Molecular Weight Iron Dextran. This is produced by PharmaCosmos (Holbaek, Denmark) and is in use in the United States (INFeD, Actavis, Dublin Ireland) and Europe (Cosmofer, Pharmacosmos, Holbaek; Ferrisat, HAC Pharma, Caen, France). It has a significant better tolerability and fewer side effects than the higher molecular weight product, which has now been removed from the US and European markets.[629–635]

Iron Sucrose. Called Venofer, this is produced by Vifor (St. Gallen, Switzerland) and marketed in the United States by American Regent Laboratories (Shirley, NY). It is used worldwide for the treatment of renal anemia and is the most used parenteral iron preparation in the United States.[636]

Table 55.1 Intravenous Iron Preparations

	FDA-Approved					Non–FDA- Approved
Generic Name	Iron Dextran	Iron Sucrose	Na Ferric Gluconate	Ferric Carboxymaltose	Ferumoxytol	Iron Isomaltoside 1000
US trade name (marketed by)	INFed (Actavis)	Venofer (American Regent, Luitpold Pharmaceuticals)	Ferrlecit (Sanofi-Aventis); Nulecit (Actavis)	Injectafer (Luitpold, American Regent)	Feraheme (AMAG Pharmaceuticals)	Monofer (Pharmacosmos A/S)
Trade name, Europe	Cosmofer, Uniferon, Ferrisat	Venofer, Idafer, FerroLogic, Ferion, Venotrix, Fermed, Netro-Fer	Ferrlecit/Ferlixit	Injectafer, Ferinject	Rienso	Monofer, Monover, Monoferro, Diafer
Carbohydrate	Dextran polysaccharide (LMW)	Sucrose	Gluconate	Carboxymaltoside	Polyglucose sorbitol carboxymethylether	Isomaltoside
Molecular weight	165 kDa	34–60 kDa	289–444 kDa	150 kDa	750 kDa	150 kDa
Iron, mg/mL	50	20	12.5	50	30	50 or 100
Hemodialysis, mg/session	100	100	125	—	510	100–200 (UK)
Peritoneal dialysis	100	1 × 300 mg; 1 × 300 mg after 14 days; 1 × 400 mg after 14 days		—	510	
CKD, nondialysis	100 mg	200 or 500 mg		750 mg	510 mg	
TDI possible	Yes	No	No	Yes	No	Yes
Maximum approved dose	100 mg	400 mg	125 mg	750 mg for body weight >50 kg	510 mg	Up to 20 mg/kg (UK)
Max safe dose	TDI over 1–4 hrs	300 mg over 2 hrs	250 mg over 1 hr	750 mg over 15 min	510 mg in >15 min	20 mg/kg over 15 min
Premedication	No	No	No	No	No	No
Test dose required	Yes	No	No	No	No	No
Black box warning (FDA)[a]	Yes	No	No	No	Yes	Na
Adverse reactions				Hypophosphatemia	Alteration of MRI scan	
Preservative	None	None	Benzyl alcohol	None	None	None

[a]FDA warnings—iron dextran and ferumoxytol, fatal allergic reactions.

CKD, Chronic kidney disease; *FDA,* U.S. Food and Drug Administration; *LMW,* low molecular weight; *MRI,* magnetic resonance imaging; *NA,* not applicable; *TDI,* tolerable daily intake.

"All intravenous iron medicines have a small risk of causing allergic reactions, which can be life-threatening if not treated promptly." European Medicines Agency. New recommendations to manage risk of allergic reactions with intravenous iron-containing medicines. http://www.ema.europa.eu/ema/index.jsp?curl=pages/news_and_events/news/2013/06/news_detail_001833.jsp&mid=WC0b01ac058004d5c1.

Allergic reactions have been reported in fewer than 1/100,000 infusions. IV injection into rats of three different commercial preparations of iron sucrose have resulted in different degrees of inflammation and oxidative stress, suggesting that the stability of the iron complex may differ from one iron sucrose preparation to another.[637]

Ferric Gluconate. This is marketed in the United States by Sanofi Aventis US (Bridgewater, NJ) as Ferrlecit. It is the second most commonly prescribed IV iron preparation in the United States and is frequently used worldwide in patients on hemodialysis.[638]

Ferric Carboxymaltose. This is marketed as either Ferinject (Vifor, St. Gallen, Switzerland) or Injectafer (American Regent, Luitpold Pharmaceuticals, Shirley, NY). It is the newest iron preparation registered in Europe and the United States.[639] A significant advantage of this preparation is the possibility of infusing up to 750 mg of iron in a short time (15 minutes), with minimal side effects.[640,641] Transient hypophosphatemia has been reported in patients without CKD and in those with ND-CKD treated with ferric carboxymaltose, possibly mediated by a decreased tubular reabsorption of phosphate.[642,643] FGF23 plays an important role in the hypophosphatemia observed with iron carboxymaltose administration. In iron deficiency, increased FGF23 production is associated with increased cleavage within osteocytes, with unchanged FGF23 plasma values and elevated values of FGF23 c-terminal fragments. Administration of iron carboxymaltose is associated with a large increase in plasma FGF23 values, possibly due to reduced cleavage. Low serum phosphate, increased phosphate fractional excretion, and decreased serum 1,25 vitamin D values are observed for 1 to 2 weeks following IV administration of 750 mg of iron carboxymaltose at week 0 and 1.[643a] In non–dialysis-dependent CKD cases, with serum ferritin values between 400 and 600 ng/mL (FIND-CKD trial), ferric carboxymaltose was better than oral iron in delaying and/or reducing ESA requirements.[626] There was no evidence of IV iron-related renal toxicity or decreased renal function in the FIND-CKD trial.[644]

Ferumoxytol. This is marketed by AMAG Pharmaceuticals (Lexington, MA) as Feraheme. It is an iron oxide nanoparticle with a polyglucose sorbitol carboxymethylether coating designed to minimize immunologic sensitivity and release of free iron, allowing a rapid injection of a large dose (510 mg) of iron, which can be repeated after 3 to 8 days.[645–648] Because infusion times of 17 to 60 seconds were associated with significant side effects, the US Food and Drug Administration (FDA) has demanded that infusion times be longer than 15 minutes.

Efficacy and adverse events in patients with CKD were found to be similar to those of iron sucrose.[649] Ferumoxytol is the only IV iron preparation possessing super magnetic properties, similar to MRI contrast agents, which may alter MRI imaging for up to 3 months owing to its uptake into the reticuloendothelial system.[646] On the basis on the studies used for FDA registration, 0.2% of treated subjects experienced anaphylaxis or anaphylactoid reactions, 3.7% had hypersensitivity-type reactions (e.g., pruritus, rash, urticaria, wheezing), 1.9% of patients had hypotension, and three patients experienced serious hypotensive reactions.[646] However, studies of larger datasets have not shown an increased incidence of adverse reactions for ferumoxytol compared with other IV iron preparations in CKD patients.[650] There were no significant differences between ferumoxytol and ferric carboxymaltose regarding hypersensitivity reactions or hypotension post-IV infusion, whereas the incidence of hypophosphatemia was greatly increased for ferric carboxymaltose (38.7% vs. 0.4%).[651] The Ferumoxytol for Anemia of CKD Trial (FACT) showed that ferumoxytol had similar efficacy and safety profile to iron sucrose in the treatment of iron deficiency anemia in patients with CKD undergoing hemodialysis.[652]

The development of nephrogenic systemic fibrosis following gadolinium use has prompted the use of ferumoxytol as an alternative MRI contrast agent in patients with CKD stage 4 or 5 and in dialysis-dependent CKD.[653,654]

Ferric Isomaltoside. This is marketed by Pharmacosmos A/S (Holbaek) as Monofer. It is based on a nonbranched carbohydrate, which does not form the typical spheroidal iron carbohydrate nanoparticle like other IV iron preparations and seems to be associated with lower immunogenic potential. Monofer can be administered in a single dose, with dosages up to 20 mg/kg, and has been shown to be equivalent to iron sucrose in CKD patients on dialysis.[655] Monofer is currently approved and marketed in 28 countries, including 21 European Union members, but not in the United States.

A US study on IV iron use for 1994 through 2002 has indicated that iron sucrose and ferric gluconate are the predominant forms of IV iron used in CKD, with 84.4% of hemodialysis and 19.3% of peritoneal dialysis patients having some form of IV iron therapy.[656] Parenteral iron administration has increased substantially in the United States, most likely because of a shift in reimbursement practices toward a bundled capitated model (see Fig. 55.7). Data from the Dialysis Outcomes and Practice Patterns Study (DOPPS) have shown IV iron use increasing from 55% to 68% of patients on hemodialysis between 2010 and 2012.[657] Similar trends have been reported for European countries, Japan, Australia, and New Zealand.[658] A ferritin threshold of 800 ng/mL is now commonly used for prompt withholding of IV iron therapy in patients on maintenance dialysis.

Side Effects of Intravenous Iron

Iron sucrose, lower molecular weight iron dextran, and ferric carboxymaltose have excellent track records for safety and tolerability. Hypersensitivity reactions (e.g., erythematous rash, urticaria) are rare, and their intensity is usually mild or moderate. Lack of recurrence after rechallenge indicates that most of these events are not due to immunologic reactions. Severe life-threatening allergic reactions have been a major problem with the higher molecular weight iron dextran, prompting its removal from the European and US markets. The use of a test dose is still required for iron dextran in the United States, but the European Medicines Agency no longer recommends it.[659] A retrospective study by Chertow and colleagues examining more than 50 million doses of IV iron has demonstrated the higher risk of reactions with higher molecular weight iron dextran; they found that rates of serious events associated with lower molecular weight iron dextran were similar to those seen with the other forms of IV iron (~1/200,000).[630,631] A study from the FDA using data obtained from the administration's Adverse Event Reporting System (AERS) and other US databases, was unable to provide firm data on the relative safety of the four IV preparations marketed in the United States owing to incomplete brand information on these reports, but it did confirm that allergic reactions have been reported for all brands.[660,661] Chertow and colleagues estimated absolute rates of life-threatening reaction per million doses of 0.6 for iron sucrose, 0.9 for sodium ferric gluconate complex, 3.3 for lower molecular weight iron dextran, and 11.3 for higher molecular weight iron dextran.[631] However, a later systematic review has highlighted the lack of properly conducted and powered studies comparing adverse event rates between lower molecular weight iron dextran and iron sucrose.[662] The amount of labile iron, which differs among the various IV iron preparations, is likely an important determinant of possible oxidative and nitrosative stress.[663]

In ND-CKD patients, the FIND-CKD study observed no adverse effects of IV ferric carboxymaltose as compared with oral iron.[664] The REVOKE trial aimed to assess the effect of IV iron on the progression of CKD by comparing IV iron sucrose, 200 mg once a week for 2 weeks, with oral iron sulfate. Although there was no difference in the slope or GFR, the authors reported a higher number of SAEs in the IV iron arm and concluded that the risk for infections and cardiovascular events was increased.[665] This unexpected result was not confirmed in FIND-CKD, and the difference between the outcomes of both studies remains unexplained so far.[666,667]

Infection Risk and Intravenous Iron Therapy. In vitro data seem to support the notion that iron can promote bacterial growth and, at the same time, impair leukocyte function.[576,668–671] IV injection of iron sucrose in dialyzed patients has been associated with the dose-dependent appearance of markers of oxidant damage in lymphocytes and a decrease in plasma ascorbate and alpha-tocopherol in some studies[672] but not in others.[673–675] In addition, studies have not accounted for the fact that the capability of leukocytes to cope with oxidant damage is markedly affected by polymorphisms in glutathione S-transferase M1 (GST M1).[676] Although there is indirect and inconclusive evidence for an association between iron stores and bacteremia,[668] most studies have failed to show an association of IV iron therapy with an increased risk of infection in dialyzed patients.[576,669–671] Many studies attempting to link iron status and risk of bacterial infection have used serum ferritin, an unreliable marker of iron status in CKD, as discussed previously.[668,677–681] One study has shown that in patients receiving more than 10 vials of 100 mg iron dextran over 6 months, there was an increased risk of death and hospitalization.[682] One uncontrolled retrospective study has reported a higher incidence of bacteremia with iron sucrose than with ferric gluconate.[683] Other studies have failed to show a significant effect of IV iron dosing or iron status (using serum ferritin) on bacteremia, mortality, infection, or hospitalization.[576,671,684,685] On the other hand, a later observational study using a very large database has found that bolus administration of higher doses of IV iron is associated with higher risks for infection-related hospitalizations and death, particularly in patients undergoing dialysis with catheters, rather than arteriovenous fistulas or grafts.[686] Higher doses of IV iron have not been shown to be associated with higher mortality risk or infections.[687]

Despite the lack of proof of significant effects on the rates of infections, cardiac events, and mortality, long-term toxicities and, in particular, the possible consequences of oxidant damage due to free radical generation are still a concern.[604,688] Unfortunately, no large outcome studies have been performed so far to test the efficacy and safety of iron replacement strategies prospectively, either in the short or long term.[689] A trial in the United Kingdom (PIVOTAL) has compared proactive high-dose IV iron sucrose therapy with reactive low-dose therapy in 2141 patients new to dialysis, and assessed the impact of both regimens on mortality and cardiovascular events.[690] The high-dose IV sucrose regimen yielded a significant decrease in ESA monthly dose, with similar effects on mortality, cardiovascular events, and infection rates.[690]

Iron Therapy in Patients With Chronic Kidney Disease. Iron therapy in CKD should be guided by iron status test results and clinical considerations and needs to take into account the potential benefits of avoiding or minimizing blood transfusions, ESA use, and anemia-related symptoms against the risk of potential harm[10] (Table 55.2). Iron tests should be performed monthly in the initial phase of ESA treatment and every 3 months thereafter.[11,12] As discussed in detail previously, a target of serum ferritin concentration higher than 200 ng/mL and TSAT value greater than 20% or CHr value greater than 29 pg/cell has been used for patients on dialysis.[11,12] The KDIGO guideline recommends using IV iron if an increase in Hgb concentration or a decrease in ESA requirements is sought if the serum ferritin value is 500 ng/mL or less and TSAT is 30% or less.[10] For patients with ND-CKD and those undergoing peritoneal dialysis, target values of more than 100 ng/mL for serum ferritin and higher than 20% for TSAT should be used. The objective of iron therapy in CKD is to abolish overt and/or functional iron deficiency because this reduces the erythropoietic response to and effectiveness of ESAs. The response to ESAs can be optimized by the simultaneous use of IV iron, which enables a significant reduction in ESA dosing.[691] Several studies conducted in the early and late 1990s have demonstrated that IV iron therapy is associated with significant ESA dose reductions.[623,674,692–700] The DRIVE (Dialysis Patients Response on IV Iron with Elevated Ferritin) study has shown that an intensive IV iron administration protocol (125 mg ferric gluconate with each of eight hemodialysis sessions) can significantly reduce ESA dosing requirements.[701,702] A Cochrane systematic review has provided additional support to the ESA-sparing effects of IV iron.[623] Shirazian and associates have noted that the ESA-sparing effects of IV iron could easily be demonstrated when there is a high prevalence of iron deficiency and low usage of IV iron.[691] However, they have suggested that most of the benefits of IV iron in reducing ESA use have already been achieved, given that 60% to 80% of US patients on dialysis are being treated with IV iron, and that it is not clear how much additional ESA dose reduction could be obtained with more intensive IV iron regimens.

Given the potential adverse effects of ESAs (see later), the latest KDIGO anemia guideline mentions explicitly that the desire to avoid or minimize ESAs can influence the decisions about iron use[10] (see Table 55.2).

As shown in Table 55.1, several forms of IV iron are available worldwide. Although they have important differences in formulation and dosing, no convincing evidence has been

Table 55.2 Current Anemia Guidelines and Position Statements Regarding Iron Administration

Guideline	Comments
KDIGO Clinical Practice Guideline (International)	2.1.1: When prescribing iron therapy, balance the potential benefits of avoiding or minimizing blood transfusions, ESA therapy, and anemia-related symptoms against the risks of harm in individual patients (e.g., anaphylactoid and other acute reactions, unknown long-term risks) (not graded) 2.1.2: For adult CKD patients with anemia not on iron or ESA therapy, we suggest a trial of IV iron (or in CKD ND patients, alternatively, a 1–3 month trial of oral iron therapy) if (2C): • An increase in Hb concentration without starting ESA treatment is desired[b] *and* • TSAT is ≤30% and ferritin is ≤500 ng/mL (≤500 µg/L) 2.1.3: For adult CKD patients on ESA therapy who are not receiving iron supplementation, we suggest a trial of IV iron (or in CKD ND patients, alternatively, a 1–3 month trial of oral iron therapy) if (2C): • An increase in Hb concentration[c] or a decrease in ESA dose is desired[d] *and* • TSAT is ≤30% and ferritin is ≤500 ng/mL (≤500 µg/L)
KDOQI Commentary (United States)	• We believe that the degree of caution expressed by KDIGO is not supported by the available evidence and could have negative effects, such as sustained iron deficiency anemia, higher ESA dose requirements, and increased blood transfusions. • We therefore believe that a therapeutic trial of IV iron could be considered when TSAT is low (≤30%), even if ferritin concentration is above 500 ng/mL. • There is insufficient evidence on which to base a recommendation for an upper ferritin limit above which IV iron must be withheld. • A decision to administer iron in the setting of high ferritin would require weighing potential risks and benefits of persistent anemia, ESA dosage, comorbid conditions, and health-related QoL. In accordance with KDIGO recommendations, Hb response to iron therapy, TSAT, and ferritin should be monitored closely and further iron therapy titrated accordingly.
CSN Commentary (Canada)[a]	• There is good evidence (1B) to support the administration of iron in adult CKD patients when the TSAT and ferritin thresholds are above 20% and 200 ng/mL. A therapeutic trial of iron can be considered in those where an increase in Hb or reduction of ESA or avoidance of ESA and transfusion is desired, while recognizing that an increase in hemoglobin is less likely when TSATs are >30% and ferritin values are >500 ng/mL. • However, as opposed to the KDIGO anemia guideline, the CSN anemia work group believes that the current evidence does not permit a clear delineation for an upper limit of TSAT or ferritin values.

[a]Quoted material is excerpted to focus on hemodialysis; ferritin units for CSN commentary converted to ng/mL.
[b]Based on patient symptoms and overall clinical goals, including avoidance of transfusion, improvement in anemia-related symptoms, and after exclusion of active infection.
[c]Consistent with recommendations 3.4.2 and 3.4.3.
[d]Based on patient symptoms and overall clinical goals, including avoidance of transfusion and improvement in anemia-related symptoms and after exclusion of active infection and other causes of ESA hyporesponsiveness.
CKD, Chronic kidney disease; *CKD ND,* non–dialysis-dependent CKD; *CSN,* Canadian Society of Nephrology; *ESA,* erythropoiesis-stimulating agent; *Hb,* hemoglobin; *IV,* intravenous; *KDIGO,* Kidney Disease: Improving Global Outcomes; *KDOQI,* Kidney Disease Outcomes Quality Initiative; *QoL,* quality of life; *TSAT,* transferrin saturation.
(Modified from Weiner DE, Winkelmayer WC. Commentary on "The DOPPS practice monitor for US dialysis care: update on trends in anemia management 2 years into the bundle": iron(y) abounds 2 years later. *Am J Kidney Dis.* 2013;62:1213–1220; quoted material from Kidney Disease Improving Global Outcomes (KDIGO) Anemia Work Group: KDIGO clinical practice guideline for anemia in chronic kidney disease. *Kidney Int Suppl.* 2012;2:279–335, 2012; Kliger AS, Foley RN, Goldfarb DS, et al. KDOQI US Commentary on the 2012 KDIGO clinical practice guideline for anemia in CKD. *Am J Kidney Dis.* 2013;62:849–859; and Moist LM, Troyanov S, White CT, et al. Canadian Society of Nephrology commentary on the 2012 KDIGO clinical practice guideline for anemia in CKD. *Am J Kidney Dis.* 2013;62: 860–873.)

provided about the superiority of one form over the others in the setting of CKD. The REPAIR-IDA trial has demonstrated the a regimen of two doses of 750 mg of ferric carboxymaltose in 1 week was not inferior to up to five infusions of iron sucrose in 14 days for anemic subjects with NK-CKD.[643] The use of larger doses of IV iron, rather than lower maintenance doses, does not appreciably affect cardiovascular morbidity and mortality in hemodialyzed patients.[703]

Some studies have suggested that the addition of ascorbic acid to the therapeutic regimen of patients treated with ESA and iron has beneficial effects, although none of these studies was rigorously conducted to provide definitive evidence.[704,705] Limited evidence has suggested that ascorbic acid may be pro-oxidant and may increase cytokine values.[706] A systematic review and meta-analysis has concluded that there is evidence in a limited number of small studies that the use of ascorbic acid results in increased Hgb concentrations, improves transferrin saturation, and reduces EPO utilization.[707] However, the use of ascorbic acid is not recommended in the KDOQI or KDIGO guidelines.[10–12]

EFFICACY AND SAFETY OF ANEMIA MANAGEMENT WITH ERYTHROPOIESIS-STIMULATING AGENTS AND IRON

The change in the condition of patients on maintenance dialysis following the advent of rhEPO has been impressive

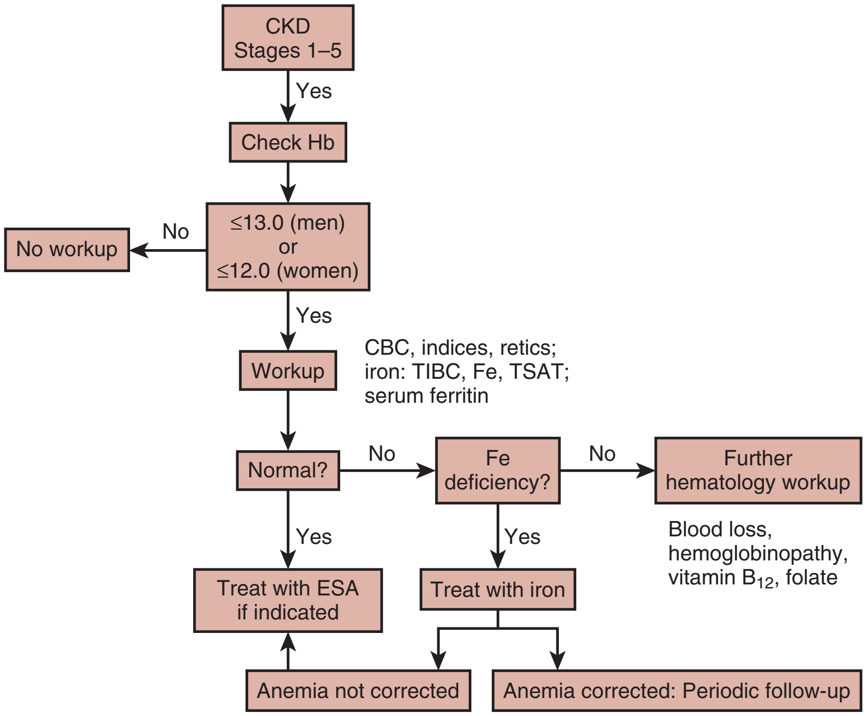

Fig. 55.9 Flow chart for the evaluation of the patient with chronic kidney disease *(CKD)* and anemia. *CBC,* Complete blood count; *ESA,* erythropoiesis-stimulating agent; *Fe,* iron; *Hb,* hemoglobin; *TIBC,* total iron-binding capacity; *TSAT,* transferrin saturation. (Modified from Lankhorst CE, Wish JB. Anemia in renal disease: diagnosis and management. Blood Rev. 2010;24:39–47.)

and obviously advantageous. Transfusion requirements declined, iron overload due to previous RBC transfusions gradually resolved, and patients could easily be maintained at Hgb values above those that had to be accepted when RBC transfusions were the only viable option of anemia management. Androgen therapy, which had been associated with significant side effects, could also be abolished. Because of these obvious benefits, the use of epoetin soon became routine, and the workup for anemia is now considered part of the management program of patients in all stages of CKD (Fig. 55.9). If the workup reveals no reasons for anemia other than EPO deficiency associated with CKD, and in particular has ruled out iron deficiency, ESA therapy provides an option to correct anemia in almost all patients. However, despite the apparent advantages, formal evidence of a positive long-term benefit has never been established.

Several lines of indirect evidence have suggested that correcting or ameliorating anemia could reduce or at least mitigate the rate of left ventricular hypertrophy, a frequent complication in CKD, that is clearly associated with poor prognosis (see preceding discussion). Together with the apparent lack of adverse effects of ESA therapy, and the contention that higher Hgb concentrations might lead to improved HRQOL and physical function, this evidence led to an increase in Hgb target values. In addition, treatment was expanded to those patients not yet on dialysis, in whom anemia is generally less severe than in those on dialysis, because avoidance of anemia rather than late correction was intuitively considered the most appropriate strategy to improve prognosis and quality of life.

Unfortunately, however, the true nature of the relationship between long-term reductions in Hgb concentrations and adverse outcomes was not adequately tested in prospective interventional trials for a long time. Few studies have actually compared ESAs against placebo, and those trials, testing two different Hgb target ranges, have usually been inadequately powered. Subsequently evidence from several RCTs has become available, suggesting that normalization of Hgb concentrations with ESAs is associated with limited benefit and relevant harm.

Trial Overview

Since 1989, slightly more than 25 RCTs using ESAs in patients with CKD have been published, in which either different target Hgb concentrations were compared or ESA treatment was compared with placebo. Approximately 50% of these trials were conducted in patients on dialysis and the other half in patients with ND-CKD. Overall, approximately 11,000 patients have been enrolled in these trials, more than 4000 of whom were involved in one study, the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT).[708] The number of patients in the other trials varied from less than 20 to approximately 1400.[11,12] Several small trials conducted until 1997 compared ESA therapy with placebo. Thereafter, only treatment strategies testing two different ESA regimens were performed until TREAT was designed as the first large trial to compare ESA therapy with placebo in patients with diabetes and CKD who were not undergoing dialysis.

Large Randomized Controlled Trials

The US Normal Hematocrit Trial was the first study to test whether normalization of Hgb concentrations improves the prognosis of patients on dialysis.[709] It was hypothesized that any presumed benefit would be most obvious in patients with cardiac disease; therefore the trial enrolled slightly more than 1200 hemodialysis recipients who had congestive heart failure or ischemic heart disease. The target Hct in the higher arm was 42% and, in the lower arm, was 30%. The primary endpoint was a composite of death and first nonfatal myocardial infarction. The study was terminated early after 29 months because more patients in the higher arm had reached the primary endpoint. Although the difference did not reach statistical significance, study termination was recommended because it was obvious that the original hypothesis—that the higher Hct target was of benefit—could not be proven. In addition, the incidence of vascular access thrombosis was significantly higher in the higher target Hct arm. Self-reported physical function score improved at higher Hct values but, importantly, there was no significant difference between the two treatment arms for this parameter. A later analysis, which included endpoint events that the data safety monitoring committee had not yet considered when recommending termination of the study, also did not reveal a significant difference, and the rates of events occurring during 1-year of follow-up after study termination were similar in the two treatment arms.[710]

A second large trial in ESKD included almost 600 patients new to hemodialysis without symptomatic heart disease and left ventricular dilation.[711] Patients were randomly assigned in a double-blind fashion to an Hgb treatment target of either 13.5 to 14.5 g/dL or 9.5 to 11.5 g/dL. The primary endpoint was a change in left ventricular volume index on the assumption that raising the Hgb concentration would prevent the progression of left ventricular hypertrophy. However, changes in left ventricular volume index were similar for the two treatment groups. The only difference among a number of secondary outcomes was a better 36-Item Short Form Health Survey (SF-36) vitality score in the higher Hgb than in the lower Hgb group. Adverse event rates were also similar, except that rates of skeletal pain, surgery, and dizziness were higher in the lower arm, whereas those of headache and cerebrovascular events were slightly higher in the higher arm.

Thus, neither trial provided any evidence in favor of normalization of Hgb concentrations in patients on dialysis. However, because the prognosis, extent of comorbidities, and hemodynamic and metabolic milieu of patients undergoing dialysis are so different from those of patients with ND-CKD, the benefit of anemia correction was further tested in ND-CKD in three larger studies.

The CREATE (Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta) trial was conducted in Europe, Mexico, and Taiwan.[712] It enrolled approximately 600 patients with an eGFR of 15 to 35 mL/min/1.73 m^2 and an Hgb concentration of 11 to 12.5 g/dL. Patients were randomly assigned to a treatment arm in which epoetin beta therapy was started immediately to achieve Hgb concentrations of 13 to 15 g/dL or to an arm in which treatment with epoetin was not initiated before Hgb had dropped to below 10.5 g/dL; the target Hgb in this second arm was 10.5 to 11.5 g/dL. The primary endpoint was a composite of eight cardiovascular events, which included the following:

….the time to a first cardiovascular event, including sudden death, myocardial infarction, acute heart failure, stroke, transient ischemic attack, angina pectoris resulting in hospitalization for 24 hours or more or prolongation of hospitalization, complication of peripheral vascular disease (amputation or necrosis), or cardiac arrhythmia resulting in hospitalization for 24 hours or more."[712]

The study did not show a significant difference in the time to event between the treatment arms. Some dimensions of HRQOL were improved in the arm with earlier treatment and higher target, but, unexpectedly, time to dialysis was significantly shorter in this treatment arm. One of the limitations of the trial was that the observed event rate was much lower than the anticipated event rate, yielding lower than expected statistical power.

The CHOIR (Correction of Hemoglobin and Outcomes in Renal Insufficiency) study, conducted in the United States, had a similar design, but enrolled patients with more comorbid disease and yielded different conclusions.[713] More than 1400 patients with eGFR values of 15 to 50 mL/min/1.73 m^2 and Hgb concentrations below 11 g/dL were randomly allocated to receive epoetin alfa to achieve one of two different target Hgb values, 13.5 or 11.3 g/dL. The primary endpoint was a composite of death, myocardial infarction, hospitalization for congestive heart failure, and stroke. The trial was terminated when significantly more patients in the higher Hgb arm experienced at least one cardiovascular event. Separate analyses of the four components of the combined endpoint revealed trends for more frequent hospitalizations for heart failure and more frequent deaths but no difference in the rates of myocardial infarction or stroke. In addition, there was a trend toward more rapid progression of kidney disease in the higher Hgb target group. However, a meta-analysis of 19 trials did not support an effect of higher Hb targets on the progression of CKD, adverse events, or mortality.[714] Unlike in the CREATE trial, twofold to threefold higher doses of epoetin were needed in the CHOIR study to achieve and maintain similar Hgb values. Interestingly, a posthoc analysis has shown that the risks associated with the higher Hgb target are not apparent among subgroups with a higher mortality risk.[715]

In contrast to the other four large trials, the TREAT study was designed to test the effect of ESA in comparison with placebo in a sufficiently powered study. More than 4000 patients with CKD (eGFR = 20–60 mL/min/1.73 m^2), type 2 diabetes, and an Hgb concentration below 11 g/dL were randomly assigned to receive darbepoetin, with a treatment target of 13 g/dL, or placebo.[708] To avoid development of severe anemia in the placebo-treated group, a rescue protocol was established, according to which darbepoetin was administered when Hgb fell below 9 g/dL. The study was double-blinded. There were two primary endpoints—a cardiovascular composite endpoint and a renal composite endpoint, including death or initiation of maintenance dialysis. The trial showed no difference in the composite renal or cardiovascular endpoints, but analysis of the components of the primary endpoint revealed a significant, twofold higher risk of stroke in the darbepoetin arm. As an additional safety signal, the number of deaths attributed to cancer tended to be higher in the treatment arm, albeit not significantly and, in a subgroup of approximately 350 patients with a history of malignancy, all-cause mortality tended to be higher, and significantly more deaths were attributed to cancer. These findings were consistent with some findings in ESA RCTs in patients with cancer, which showed higher mortality and more rapid progression of malignancy in patients treated with ESA for chemotherapy-related anemia.[716] Patients in the darbepoetin arm of the TREAT study received fewer transfusions and showed a larger mean change in the Functional Assessment of Cancer Therapy: Fatigue (FACT) score.

Risk-Benefit Relationship and Target Hemoglobin Recommendations

In summary, evidence from well-designed, larger RCTs has indicated that raising Hgb to normal or near-normal values with ESAs does not enhance survival or reduce the rate of cardiovascular events in patients with ND-CKD or ESKD, but is associated with risk for harm.[717] These results are also consistent with another large trial in patients with heart failure, many of whom had CKD.[633] Almost all studies showed increased rates of thromboembolic events but, for unknown reasons, other risks are not consistent across different studies. Although the CHOIR study, for example, suggested a mortality risk,[713] this finding was not confirmed in TREAT.[708] Also, a negative impact on the time to dialysis, as found in the CREATE trial,[712] was not found in TREAT. TREAT, on the other hand, found an increased incidence of stroke[708] and, although another study had previously reported a slightly higher number of strokes in a higher Hgb treatment arm,[718] neither the CREATE trial nor the CHOIR study found differences in stroke rates.[712,713] These inconsistencies may indicate important yet unrecognized factors that determine the side-effect profile of ESAs. Despite intensive investigation, it has not been possible so far to identify characteristics that distinguish patients in whom stroke developed during ESA therapy in TREAT.[719] Whether any of the observed adverse events is related to the actual achieved Hgb values, to indirect effects of an increase in erythropoiesis, or to direct, Hgb-independent effects of ESAs is unknown.[715]

Any benefit of higher Hgb values for HRQOL appears modest, on average, once Hgb concentrations above about 10 g/dL are reached. Transfusion rates are lower with higher Hgb values[720,721] but it is also clear that attempts to normalize Hgb by no means eliminate transfusion requirements. Moreover, the actual benefit from avoiding RBC transfusions is difficult to determine in individual patients, although the risk of sensitization in prospective transplant recipients should be strongly considered.[722]

Whether the balance of the risks and benefits of ESA therapy depends on the patient's responsiveness remains unclear. In treatment protocols driven by a target Hgb range, hyporesponsiveness leads to the use of higher doses and is associated with a greater likelihood of adverse events, but whether ESAs play a causal role remains unclear. A secondary analysis of the CHOIR study has suggested that high ESA doses rather than high Hgb concentrations are associated with poor outcomes.[411,715] In TREAT, the response to the first two weight-based doses of darbepoetin was a significant predictor of poor prognosis, with patients in the lowest quartile of ESA responsiveness having higher rates of the composite cardiovascular endpoint or death.[723] However, because "hyporesponders" could be identified only among the treated patients, it is unclear whether their poor prognosis was affected by ESA therapy.

The global KDIGO anemia guideline for anemia in CKD takes these considerations into account.[10] Careful balancing of the risks and benefits of ESA and iron therapy is an overarching recommendation. In patients not undergoing dialysis, the guideline recommends that a decrease of Hgb values to less than 9 g/dL be avoided by the initiation of ESA when Hgb is between 9 and 10 g/dL. In general, ESAs should not be used to maintain Hgb concentrations above 11.5 g/dL, and there is a strong recommendation against intentionally raising Hgb above 13 g/dL. However, individualization appears appropriate because some patients may have improvements in quality of life with Hgb values above 11.5 g/dL and are prepared to accept an increased risk.[724]

In the United States, a major change in payment for dialysis and related services has resulted in bundling of payments for laboratory services and IV medications and their oral equivalents. Together with the previously mentioned results from clinical trials and subsequent changes in drug labelling from the FDA, these payment changes have produced measurable reductions in the use of ESAs and Hgb values and have resulted in higher rates of transfusion in patients on maintenance dialysis.[57,725] The US experience may not be directly applicable to other countries.[726] A European study in 1679 patients on hemodialysis has identified weekly ESA doses more than 8000 IU as an independent risk factor for all-cause mortality and hospitalization.[727]

Ferric Citrate–Based Phosphate Binding and Iron Metabolism in Chronic Kidney Disease

Ferric citrate (Zerenex, Keryx Biopharmaceuticals, NY), an iron-containing phosphate binder, has been shown to reduce serum phosphate values increase TSAT values, and moderately increase Hgb values in comparison with placebo in patients with ND-CKD, providing further support for the potential use of this product as an oral iron supplement in this patient group.[728–730] However, the FDA has approved use of the product only as a phosphate binder and placed a safety warning for a potentially excessive elevation of iron stores, suggesting regular monitoring with serum ferritin and TSAT values.[731] An increased incidence of gastrointestinal side effects was also noted for ferric citrate compared with placebo.[732] Control of serum phosphorus with another ferric citrate–based phosphate binder has been shown to be associated with lower ESA requirements and lower use of IV iron.[733] Sucroferric oxyhydroxide, another recently FDA-approved phosphate binder, has shown no significant effects on IV iron metabolism and anemia.[734]

RED BLOOD CELL TRANSFUSION

When large RCTs questioned the safety and overall benefit of ESAs and of targeting higher Hgb concentrations, an appreciable increase was observed in the proportion of patients having Hgb values less than 10 g/dL, which, in conjunction with stable transfusion rates for this patient subgroup, translated into an increase in the absolute number of transfusions.[720] As shown in Fig. 55.10 (and Fig. 55.3), transfusion rates for US patients on maintenance dialysis were 2.9% in 2011 and 3.0% in 2012. Interestingly, transfusion rates in ND-CKD patients also rose significantly from 2002–2003 to 2008 (see Fig. 55.10).[721]

Transfusion is associated with the development of alloantibodies and human leukocyte antigen (HLA) sensitization,[735] which has important negative consequences for donor matching of patient candidates for a renal transplantation.

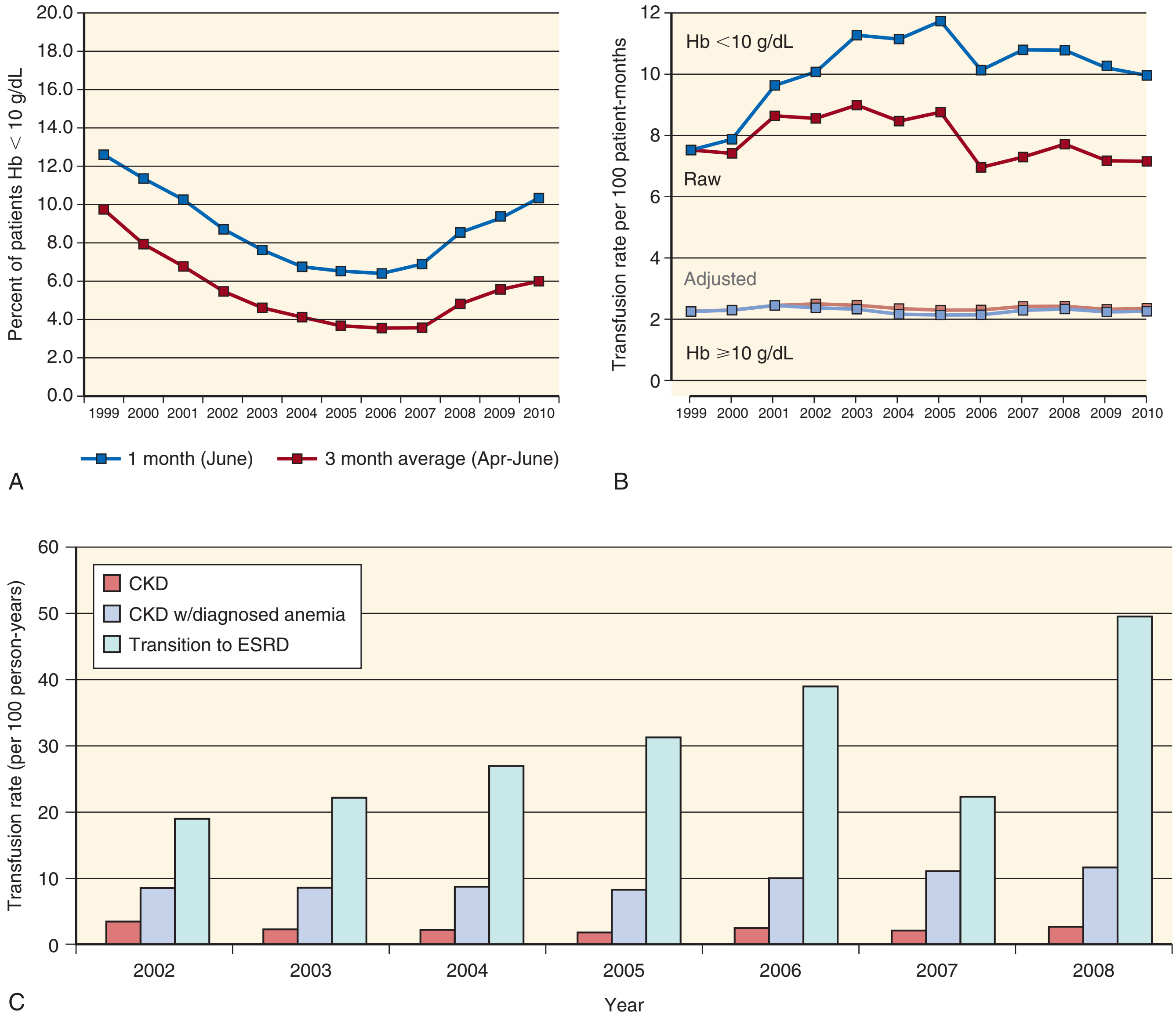

Fig. 55.10 Red blood cell transfusion in chronic kidney disease *(CKD)*. (A) Proportion of patients with a single monthly or 3-month average hemoglobin *(Hb)* concentration less than 10 g/dL, 1999–2010. (B) Unadjusted and adjusted (for age, gender, race, primary cause of end-stage kidney disease, hospitalization days, and dose of erythropoiesis-stimulating agent) 6-month transfusion rates for patients with Hb values less than 10 and 10 g/dL or more (1999–2010). (C) Annual red blood cell transfusion rates/100 person-years (2002–2008) for patients with CKD, patients with CKD and diagnosed anemia, and patients with CKD transitioning to end-stage kidney disease (ESKD) during each follow-up year. *ESRD,* End-stage renal disease. (A and B from Gilbertson DT, Monda KL, Bradbury BD, et al. RBC transfusions among hemodialysis patients (1999–2010): influence of hemoglobin concentrations below 10 g/dL. Am J Kidney Dis. 2013;62:919–928; C from Gill KS, Muntner P, Lafayette RA, et al. Red blood cell transfusion use in patients with chronic kidney disease. Nephrol Dial Transpl. 2013;28:1504–1515.)

HLA sensitization also increases graft rejection and diminishes graft survival. It is unlikely that blood transfusion has any benefit on subsequent allograft function. Although the published literature seems to indicate mixed results, patients who received transfusions and developed alloantibodies were less likely to undergo transplantation. It is unreasonable and potentially misleading to compare patients who received transfusion and did not develop alloantibodies with all patients who did not receive a transfusion. We agree with the current consensus that blood transfusion should be avoided, if possible.[722] Because transplantation waiting times exceed life expectancy throughout most of the United States and much of the developed world, patients can ill afford a procedure (i.e., transfusion) that offers modest if any benefit and that can further lower the likelihood of ever receiving a kidney transplant.

Relatively common complications of RBC transfusions are febrile, urticarial, and/or allergic (immediate hypersensitivity) reactions. Less common complications include acute and delayed hemolytic transfusion reactions, hypotensive transfusion reactions, transfusion-associated dyspnea, transfusion-associated circulatory overload, transfusion-related acute lung injury, posttransfusion purpura, and transfusion-associated graft versus host disease. Additional complications of RBC transfusion include potential transmission of known and unknown infectious agents and iron overload.

DISORDERS OF HEMOSTASIS IN CHRONIC KIDNEY DISEASE

BLEEDING AND CHRONIC KIDNEY DISEASE

Excessive bleeding has long been recognized as an important complication of the uremic state.[736–738] This was particularly true prior to the advent of dialysis and the availability of rhEPO. Events may be as minor as epistaxis, excessive bleeding with toothbrushing, or easy bruisability. More severe, clinically relevant bleeding episodes tend to occur with trauma or after invasive procedures, rather than spontaneously.[739] Before the availability of routine dialysis, catastrophic gastrointestinal hemorrhage was the major cause of death with uremia.[736] Bleeding is frequently a predictor of increased mortality risk or complications.[740,741]

PATHOPHYSIOLOGY

Traumatic disruption of the endothelial lining of blood vessels results in a complex and coordinated response aimed to maintain vascular integrity and prevent bleeding. The first line of defense in hemostasis is represented by platelets, which specifically interact with ligands exposed as a consequence of endothelial damage. These ligands, which include collagen, fibronectin, laminin, thrombospondin, and von Willebrand factor (vWF) promote the adhesion of platelets to subendothelium and their activation. Activated platelets further release adhesive ligands stored in their alpha granules, such as vWF, fibrinogen, thrombospondin, fibronectin, and vitronectin and promote the activation of additional platelets by releasing aggregating agents such as thromboxane A2 (TXA2) and adenosine diphosphate (ADP). An occlusive plug is eventually formed by the deposition of platelets on collagen fibers. The surfaces of platelets play an essential role in supporting the coagulation cascade in plasma, which results in the activation of thrombin, conversion of fibrinogen to fibrin, and formation of the fibrin plug, which is stabilized by factor XIIIa. The generation of thrombin further enhances the activation of platelets and upregulates glycoprotein (GP) receptors, such as those for GPIb-IX-V and GPIIb/IIIa. Several systems play an important role in limiting the extent of coagulation activation and thrombus formation. Nitric oxide (NO) and prostacyclin limit the activation of platelets. Tissue factor pathway inhibitor (TFPI), the protein C and S system, and antithrombin turn off activated coagulation factors at various steps of the coagulation cascade. The fibrinolytic system is also crucial in both limiting the growth of thrombi and promoting their organization and removal. Fibrin digestion is mediated by plasmin, which is circulating in plasma as plasminogen, an inactive precursor. Conversion of plasminogen to plasmin is promoted by tissue plasminogen activator (tPA) and inhibited by plasminogen activator inhibitors (PAI-1 and PAI-2).[742,743]

Several factors contribute to increase the risk of bleeding in patients with CKD (Fig. 55.11).[744] It has long been noted that bleeding in uremic patients occurs despite normal or elevated circulating values of coagulation factors.[736] This observation has suggested that platelet abnormalities are the primary cause of the bleeding diathesis. The function of platelets is often impaired (thrombasthenia), whereas the number of circulating platelets is generally normal, with perhaps a tendency to decrease the longer that patients have been undergoing dialysis.[745] Thrombopoietin values are elevated in patients who are on maintenance hemodialysis and being treated with rhEPO but do not correlate with platelet counts.[745,746] Evidence for platelet dysfunction includes elevated bleeding time,[736] diminished aggregation response to ADP and epinephrine,[747] reduced ristocetin-induced platelet agglutination,[748] and prolonged closure time with the Platelet Function Analyzer (PFA, Siemens Medical Solutions).[749–751]

The most consistent abnormality in platelet function in uremia is an impaired interaction of platelets with the vascular subendothelium.[736] As a result, platelet adhesion and aggregation are hindered. The cause of this dysfunction is incompletely understood and could be related to abnormalities of the vessel wall, platelets, or plasma constituents. As for the vessel wall, it appears that its function may be altered in uremia. In particular, endothelial production of NO, a powerful platelet inhibitor, has been noted to be increased,[752–754] resulting in higher concentrations of cyclic guanosine monophosphate (cGMP) and reduction of platelet responsiveness. In uremic rats, treatment with an NO inhibitor partially restores platelet function.[755] Interestingly, guanidinosuccinic acid (GSA), long postulated to play a role in uremic platelet dysfunction, has been found to upregulate NO production by the vascular endothelium.[756] Prostaglandin I2 (PGI2), which is released by endothelium, is increased in patients with CKD who exhibit increased bleeding times[757] and probably plays a role in reducing platelet aggregability.[757]

The platelet itself is intrinsically altered in uremia. For example, the content of serotonin and ADP is reduced in uremic platelet granules.[747] Secretion of mediators may also be impaired, although this may be a function of repeated activation during hemodialysis.[758] Platelet receptors that play a critical role in adhesion to the vessel wall and aggregation, such as those for GP1b and GPIIb-IIIa, are probably not significantly reduced in quantity in uremia.[759] However, interaction of the receptors with vessel wall proteins may be abnormal.[760] In particular, activation of GPIIb-IIIa to facilitate its adhesion to vWF may be impaired.[761] The platelet cytoskeleton may be altered, with diminished actin incorporation and suboptimal intracellular trafficking of molecules.[762–764]

Although the platelet itself is not entirely normal in uremia, it appears that a more important pathogenic factor in platelet dysfunction may be the effect of uremic plasma on platelet responsiveness. Platelets from normal individuals develop impaired adhesive function on exposure to uremic plasma.[765] In contrast, platelets from uremic subjects regain some function on exposure to normal plasma.[765] Certain molecules with molecular weights that preclude adequate clearance with hemodialysis accumulate in uremia and may contribute to platelet dysfunction.[766] A variety of toxins, including quinolinic acids and guanidine substances, have been implicated.[736] In addition, a role for hyperparathyroidism has been suggested. Benigni and colleagues have found that PTH impairs platelet aggregation induced by a variety of substances.[767] Hyperparathyroidism may affect platelet function by elevating intracellular calcium concentrations via channels that are sensitive to calcium channel blockers such as nifedipine.[768]

It is generally accepted that dialysis reduces uremic platelet dysfunction and the risk for bleeding. However, dialysis does not completely eliminate the problem. Moreover, hemodialysis may induce a transient worsening in platelet function. Sloand

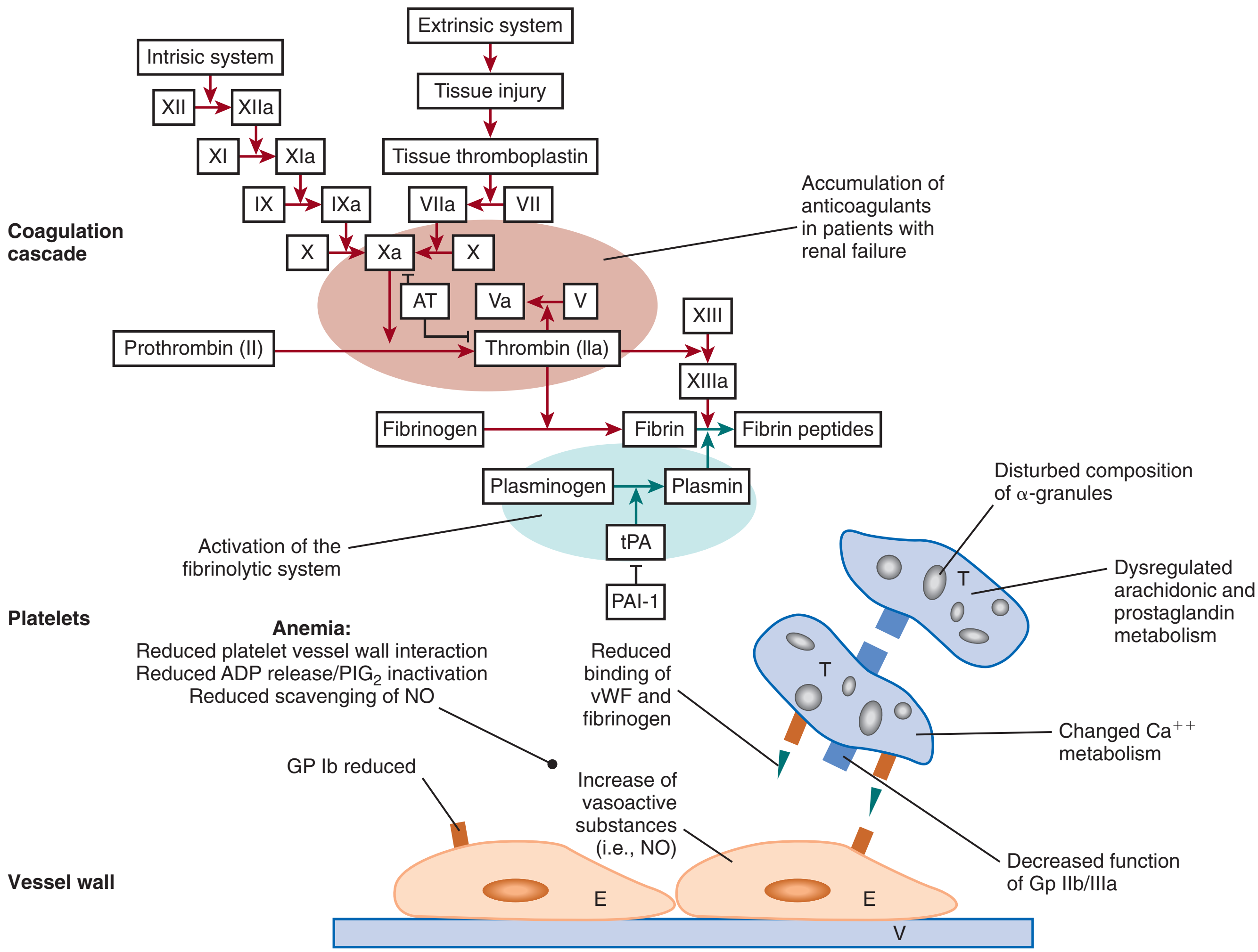

Fig. 55.11 Factors involved in the increased risk of bleeding in patients with renal failure. *Roman numerals with or without lower case letters* indicate clotting factors; *ADP,* Adenosine diphosphate; *AT,* antithrombin; *Ca^{2+},* calcium ion; *E,* endothelium; *GP,* glycoprotein; *NO,* nitric oxide; *PGI2,* prostaglandin I2; *T,* thrombocyte; *tPA,* tissue-type plasminogen activator; *V,* vessel; *vWF,* von Willebrand factor. (From Lutz J, Menke J, Sollinger D, et al. Haemostasis in chronic kidney disease. Nephrol Dial Transpl. 2014;29:29–40.)

and Sloand have measured a variety of indicators of platelet function immediately before and after treatments and noted a transient decrease of platelet membrane expression of GPIb after hemodialysis.[769] Ristocetin responsiveness was impaired after hemodialysis and normalized the day after treatment. Other potential detrimental consequences of hemodialysis include the enervating effect of repeated platelet activation,[770,771] removal of younger platelets with greater function,[772–774] and impairment of platelet function from a secondary effect of activated leukocytes.[775]

Anemia is an important contributor to uremic platelet dysfunction.[776] During normal circulation, erythrocytes tend to force the flow of platelets radially, away from the center of flow and toward the endothelial surfaces. When vascular injury occurs, platelets are in closer apposition to the vessel wall, facilitating platelet adherence and activation by vessel wall constituents such as collagen. With anemia, more platelets circulate in the center of the vessel, further from endothelial surfaces, hindering efficient platelet activation.[776] In addition, anemia may contribute to platelet dysfunction because release of ADP by erythrocytes normally stimulates platelet interaction with collagen.[777,778] Anemia has also been associated with increased urinary platelet activation markers in patients with CKD stages 1 to 4.[779] Treatment of anemia may help reverse platelet dysfunction because both transfusion of blood[776,780] and rhEPO therapy[781] have been found to be beneficial. Anemia is an important risk factor for hemorrhagic stroke—but not ischemic stroke—in patients undergoing hemodialysis.[782]

The plasma content of the major adhesive proteins, vWF and fibrinogen, are normal in uremia. One study has shown a normal distribution of vWF multimers,[748] but another reported a reduction in high-molecular-weight vWF multimers.[783] The functional properties of vWF are altered, however, mostly at the level of the interaction with the GPIb/IX-V platelet receptors, a key step in the signaling pathways, which ultimately lead to TxA2 production.[784–787]

Platelet-derived procoagulant microparticles have been described in CKD,[788,789] but inconsistent and unreliable measurement methodologies have hampered the assessment of their clinical relevance. Circulating endothelial microparticles are increased in CKD, and their baseline values (but not longitudinal changes) show an association with mortality risk.[790]

DIAGNOSTIC STUDIES

Despite abundant evidence that the bleeding time is an unreliable test, with limited value in predicting bleeding complications,[791] use of this test is still reported in studies of CKD.[744] More reliable tests are available, such as platelet aggregation and platelet function analyser (PFA), although their value in predicting and managing bleeding complications has been questioned.[792] Thrombin generation assays may help in assessing both hypocoagulable and hypercoagulable states, but so far there are only limited studies in patients with CKD.[793]

TREATMENT

The treatment of patients with renal failure experiencing bleeding episodes requires the following:

1. An assessment of the severity of blood loss
2. Hemodynamic stabilization
3. Replacement of blood products as needed
4. Identification of the bleeding source and cause
5. Correction of platelet dysfunction and other factors contributing to the bleeding diathesis (Fig. 55.12).

The first four factors are routine components of clinical care and are not discussed further here; the fifth extends from the previous discussion on the pathobiology of uremic bleeding. It should be clear, however, that the intensity of interventions to correct uremic platelet dysfunction hinges on the degree of bleeding severity.

The first aspect of treatment to correct uremic platelet dysfunction is provision of adequate dialysis. Initiation of dialysis will lead to some improvement in thrombasthenia and bleeding risk.[747,794] The PFA closure time improves in 25% of patients after a dialysis session.[795] No studies have fully elucidated the relative effectiveness of hemodialysis versus other dialytic modalities, but platelet activation measured by CD62 expression was increased by hemodiafiltration, whereas platelet degranulation products were increased in hemodialysis.[796] In any case, anticoagulation must be minimized. The

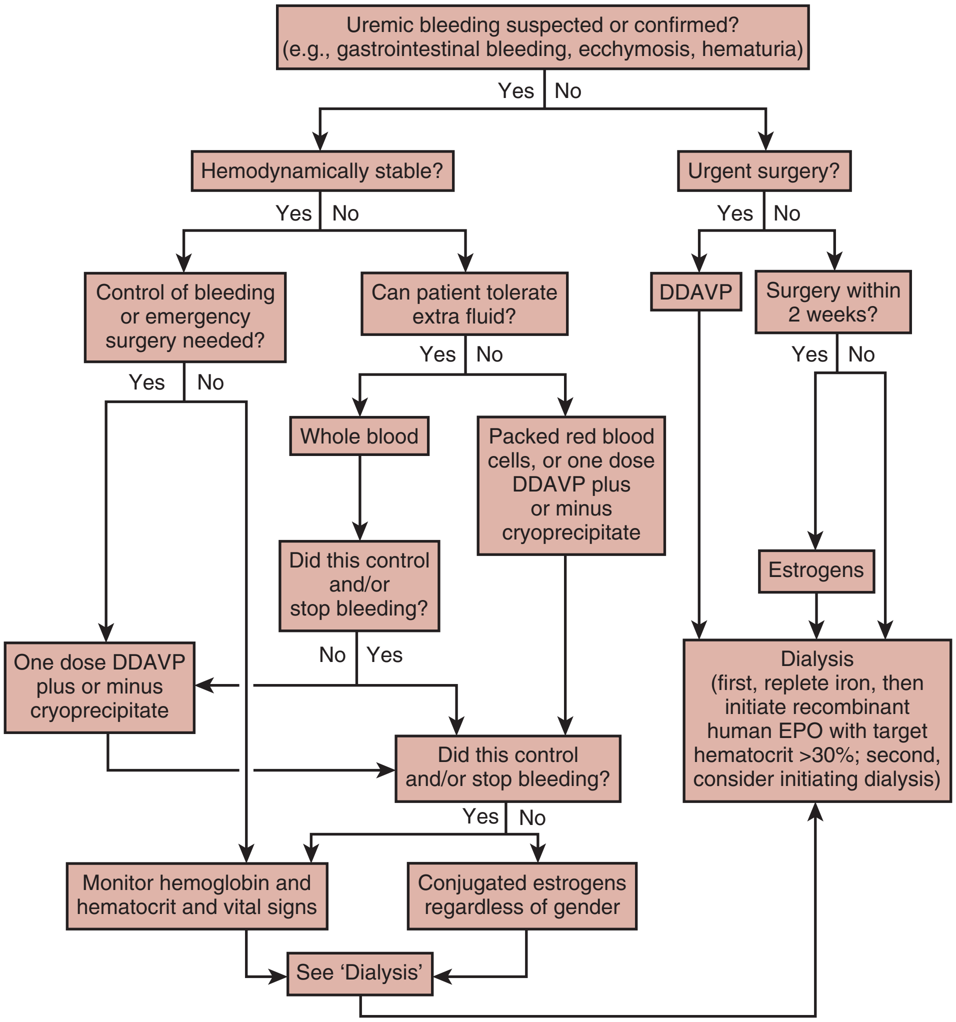

Fig. 55.12 Algorithm for the management of patients with uremic platelet dysfunction. If at any stage in the algorithm the patient with uremic platelet dysfunction starts to bleed actively, the clinician should return to the top of the algorithm. This algorithm is not intended to replace sound clinical judgment or prevent additional consideration of patient factors that could influence management decisions. *DDAVP,* Desmopressin (1-deamino-8-D-arginine vasopressin; single doses of 0.3–0.4 µg/kg body weight IV); *EPO,* erythropoietin. (From Hedges SJ, Dehoney SB, Hooper JS, et al: Evidence-based treatment recommendations for uremic bleeding. Nat Clin Pract Nephrol. 2007;3:138–153.)

relationship of dose of dialysis with improvement of platelet function has not been well studied.

Treatment of anemia with rhEPO may be the most effective treatment of uremic platelet dysfunction (see earlier). Cases and associates have found that treatment with epoetin alfa, 40 U/kg IV, results in improvement in several parameters of platelet function as Hgb increases.[797] Others have found the same salutary effect of ESA treatment.[110,781] Improved platelet function following EPO treatment is most likely related to the associated changes in blood flow, with platelets moving closer to the vessel walls. However, it is also possible that EPO treatment itself may directly affect platelet function. Tassies and colleagues have found that platelet function improves in some patients after epoetin treatment is initiated, before Hgb values increase.[798] They attributed this effect to an increase in young circulating forms of platelets, with improved functional characteristics. Other potential direct beneficial effects of ESA include improved platelet intracellular calcium mobilization,[799] increased expression of GPIb,[736,800] and repaired platelet signal transduction.[801]

Desmopressin (1-deamino-8-D-arginine vasopressin; DDAVP) is a synthetic form of antidiuretic hormone that is often used to treat uremic bleeding. The drug has little vasopressor activity and only rarely induces hyponatremia. The mechanism of improved platelet function is not completely known, but enhanced release of larger vWF multimers by endothelial cells probably plays an important role.[802,803] Other factors may include improved platelet aggregation on contact with collagen[804] and increased concentrations of platelet GPIb/IX.[805] Given the unreliability of the bleeding time, it is no surprise that IV infusion of DDAVP (0.3 μg/kg [3.0 μg/kg subcutaneously]) produced inconsistent results.[802,806–808] DDAVP infusion improves platelet function in vitro and increases plasma concentrations for both vWF and factor VIII.[749] DDAVP may also be administered by the intranasal route, at a dose approximately tenfold greater than that given intravenously.[809–811] Repeated administrations of DDAVP may result in a diminished response, with the development of tachyphylaxis caused by the depletion of the endothelial stores of vWF multimers.[812,813]

Other treatments for uremic bleeding include infusion of cryoprecipitate, a plasma product rich in vWF and fibrinogen.[814,815] There has been very little published evidence to support the use of cryoprecipitate, and responses appear to be highly variable. In one study of five patients with active bleeding, only two had normalization of bleeding time and a favorable clinical outcome after treatment.[816] Cryoprecipitate use should be reserved for life-threatening bleeding due to the risk for infectious complications and limited availability.

Estrogens improve platelet function in men and women.[817–819] After IV infusion, Livio and associates have found the beneficial effect of conjugated estrogens to begin early and last for up to 2 weeks.[820] The mechanism of action of estrogen treatment is not fully known, but it may be related to the inhibition of vascular NO production by decreasing production of its precursor, L-arginine.[821] Transdermal estrogens at low doses (≤50 μg/day) are recommended for long-term therapy to balance the hemostatic benefit with thrombotic risks and are preferable to oral ones.[822,823]

Short-term (6 days) and long-term (3 months) treatments with the fibrinolytic inhibitor tranexamic acid have been associated with a reduction in bleeding time and improved platelet function.[824,825] Tranexamic acid may also be beneficial in the treatment of acute upper gastrointestinal bleeding episodes.[826]

HYPERCOAGULABILITY AND CHRONIC KIDNEY DISEASE

Although bleeding is the most clinically relevant manifestation of the effects of advanced CKD on hemostasis, several lines of evidence indicate the presence of a prothrombotic, hypercoagulable state, which may play a role in the atherosclerotic/cardiovascular complications. Deep venous thrombosis (DVT) seems to affect predominantly CKD patients in younger age, of African-American or Hispanic background, in association with cardiovascular disease and prior surgical interventions.[827] The incidence of symptomatic venous thromboembolism is moderately increased in mild-to-moderate CKD (based on eGFR and albuminuria) as shown by a study pooling three European and two U.S.-based community cohorts,[828,829] as well as by a large population-based study in Denmark[830] and by smaller patient series.[831] The incidence of pulmonary embolism in CKD and ESKD is not precisely known (it may be particularly common after vascular access procedures – see Chapter 65), but mortality rates for pulmonary embolism are substantially higher in patients on dialysis than in the general population.[832] As described above, ESA therapy may further increase thromboembolic complications. Vascular calcification and cutaneous necrosis are the key features of calciphylaxis, a serious cutaneous disease affecting 1% to 4% of ESKD patients, which has been linked to a hypercoagulable state.[833]

EVIDENCE FOR HYPERCOAGULABILITY IN CHRONIC KIDNEY DISEASE

As outlined in Fig. 55.13, several pathways are altered—toward hypercoagulability and increased risk of thrombosis in CKD.[744] Activated hypercoagulable platelets have been reported in patients with impaired or declining kidney function,[834,835] whereas other studies have shown increases in soluble markers of activated coagulation and fibrinolysis.[836,837] The values of several markers for thrombin activation (prothrombin fragment F1.2 and thrombin-antithrombin complex) and fibrinolysis (D-dimer and plasmin-antiplasmin complex) are abnormally elevated in dialyzed CKD patients, with erythrocyte membrane phosphatidylserine externalization possibly playing a role in this procoagulant state.[838–841] Phosphatidylserine externalization may be mediated by uremic toxins because it improves after dialysis treatment.[842]

Despite the functional platelet defects described previously, abnormalities in the soluble coagulation cascade and in some of the natural anticoagulant systems (e.g., fibrinolysis) generate a hypercoagulable state, which may facilitate cardiovascular and thrombotic complications in patients undergoing dialysis.[843,844] Complement activation may take place during dialysis, with increased expression of tissue factor on peripheral neutrophils and increased production of granulocyte colony-stimulating factor (G-CSF) resulting in a hypercoagulable state.[845] Increased prothrombotic markers and platelet activation have been reported following dialysis,[846] with some variability based on the type of membranes used.[847] Thromboelastography has shown delayed clot formation and

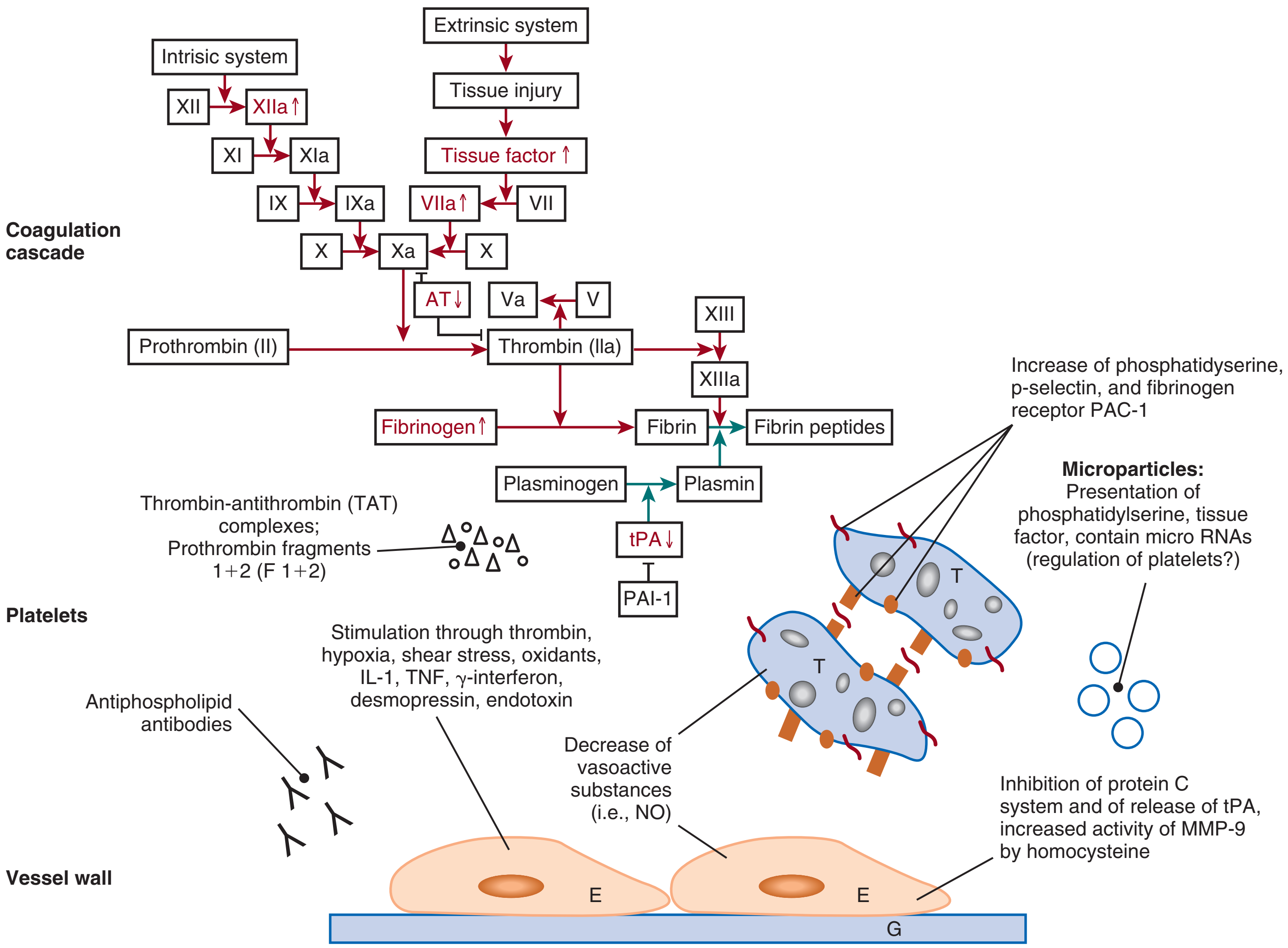

Fig. 55.13 Factors involved in the increased risk of thrombosis in patients with renal failure. *Roman numerals with/without lower case letters* indicate clotting factors; *AT,* Antithrombin; *E,* endothelium; *G,* subendothelial connective tissue; *IL-1,* interleukin-1; *MMP-9,* matrix metalloproteinase 9; *NO,* nitric oxide; *PAC-1,* monoclonal antibody specific for the activated form of GPIIb-IIIa; *PAI-1,* plasminogen activator inhibitor-1; *T,* thrombocytes; *TNF,* tumor necrosis factor; *tPA,* tissue-type plasminogen activator; *↓,* decreased; *↑,* increased. (From Lutz J, Menke J, Sollinger D, et al. Haemostasis in chronic kidney disease. Nephrol Dial Transpl 29:29–40, 2014.)

decreased clot breakdown in patients with CKD, with increased plasma fibrinogen values playing a potential role.[848] Indolic uremic solutes have also been implicated in the procoagulant phenotype of uremia in animal models[849] and in patients with CKD.[850] Oxidized plasma albumin has also been implicated in generating a prothrombotic state via CD-36-mediated platelet activation.[851]

PHARMACOLOGIC INTERVENTIONS

Treatment of hypercoagulability may expose patients to additional bleeding complications. A systematic review of bleeding rates in patients with CKD treated with antiplatelet drugs has shown that these agents are effective in reducing arteriovenous fistula and central venous catheter, but not arteriovenous graft thrombosis.[852] No firm conclusions could be reached about possible increases in bleeding rates in patients treated with a single agent, whereas there was an apparent increase in bleeding risk for combination therapy.[852] In patients treated for ischemic stroke, the presence of CKD was associated with a twofold increased frequency of clopidogrel resistance (by the VerifyNow P2Y12 Assay, Instrumentation Laboratory, Bedford, MA).[853]

Atrial fibrillation is a relatively common occurrence in patients with CKD and ESKD. The optimal approach to prevention of stroke and other embolic complications is unknown. There are concerns that chronic treatment with vitamin K antagonists may worsen vascular calcification.[854] Warfarin dosing is complicated in CKD and ESKD by drug-drug interactions, variability in dietary intake, and frequent administration of antibiotics, resulting in poor anticoagulation control.[855] The risk-benefit balance of warfarin for stroke prevention in advanced CKD and ESKD is unknown. The increased bleeding risk should be carefully considered,[856,857] especially in view of conflicting effects observed in older patients with atrial fibrillation and an eGFR less than 45 to 50 mL/min/1.73^2, which showed lower mortality despite a higher incidence of hemorrhage and higher or unchanged risk of ischemic stroke.[858,859]

One study has shown no reduction in the risk of stroke and higher bleeding risk in dialyzed CKD patients with atrial fibrillation treated with warfarin.[860] However, warfarin use to treat atrial fibrillation in patients with CKD post–myocardial infarction (MI) was associated with lower mortality and a lower incidence of MI and ischemic stroke, with no substantial higher risk of bleeding complications.[861] A Danish registry study has shown that in high-risk CKD patients, treatment of atrial fibrillation with warfarin results in measurable reductions in all-cause mortality and in hospitalization for stroke and bleeding.[862] This was contradicted by a small retrospective study that showed a substantially higher risk of death, bleeding, and MI for patients with CKD stages 3 to 5 and ESKD treated with warfarin.[863] A meta-analysis study has shown that warfarin therapy for atrial fibrillation in patients with CKD fails to reduce the risk of stroke and mortality, but it moderately increased the risk of bleeding.[864] A similar study in patients on chronic dialysis with atrial fibrillation and treated with warfarin has shown neither clear benefit nor clear harm.[865]

Novel oral anticoagulants (NOACs) have been approved for use in the general population; studies included a variable fraction (7%–21%) of subjects with impaired kidney function (eGFR <50 mL/min/1.73 m^2).[866] A systematic review and meta-analysis has shown no significant differences in thromboembolic or hemorrhagic complications in CKD patients treated with warfarin or NOACs.[867] Compared with warfarin, a similar study has shown a significant reduction for bleeding complications in patients with impaired renal function only for agents with lower renal excretion (<50%; e.g., apixaban, rivaroxaban, edoxaban).[866] In a primary care setting, bleeding risks and occurrence of stroke were similar in CKD patients with atrial fibrillation treated with an NOAC or warfarin.[868] Large randomized trials have included patients with reduced renal function and have shown efficacy for stroke prevention, with no increased bleeding risk.[856,869] However, due to the lack of specific trials in patients with CKD or ESKD, recommendations have been made to use NOAC in CKD stage G3,[870] not to use NOAC in advanced CKD.[871] and to perform proper studies in CKD.[872]

Dabigatran,[873] a direct thrombin inhibitor and apixaban[874] and rivaroxaban,[875] two factor Xa inhibitors, have been used in patients with CKD. Limited experience is available for newer agents, such as the indirect factor Xa inhibitor fondaparinux.[876,877] Although low-molecular- weight (LMW) and unfractionated heparin (UFH) can be reversed with the administration of protamine sulfate, newer agents such as fondaparinux have no specific antidotes, although recombinant factor VIIA and antithrombin have been used to reverse novel anticoagulant overdoses.[878–880] Table 55.3 provides some suggestions for the selection of an oral anticoagulant in patients with atrial fibrillation and CKD.[856] An essential part of this approach is the proper identification of stroke risk and bleeding risk using established tools such as the CHA2DS2-VASc and HAS-BLED scores, respectively,[856] although their utility has been questioned by a large Canadian study.[881]

The safety of omitting heparinization when dialyzing patients on chronic anticoagulation therapy with vitamin K antagonists has been demonstrated.[882] The PROMETHEUS study has shown no clinically significant difference in outcomes between prasugrel and clopidogrel treatment in patients with CKD undergoing percutaneous coronary intervention; it confirmed a higher adjusted risk for major adverse cardiac events and bleeding complications in the presence of CKD.[883]

HEPARIN-INDUCED THROMBOCYTOPENIA

Heparin-induced thrombocytopenia (HIT) can be seen in patients undergoing hemodialysis due to their repeated and frequent exposure to heparin.[884,885] The presence of antibodies to the platelet factor 4–heparin (PF4-H) complex has been associated with arterial and venous thrombosis and increased mortality,[886,887] but other studies have found no correlation between the presence of these antibodies and reduction in platelet counts,[888] clinical complications,[885,889] or vascular access thrombosis.[890] An acute thrombotic event in a thrombocytopenic patient on maintenance hemodialysis or unexpected occlusions of the extracorporeal circuit[891] should prompt a search for possible HIT. However, the isolated presence of PF4-H antibodies should not by itself lead to a diagnosis of HIT or institution of specific anti-HIT therapies. The presence of oversulfated chondroitin sulfate as a purposeful contaminant of heparins produced in China, which resulted in a large number of adverse events, has also been associated with an increased prevalence of PF4-H antibodies but no thrombocytopenia.[892] If the presence of HIT is confirmed based on established criteria,[885] all heparin-based therapies should be discontinued, and the use of direct thrombin inhibitors or factor Xa inhibitors should be considered. Warfarin should not be considered until the resolution of thrombocytopenia and neither should prophylactic platelet transfusions.

WHITE CELL FUNCTION IN CHRONIC KIDNEY DISEASE

CKD is accompanied by a chronic inflammatory state of complex pathogenesis, which is believed to be at least, in part, due to an increased generation of oxygen radicals and associated activation of monocytes. Uremic toxins have been blamed as a likely cause of this dysfunctional state, but with no specifically proven connections.[264,265,893,894] For more specific information on the biologic significance of uremic toxins; also, the European Uremic Solutes database (EUTox-db) can be found at http://eutoxdb.odeesoft.com/index.php. The use of particular dialyzers and dialysates has been associated with intradialytic leukocyte activation and enhanced oxidant stress, which may exacerbate the underlying activated inflammatory state. Activation of platelets adhering to dialysis membranes may contribute to leukocyte activation and the production of ROS.[895–901] Different types of synthetic dialysis membranes have been shown to induce different degrees of oxidative stress (measured in serum with the surrogate marker malondialdehyde or as ROS).[902,903]

LEUKOCYTE (MONOCYTE) ACTIVATION

Several studies have shown elevations in markers of leukocyte and monocyte activation in patients undergoing dialysis,[900,904–906] as well as increased heterotypic aggregation for leukocytes and lymphocytes.[907] Advanced oxidation protein products (AOPPs) carried mostly by serum albumin have been identified in the serum of patients undergoing dialysis.[908] These AOPPs

Table 55.3 Selection of Appropriate Oral Anticoagulant for Patients With Atrial Fibrillation and Chronic Kidney Disease

Drug (Renal Excretion[a])	CrCl (mL/min)[b] ≥50	30–49	15–29	<15	ESKD on RRT
Preferred class	NOAC	NOAC	VKA or NOAC	VKA or NOAC (use with caution)	VKA or NOAC (use with caution)
VKA (NA)	Maintain TTR ≥70%	Maintain TTR ≥70%	Maintain TTR ≥70%	Maintain TTR ≥70%	Maintain TTR ≥70%
Dabibatran (80%)	150 mg bid or 110 mg bid if patient is ≥80 years or receiving veraparamil or at increased bleeding risk	150 mg bid or 110 mg bid if patient is ≥80 years or receiving verapamil or at increased bleeding risk	United States, 75 mg; other country, do not use.	Do not use.	Do not use.
Rivaroxaban (35%)	20 mg once daily	15 mg once daily	15 mg once daily	Do not use.	Do not use.
Apixaban (27%)	5 mg bid or 2.5 mg bid if two or more of the following criteria are fulfilled: age ≥80 years, body weight ≤60 kg, and sCr ≥1.5 mg/dL (133 µmol/L)	5 mg bid or 2.5 mg bid if two or more of the following criteria are fulfilled: age ≥80 years, body weight ≤60 kg and sCr ≥1.5 mg/dL (133 µmol/L)	2.5 mg bid	United States, 5 mg bid; other country, do not use.	United States, 5 mg bid; other country, do not use.
Edoxaban (50%)	60 mg once daily or 30 mg once daily if two or more of the following criteria are fulfilled: body weight ≤60 kg; CrCl, 30–50 mL/min, and concomitant therapy with verapamil, dronedarone, or quinidine	30 mg once daily	30 mg once daily	Do not use.	Do not use.

[a]Fraction of absorbed dose.
[b]Estimated using the Cockcroft-Gault criteria.
CrCL, Creatinine clearance; *NA,* not available; *NOAC,* non–vitamin K antagonist oral anticoagulant; *sCr,* serum creatinine; *TTR,* time in therapeutic range; *VKA,* vitamin K antagonist.
Adapted from January CT, Wann LS, Alpert JS, et al. 2014 AHA/ACC/HRS guideline for the management of patients with atrial fibrillation: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society. *J Am College Cardiol.* 2014; 64:e1–e76; Kirchhof P, Benussi S, Kotecha D, et al. 2016 ESC guidelines for the management of atrial fibrillation developed in collaboration with EACTS *Eur. Heart J.* 2016; 37:2893–2962; and Heidbuchel H, Verhamme P, Alings M, et al. Updated European Heart Rhythm Association practical guide on the use of non-vitamin K antagonist anticoagulants in patients with non-valvular atrial fibrillation. *Europace.* 2015; 17:1467–1507.

are believed to be end products of protein oxidation. Their concentration is correlated with the severity of uremia, extent of monocyte activation (assessed by serum neopterin),[909,910] and generation of myeloperoxidase by neutrophils in dialysis patients but not in predialysis conditions.[911] AOPPs can trigger neutrophil activation and respiratory burst, which can be reduced in vitro by N-acetylcysteine.[912] Leukocyte 8-hydroxy-2′-deoxyguanosine (8-OHdG) is a marker of oxidant-induced DNA damage, which is particularly elevated in patients carrying a GST M1 polymorphic dysfunctional variant.[676] Unique functional features of monocytes collected from patients on hemodialysis include increased transendothelial migration and induction of multiple activation markers on cultured human endothelial cells, which depend on superoxide generation.[913]

Evidence for leukocyte activation and ROS generation has also been found in patients with ND-CKD.[914] Degranulation of neutrophils results in release of a variety of enzymes and proinflammatory mediators[915,916]; some of these mediators, such as heparanase,[917] an endoglycosidase involved in the degradation of extracellular matrix, have been linked to the generation of atherosclerotic lesions; others, such as myeloperoxidase, generate hypochlorous acid, a potent microbicidal and oxidant compound. Hypochlorous acid may play a role in activating monocytes, which produce a whole array of inflammatory cytokines (e.g., IL-6, TNF-α, IL-1β).

LEUKOCYTE FUNCTIONAL IMPAIRMENT

Granulocytes of patients undergoing hemodialysis exhibit impaired adhesion to fibronectin, especially in conditions of malnutrition.[918] Prominent apoptosis is observed in monocytes of patients with CKD,[919] and changes in monocyte subpopulations (CD16+) are associated with increased, soluble, proinflammatory markers, such as chemokine (C-X3-C motif) ligand 1 or CX(3)CL1.[920] Increased production of ROS and the accumulation of toxic products associated with uremia[264,265] are likely but not yet proven culprits in the generation of a dysfunctional immune response in patients with CKD. Phagocytosis by neutrophils is also impaired in patients on hemodialysis and is associated with increased serum endotoxin values.[921] Enhanced susceptibility to bacterial or viral infections and reduced response to hepatitis B vaccine have also been described.[922] Functional abnormalities in monocytes and T lymphocytes,[923–925] as well as in natural killer (NK) cells,[926] have been reported. It has been suggested that such abnormalities may be representative of a myeloid shift of erythropoiesis, similar to that observed with aging.[927] New dialysis membranes designed to reduce immune dysfunction are being developed, with encouraging but still preliminary results.[928]

MARKERS OF LEUKOCYTE ACTIVATION

Elevations in serum CRP or myeloperoxidase have been associated with a higher mortality risk in hemodialyzed patients.[929] Expression studies with oligonucleotide microarray chips have identified distinct patterns of inflammatory and oxidative stress responses in dialyzed patients,[930] with some evidence suggesting a possible pathogenetic role of an impairment of the mitochondrial respiratory system.[931] IV administration of vitamin C in a small cohort of patients has produced changes in markers of oxidant stress.[932] Vitamin C supplementation in hemodialysis is a controversial issue because of the requirements for IV administration, prolonged therapy, and risk of hyperoxaluria.[933]

A better identification of the pathogenesis and clarification of disease modifier genes will allow us to design better focused and more personalized treatment approaches for the inflammatory state associated with CKD.[934,935]

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A clinically relevant side effect observed following IV administration of ferric carboxymaltose is:
   a. Hyperphosphatemia
   b. Hypophosphatemia
   c. Hypernatremia
   d. Hyponatremia
   Answer: b

2. Chuvash congenital polycythemia, an autosomal recessive disorder endemic in the mid-Volga River region, is due to a mutation in:
   a. Hypoxia Inducible Factor (HIF)-a
   b. Erythropoietin (EPO)
   c. Erythropoietin (EPO)-receptor
   d. Von Hippel-Lindau (VHL) protein
   Answer: d

3. Hepcidin is a peptide hormone produced by the liver which affect iron metabolism by:
   a. Negatively regulating both duodenal iron absorption and iron release from macrophages
   b. Positively regulating both duodenal iron absorption and iron release from macrophages
   c. Negatively regulating only duodenal iron absorption and not iron release from macrophages
   d. Positively regulating only duodenal iron absorption and not iron release from macrophages
   Answer: a

4. Identify the test which most closely reflect the functional availability of iron for erythropoiesis in CKD.
   a. Serum ferritin
   b. Bone marrow iron stain
   c. Serum transferrin receptor
   d. Reticulocyte hemoglobin content
   Answer: d

5. Please identify the correct statement regarding the bleeding diathesis observed in patients with CKD.
   a. The platelet itself is normal in uremia.
   b. The interaction of platelets with the vascular endothelium is impaired in uremia.
   c. Uremic platelet dysfunction is independent of anemia.
   d. Adhesive proteins like von Willebrand factor (VWF) and fibrinogen are decreased in uremia.
   Answer: b